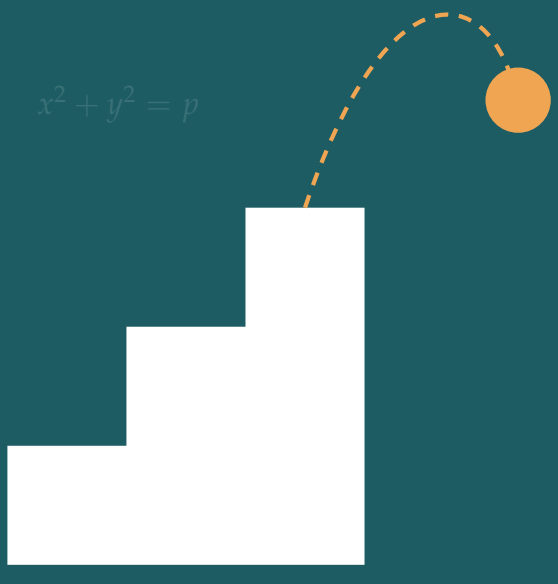
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

STEP I, II AND III

1994 Worked Solutions



KEITH McNULTY

Beyond STEP

Past STEP questions as a doorway
to university mathematics

STEP I, II AND III

1994 Worked Solutions

Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

1994 Worked Solutions

Worked solutions to all 38 questions of the 1994 STEP papers (STEP I, II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

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Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

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WORKED SOLUTION

STEP I 1994, Question 2

Pure • Logarithmic Differentiation

> **STEP I 1994, Question 2**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/94-s1-q2

The key idea

The power rule works when the variable is in the base and the exponent is fixed; the exponential rule works when the base is fixed and the variable is in the exponent. As soon as x appears in both places, neither rule applies directly. The cure is to write everything as a power of e, or equivalently to take logarithms and differentiate implicitly: logarithms turn awkward powers into products, and products we can handle.

Throughout, $x > 0$ (otherwise x^x and $\ln x$ make no sense), and in (ii) the base a must be positive.

Solution

The one tool we use repeatedly: for $u > 0$,

$$u^v = \mathrm{e}^{v \ln u},$$

so if $y = u^v$ then $\ln y = v \ln u$, and differentiating with the chain rule gives $\dfrac{1}{y}\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{\mathrm{d}}{\mathrm{d}x}(v \ln u)$.

Part (i)

Here the exponent is constant, so this is the ordinary power rule:

$$\frac{\mathrm{d}}{\mathrm{d}x}\, x^a = a\, x^{a-1}.$$

If you want to see why it holds for any real a (not just integers), write $x^a = \mathrm{e}^{a \ln x}$; the chain rule gives $\mathrm{e}^{a \ln x} \cdot \frac{a}{x} = a\, x^a / x = a\, x^{a-1}$.

Part (ii)

Now the base is constant. Writing $a^x = \mathrm{e}^{x \ln a}$, where $\ln a$ is just a constant,

$$\frac{\mathrm{d}}{\mathrm{d}x} a^x = \ln a \, \mathrm{e}^{x \ln a} = a^x \ln a.$$

Note how different this is from (i): it is *not* $x\, a^{x-1}$.

Part (iii)

Let $y = x^x$. Taking logarithms, $\ln y = x \ln x$. Differentiating both sides, using the product rule on the right,

$$\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}x} = \ln x + x \cdot \frac{1}{x} = 1 + \ln x,$$

so

$$\frac{\mathrm{d}}{\mathrm{d}x} x^x = x^x \, (1 + \ln x).$$

? **Ask yourself** Why can't we just use the power rule and get $x \cdot x^{x-1} = x^x$?

The power rule assumes the exponent is constant, and the exponential rule assumes the base is constant. Here both vary. In fact the answer is exactly the sum of the two "wrong" answers: treating the exponent as fixed gives $x \cdot x^{x-1} = x^x$, treating the base as fixed gives $x^x \ln x$, and the true derivative is $x^x + x^x \ln x$. This is the multivariable chain rule in disguise: when $y = u^v$, each variable contributes its own rate of change.

Part (iv)

Let $y = x^{(x^x)}$, a tower in which the exponent is itself x^x. Taking logarithms,

$$\ln y = x^x \ln x.$$

Differentiate the right-hand side with the product rule, using (iii) for the derivative of x^x:

$$\begin{aligned}\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}x} &= x^x(1 + \ln x) \cdot \ln x + x^x \cdot \frac{1}{x} \\ &= x^x \left(\frac{1}{x} + \ln x + (\ln x)^2 \right).\end{aligned}$$

Multiplying by $y = x^{(x^x)}$ and combining the powers of x,

$$\frac{\mathrm{d}}{\mathrm{d}x} x^{(x^x)} = x^{(x^x)} \, x^x \left(\frac{1}{x} + \ln x + (\ln x)^2 \right) = x^{x^x + x} \left(\frac{1}{x} + \ln x + (\ln x)^2 \right).$$

Equivalently, taking the $\frac{1}{x}$ inside, $x^{x^x + x - 1}(1 + x \ln x + x(\ln x)^2)$.

Part (v)

? Ask yourself Isn't $(x^x)^x$ the same as the expression in (iv)?

No. By the law of indices $(u^m)^n = u^{mn}$, we have $(x^x)^x = x^{x\cdot x} = x^{x^2}$, whereas in (iv) the exponent is x^x, not x^2. They happen to agree at $x = 1$ and $x = 2$ (both give 16 at $x = 2$), but at $x = 3$ one is 3^{27} and the other only 3^9.

So let $y = x^{x^2}$. Then $\ln y = x^2 \ln x$, and by the product rule

$$\frac{1}{y}\frac{\mathrm{d}y}{\mathrm{d}x} = 2x\ln x + x^2 \cdot \frac{1}{x} = x\,(1 + 2\ln x).$$

Hence

$$\frac{\mathrm{d}}{\mathrm{d}x}\,(x^x)^x = x^{x^2} \cdot x\,(1 + 2\ln x) = x^{x^2+1}\,(1 + 2\ln x).$$

(You could instead differentiate $(x^x)^x$ directly: $\ln y = x\ln(x^x) = x^2 \ln x$, the same thing.)

Checks

- At $x = 1$ every one of (iii), (iv) and (v) gives derivative 1: for instance in (iv), $1^{1+1}(1 + 0 + 0) = 1$. A central difference $\big(f(1.000001) - f(0.999999)\big)/0.000002$ agrees.
- x^x has a minimum at $x = \mathrm{e}^{-1}$, where $1 + \ln x = 0$; and $(x^x)^x = x^{x^2}$ has a minimum at $x = \mathrm{e}^{-1/2}$, where $1 + 2\ln x = 0$. Both are visible on a graphical calculator.
- We compared every formula with a numerical derivative at 2000 random points with $0.2 \le x \le 2.5$ (and $0.2 \le a \le 4$); the relative error was always below 10^{-9}.

Common slips

- Writing $\frac{\mathrm{d}}{\mathrm{d}x}x^x = x \cdot x^{x-1}$ or $\frac{\mathrm{d}}{\mathrm{d}x}a^x = x\,a^{x-1}$: the power rule needs a constant exponent.
- Treating (iv) and (v) as the same function. Powers are not associative: $x^{(x^x)} \neq (x^x)^x$ in general.
- Forgetting to multiply back by y after differentiating $\ln y$.

Beyond STEP

What does x^a even mean when a is irrational, say $x^{\sqrt{2}}$? At university the answer is to *define* $u^v = \mathrm{e}^{v\ln u}$ for $u > 0$, after first defining exp by its power series and ln as its inverse; all the rules of indices and the derivatives above then become theorems. This is done carefully in Abbott, *Understanding Analysis*, or in Bartle and Sherbert, *Introduction to Real Analysis*. The "sum of the two wrong answers" in (iii) is a first glimpse of partial derivatives and the multivariable chain rule.

WORKED SOLUTION

STEP I 1994, Question 4

Pure • Inverse Tangent Integrals

STEP I 1994, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s1-q4

The key idea

Part (i) is a half-angle identity, and part (ii) is the standard "complete the square, then use $\tan^{-1}$" integral. The final integral is part (ii) with $k = \cos\alpha$, which makes $\sqrt{1-k^2} = \sin\alpha$ and cancels the factor $\sin\alpha$ on top. The only real care needed is at the end: evaluating $\tan^{-1}$ of a tangent gives back the angle only when that angle lies in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

Solution

Part (i)

Using the double-angle formulae $1 - \cos\alpha = 2\sin^2\frac{1}{2}\alpha$ and $\sin\alpha = 2\sin\frac{1}{2}\alpha\cos\frac{1}{2}\alpha$,

$$\frac{1-\cos\alpha}{\sin\alpha} = \frac{2\sin^2\frac{1}{2}\alpha}{2\sin\frac{1}{2}\alpha\cos\frac{1}{2}\alpha} = \frac{\sin\frac{1}{2}\alpha}{\cos\frac{1}{2}\alpha} = \tan\tfrac{1}{2}\alpha,$$

which holds whenever $\sin\alpha \neq 0$ (so both sides are defined). This is what we had to show.

Part (ii)

Complete the square in the denominator:

$$1 - 2kx + x^2 = (x-k)^2 + (1-k^2).$$

Since $|k| < 1$, the constant $1 - k^2$ is positive, so we may write $1 - k^2 = a^2$ with $a = \sqrt{1-k^2} > 0$. (This is exactly where the condition $|k| < 1$ is used: otherwise the denominator would factorise and we would get logarithms instead.) Substituting

$x - k = a\tan\theta$, so that $\mathrm{d}x = a\sec^2\theta\,\mathrm{d}\theta$,

$$\int \frac{\mathrm{d}x}{(x-k)^2+a^2} = \int \frac{a\sec^2\theta}{a^2\tan^2\theta + a^2}\,\mathrm{d}\theta = \int \frac{1}{a}\,\mathrm{d}\theta = \frac{\theta}{a} + C$$
$$= \frac{1}{\sqrt{1-k^2}}\tan^{-1}\left(\frac{x-k}{\sqrt{1-k^2}}\right) + C,$$

as required. (Equivalently, differentiate the right-hand side: the chain rule gives $\frac{1}{a}\cdot\frac{1/a}{1+(x-k)^2/a^2} = \frac{1}{(x-k)^2+a^2}$.)

The final integral

Take $k = \cos\alpha$. Since $0 < \alpha < \pi$, we have $|k| < 1$, so part (ii) applies, and $\sin\alpha > 0$ gives

$$\sqrt{1-k^2} = \sqrt{\sin^2\alpha} = \sin\alpha$$

(not $-\sin\alpha$, because $\sin\alpha > 0$ on this interval). The factor $\sin\alpha$ in the numerator therefore cancels the factor $1/\sqrt{1-k^2}$ from part (ii):

$$\int_0^1 \frac{\sin\alpha}{1-2x\cos\alpha+x^2}\,\mathrm{d}x = \left[\tan^{-1}\left(\frac{x-\cos\alpha}{\sin\alpha}\right)\right]_0^1$$
$$= \tan^{-1}\left(\frac{1-\cos\alpha}{\sin\alpha}\right) - \tan^{-1}\left(\frac{-\cos\alpha}{\sin\alpha}\right)$$
$$= \tan^{-1}\left(\tan\tfrac{1}{2}\alpha\right) + \tan^{-1}(\cot\alpha),$$

using part (i) for the first term and the fact that $\tan^{-1}$ is an odd function for the second.

? **Ask yourself** Is $\tan^{-1}(\tan\theta)$ always equal to θ?

No: $\tan^{-1}$ takes values in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, so $\tan^{-1}(\tan\theta) = \theta$ only for θ in that interval. We check each term.

- Since $0 < \alpha < \pi$, we have $0 < \frac{1}{2}\alpha < \frac{\pi}{2}$, so $\tan^{-1}\left(\tan\frac{1}{2}\alpha\right) = \frac{1}{2}\alpha$.
- Write $\cot\alpha = \tan\left(\frac{\pi}{2} - \alpha\right)$. Since $0 < \alpha < \pi$, the angle $\frac{\pi}{2} - \alpha$ lies in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, so $\tan^{-1}(\cot\alpha) = \frac{\pi}{2} - \alpha$.

Adding,

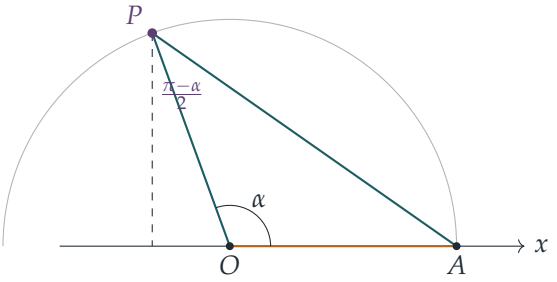
$$\int_0^1 \frac{\sin\alpha}{1-2x\cos\alpha+x^2}\,\mathrm{d}x = \frac{\alpha}{2} + \frac{\pi}{2} - \alpha = \frac{\pi-\alpha}{2},$$

which is the required result.

Otherwise: a picture

There is a lovely geometric reason for the answer. Let $P = (\cos\alpha, \sin\alpha)$ on the unit circle, O the origin and $A = (1, 0)$. As a point $X = (x, 0)$ moves along the x-axis, the angle θ between PX and the downward vertical through P satisfies $\tan\theta = (x - \cos\alpha)/\sin\alpha$,

and the integrand is exactly $\mathrm{d}\theta/\mathrm{d}x$. So the integral is the angle swept out as X moves from O to A, namely the angle OPA. Triangle OPA is isosceles with $OP = OA = 1$ and apex angle α at O, so its base angles are each $\frac{1}{2}(\pi - \alpha)$.



Checks

- At $\alpha = \frac{\pi}{2}$ the integral is $\int_0^1 \frac{\mathrm{d}x}{1+x^2} = \frac{\pi}{4}$, and the formula gives $\frac{1}{2}(\pi - \frac{\pi}{2}) = \frac{\pi}{4}$.
- Numerically, with $\alpha = 1$, Simpson's rule gives $1.0707963\ldots$, matching $\frac{1}{2}(\pi - 1) = 1.0707963\ldots$.
- As $\alpha \to \pi^-$ the integrand tends to 0 for each x in $[0, 1]$, and so does $\frac{1}{2}(\pi - \alpha)$.

Common slips

- Writing $\tan^{-1}(\cot\alpha) = \frac{\pi}{2} - \alpha$ without checking that $\frac{\pi}{2} - \alpha$ lies in the range of $\tan^{-1}$; the same move fails for α outside $(0, \pi)$.
- Taking $\sqrt{1 - \cos^2\alpha} = |\sin\alpha|$ to be $\sin\alpha$ without saying why; here it is because $0 < \alpha < \pi$.

Beyond STEP

As $\alpha \to 0^+$ the answer tends to $\frac{\pi}{2}$, even though at $\alpha = 0$ the integrand is identically zero: the integrand becomes a tall, narrow spike near $x = 1$ whose area does not shrink. This is the Poisson kernel for the half-plane, and swapping limits and integrals carelessly gives the wrong answer. When such a swap is allowed is a central question of first-year analysis; see Abbott, *Understanding Analysis*. The geometric "angle subtended" picture is developed beautifully in Needham, *Visual Complex Analysis*.

WORKED SOLUTION

STEP I 1994, Question 5

Pure • Loci on a Parabola

STEP I 1994, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s1-q5

The key idea

Turn the geometric condition into algebra first. A right angle at the origin means the position vectors $\overrightarrow{OP}$ and $\overrightarrow{OQ}$ are perpendicular, and for points on $y = x^2$ this collapses to a single neat relation between p and q. Both points R and T then have coordinates that are symmetric in p and q, so we write everything in terms of $p+q$ and pq and eliminate.

Solution

The condition for a right angle

Neither P nor Q can be the origin, or the angle would not be defined, so $p \neq 0$ and $q \neq 0$. The angle POQ is a right angle exactly when the scalar product of $\overrightarrow{OP} = (p, p^2)$ and $\overrightarrow{OQ} = (q, q^2)$ is zero:

$$pq + p^2q^2 = 0 \quad \Longleftrightarrow \quad pq(1+pq) = 0 \quad \Longleftrightarrow \quad pq = -1,$$

dividing by $pq \neq 0$. (Equivalently, the gradients p and q of OP and OQ multiply to -1.)

So the pairs we want are exactly $q = -1/p$ with $p \neq 0$. As p runs over the non-zero reals, the sum

$$s = p + q = p - \frac{1}{p}$$

takes *every* real value: given any s, the quadratic $p^2 - sp - 1 = 0$ has discriminant $s^2 + 4 > 0$, so it has real roots, and their product is -1, so neither is zero. We will need this to say that the loci below are complete curves rather than parts of them.

Part (i)

The midpoint of PQ is

$$R = \left(\frac{p+q}{2}, \frac{p^2+q^2}{2} \right).$$

Write $R = (x, y)$, so $p + q = 2x$. Using $pq = -1$,

$$y = \frac{(p+q)^2 - 2pq}{2} = \frac{4x^2 + 2}{2} = 2x^2 + 1.$$

Since $p + q$ takes every real value, so does x, and the locus of R is the whole parabola

$$y = 2x^2 + 1,$$

with vertex $(0, 1)$ and axis the y-axis. It is narrower than $y = x^2$ and lies entirely above it, since $2x^2 + 1 - x^2 = x^2 + 1 > 0$; this fits with R being the midpoint of a chord of a convex curve.

Part (ii)

Differentiating $y = x^2$, the tangent at P has gradient $2p$, so its equation is

$$y - p^2 = 2p(x - p), \qquad \text{that is,} \qquad y = 2px - p^2,$$

and similarly the tangent at Q is $y = 2qx - q^2$. These are not parallel, since $p \neq q$ (their product is -1, so they cannot be equal). Subtracting the two equations,

$$2(p - q)x = p^2 - q^2 \quad \Longrightarrow \quad x = \frac{p+q}{2},$$

dividing by $p - q \neq 0$. Then

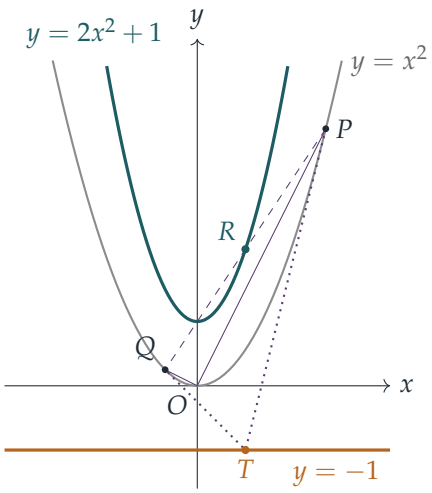
$$y = 2p \cdot \frac{p+q}{2} - p^2 = pq = -1.$$

Ask yourself The y-coordinate of T is constant. Is the locus a single point, a segment, or the whole line?

The x-coordinate of T is $\frac{1}{2}(p + q)$, which we showed takes every real value. So T can be anywhere on the line, and the locus of T is the whole straight line

$$y = -1.$$

Notice also that T always lies directly below R: both have x-coordinate $\frac{1}{2}(p + q)$.



The figure shows one position, $p = 2$, $q = -\frac{1}{2}$: here $R = (\frac{3}{4}, \frac{17}{8})$ on the teal parabola and $T = (\frac{3}{4}, -1)$ on the amber line.

Checks

Take $p = 1$, $q = -1$: then $P = (1, 1)$, $Q = (-1, 1)$, and OP, OQ are the lines $y = \pm x$, clearly perpendicular. The midpoint is $(0, 1)$, the vertex of $y = 2x^2 + 1$, and the tangents $y = 2x - 1$ and $y = -2x - 1$ meet at $(0, -1)$, on $y = -1$. With $p = 2$, $q = -\frac{1}{2}$ as in the figure, $2 \cdot (\frac{3}{4})^2 + 1 = \frac{17}{8}$, as it should be. A numerical check over 10^5 random values of p confirmed that R satisfies $y = 2x^2 + 1$ to within about 10^{-10} and that T always has $y = -1$.

Common slips

- Stopping at $pq(1 + pq) = 0$ without explaining why $pq = 0$ is ruled out: it would put P or Q at the origin.
- Giving the equations of the loci without checking that every x is attained, so that the sketch is of the whole curve rather than part of it.
- Assuming T lies on the directrix. The directrix of $y = x^2$ is $y = -\frac{1}{4}$; tangents meet there when PQ passes through the focus, which is a different condition from $\angle POQ = 90°$.

Beyond STEP

The chord PQ here always passes through the fixed point $(0, 1)$: its equation is $y = (p + q)x - pq = (p + q)x + 1$. Families of lines through a fixed point, and the matching "poles" where tangents meet (here on the line $y = -1$, the polar of $(0, 1)$), are the start

of projective geometry, where conics and their pole–polar relationships are studied systematically. For more problems in this spirit, try Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP I 1994, Question 6

Pure • Complex Function Composition

STEP I 1994, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s1-q6

The key idea

To find the real and imaginary parts of a quotient, multiply top and bottom by the conjugate of the denominator; with x real, that conjugate is easy to write down. For the repeated function, it is much cleaner to compose f with itself *algebraically*, for a general z, than to push the messy real and imaginary parts from (i) through the formula a second time. Once $\mathrm{f}(\mathrm{f}(z))$ is in a tidy form, the last part is a one-line substitution.

Solution

Since x is real, $\mathrm{i}x+1$ has real part 1, so it is never zero and $\mathrm{f}(x)$ is defined for every real x.

Part (i)

The conjugate of $\mathrm{i}x+1$ is $1-\mathrm{i}x$, because x is real. Multiplying top and bottom by it,

$$\begin{aligned}\mathrm{f}(x) = \frac{(\mathrm{i}x-1)(1-\mathrm{i}x)}{(1+\mathrm{i}x)(1-\mathrm{i}x)} &= \frac{\mathrm{i}x+x^2-1+\mathrm{i}x}{1+x^2} \\ &= \frac{x^2-1}{x^2+1} + \mathrm{i}\,\frac{2x}{x^2+1}.\end{aligned}$$

Both fractions are real, so these are the real and imaginary parts, as required.

Part (ii)

For any complex w, $ww^* = (\operatorname{Re} w)^2 + (\operatorname{Im} w)^2$. Using part (i),

$$\mathrm{f}(x)\mathrm{f}(x)^* = \frac{(x^2-1)^2+4x^2}{(x^2+1)^2} = \frac{x^4+2x^2+1}{(x^2+1)^2} = 1.$$

So $|\mathrm{f}(x)| = 1$: f sends every real number to a point on the unit circle.

Part (iii)

Rather than substituting the answer to (i) into f, we compose for a general z. Put $w = \mathrm{f}(z) = \dfrac{\mathrm{i}z - 1}{\mathrm{i}z + 1}$ and multiply the top and bottom of $\mathrm{f}(w) = \dfrac{\mathrm{i}w - 1}{\mathrm{i}w + 1}$ by $\mathrm{i}z + 1$:

$$\begin{aligned}\mathrm{f}(\mathrm{f}(z)) &= \frac{\mathrm{i}(\mathrm{i}z - 1) - (\mathrm{i}z + 1)}{\mathrm{i}(\mathrm{i}z - 1) + (\mathrm{i}z + 1)} = \frac{-z - \mathrm{i} - \mathrm{i}z - 1}{-z - \mathrm{i} + \mathrm{i}z + 1} \\ &= \frac{-(1+\mathrm{i})(z+1)}{(1-\mathrm{i})(1-z)} = \frac{1+\mathrm{i}}{1-\mathrm{i}} \cdot \frac{z+1}{z-1} = \mathrm{i}\,\frac{z+1}{z-1},\end{aligned}$$

since $\dfrac{1+\mathrm{i}}{1-\mathrm{i}} = \dfrac{(1+\mathrm{i})^2}{2} = \mathrm{i}$.

Ask yourself Is $\mathrm{f}(\mathrm{f}(x))$ defined for every real x?

No. We need $\mathrm{i}w + 1 \neq 0$, that is, $\mathrm{f}(x) \neq \mathrm{i}$. From part (i), $\mathrm{f}(x) = \mathrm{i}$ needs $x^2 - 1 = 0$ and $2x = x^2 + 1$, which happens only for $x = 1$. This matches the factor $x - 1$ in the denominator above. So for real $x \neq 1$,

$$\mathrm{f}(\mathrm{f}(x)) = \mathrm{i}\,\frac{x+1}{x-1}, \qquad \text{so} \qquad \operatorname{Re}\mathrm{f}(\mathrm{f}(x)) = 0, \quad \operatorname{Im}\mathrm{f}(\mathrm{f}(x)) = \frac{x+1}{x-1}.$$

The real line is sent to the unit circle by the first application of f, and the unit circle is sent to the imaginary axis by the second.

Part (iv)

Write $\mathrm{f}(\mathrm{f}(x)) = \mathrm{i}y$ with $y = \dfrac{x+1}{x-1}$ real. Then $\mathrm{i}(\mathrm{i}y) = -y$, so

$$\mathrm{f}(\mathrm{i}y) = \frac{-y-1}{-y+1} = \frac{y+1}{y-1}.$$

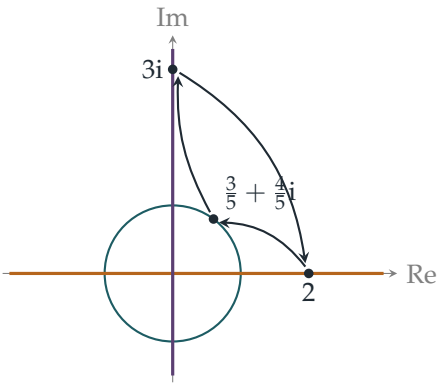
This is defined because $y = 1$ would need $x + 1 = x - 1$, which is impossible. Now

$$y + 1 = \frac{2x}{x-1}, \qquad y - 1 = \frac{2}{x-1},$$

so

$$\mathrm{f}(\mathrm{f}(\mathrm{f}(x))) = x \qquad \text{for every real } x \neq 1.$$

Applying f three times brings us back to where we started: the third application sends the imaginary axis back to the real line. The picture shows the cycle for $x = 2$, which goes $2 \mapsto \frac{3}{5} + \frac{4}{5}\mathrm{i} \mapsto 3\mathrm{i} \mapsto 2$.



Checks

- $x = 0$: $\mathrm{f}(0) = -1$, which is on the unit circle and agrees with (i). Then $\mathrm{f}(-1) = \dfrac{-\mathrm{i}-1}{-\mathrm{i}+1} = -\mathrm{i}$, matching $\mathrm{i}\frac{0+1}{0-1}$, and $\mathrm{f}(-\mathrm{i}) = \frac{1-1}{1+1} = 0$.
- $x = 2$: part (i) gives $\frac{3}{5}+\frac{4}{5}\mathrm{i}$, part (iii) gives 3i, and part (iv) returns 2. A numerical check on 10^4 random real x (and random complex z) confirmed every formula above to rounding error.

Common slips

- Substituting the real and imaginary parts from (i) into f directly. It works, but the algebra is heavy and easy to get wrong. Composing in terms of z is far cleaner.
- Forgetting that $x = 1$ must be excluded from (iii) and (iv), because $\mathrm{f}(1) = \mathrm{i}$ and $\mathrm{f}(\mathrm{i})$ has a zero denominator.

Beyond STEP

A function of the form $z \mapsto \dfrac{az+b}{cz+d}$ is a *Möbius transformation*. Such maps send circles and lines to circles and lines, as we saw here: real axis to unit circle to imaginary axis and back again. Composing two of them corresponds to multiplying the 2×2 matrices $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$. In fact $\mathrm{f}(\mathrm{f}(\mathrm{f}(z))) = z$ for every complex z, so f has order 3 in the group of Möbius transformations. Also, putting $x = \tan\frac{\theta}{2}$ in (i) gives $\mathrm{f}(x) = -\cos\theta + \mathrm{i}\sin\theta$, the familiar "$t$-substitution" parametrisation of the circle. Needham's *Visual Complex Analysis* has a beautiful chapter on Möbius transformations and their geometry.

WORKED SOLUTION

STEP I 1994, Question 8

Pure • Integration by Substitution

STEP I 1994, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s1-q8

The key idea

The substitution $\theta = \frac{1}{4}\pi - \phi$ reflects the interval $[0, \frac{1}{4}\pi]$ onto itself, and the addition formula turns $1 + \tan(\frac{1}{4}\pi - \phi)$ into $2/(1 + \tan\phi)$. So the integral reappears on the right-hand side with a minus sign, and we can solve for it rather than integrate anything difficult. The two later integrals are this same integral in disguise: one after $x = \tan\theta$, the other after half-angle formulae.

Solution

The first integral

Call the integral I. Put $\theta = \frac{1}{4}\pi - \phi$, so $\mathrm{d}\theta = -\mathrm{d}\phi$; when $\theta = 0$, $\phi = \frac{1}{4}\pi$, and when $\theta = \frac{1}{4}\pi$, $\phi = 0$. Hence

$$I = \int_{\frac{1}{4}\pi}^{0} \ln\bigl(1 + \tan(\tfrac{1}{4}\pi - \phi)\bigr)\,(-\mathrm{d}\phi) = \int_{0}^{\frac{1}{4}\pi} \ln\bigl(1 + \tan(\tfrac{1}{4}\pi - \phi)\bigr)\,\mathrm{d}\phi.$$

By the addition formula, using $\tan\frac{1}{4}\pi = 1$,

$$1 + \tan(\tfrac{1}{4}\pi - \phi) = 1 + \frac{1 - \tan\phi}{1 + \tan\phi} = \frac{2}{1 + \tan\phi}.$$

(On $[0, \frac{1}{4}\pi]$ we have $\tan\phi \geq 0$, so nothing vanishes and every logarithm is of a positive number.) Therefore

$$I = \int_{0}^{\frac{1}{4}\pi} \bigl(\ln 2 - \ln(1 + \tan\phi)\bigr)\,\mathrm{d}\phi = \tfrac{1}{4}\pi \ln 2 - I,$$

because the last integral is I again, with the dummy variable renamed.

Ask yourself We never found an antiderivative. Why is that allowed?

We don't need one. I is just a number, and the substitution gives an equation $I = \frac{1}{4}\pi \ln 2 - I$ which that number must satisfy. Solving it,

$$2I = \tfrac{1}{4}\pi \ln 2 \quad \Longrightarrow \quad \int_0^{\frac{1}{4}\pi} \ln(1+\tan\theta)\,\mathrm{d}\theta = \tfrac{1}{8}\pi \ln 2,$$

as required.

The integral with $1 + x^2$

The factor $\dfrac{\mathrm{d}x}{1+x^2}$ suggests $x = \tan\theta$, so $\mathrm{d}x = \sec^2\theta\,\mathrm{d}\theta = (1+x^2)\,\mathrm{d}\theta$. The limits $x = 0$ and $x = 1$ become $\theta = 0$ and $\theta = \frac{1}{4}\pi$. Thus

$$\int_0^1 \frac{\ln(1+x)}{1+x^2}\,\mathrm{d}x = \int_0^{\frac{1}{4}\pi} \ln(1+\tan\theta)\,\mathrm{d}\theta = \tfrac{1}{8}\pi \ln 2.$$

The integral with $\sin x$ and $\cos x$

Call it J. The quickest route is symmetry: substituting $x = \frac{1}{2}\pi - u$ swaps sin and cos and maps $[0, \frac{1}{2}\pi]$ to itself, so

$$J = \int_0^{\frac{1}{2}\pi} \ln\left(\frac{1+\cos u}{1+\sin u}\right)\mathrm{d}u = -J,$$

and hence $J = 0$. (Equivalently, $\int_0^{\frac{1}{2}\pi} \ln(1+\sin x)\,\mathrm{d}x$ and $\int_0^{\frac{1}{2}\pi} \ln(1+\cos x)\,\mathrm{d}x$ are equal, so their difference is zero.)

> **Ask yourself** Where does the first result come in?

It is worth seeing how J is linked to I, which is presumably why the question places it here. With $t = \frac{1}{2}x$, the half-angle formulae give

$$1+\cos x = 2\cos^2 t, \qquad 1+\sin x = (\cos t + \sin t)^2,$$

so

$$\frac{1+\sin x}{1+\cos x} = \frac{(\cos t+\sin t)^2}{2\cos^2 t} = \tfrac{1}{2}(1+\tan t)^2.$$

As x runs over $[0, \frac{1}{2}\pi]$, t runs over $[0, \frac{1}{4}\pi]$, and $\mathrm{d}x = 2\,\mathrm{d}t$. Hence

$$J = \int_0^{\frac{1}{4}\pi} \big(2\ln(1+\tan t) - \ln 2\big)\,2\,\mathrm{d}t = 4I - \tfrac{1}{2}\pi\ln 2 = \tfrac{1}{2}\pi\ln 2 - \tfrac{1}{2}\pi\ln 2 = 0,$$

which agrees with the symmetry argument. So

$$\int_0^{\frac{1}{2}\pi} \ln\left(\frac{1+\sin x}{1+\cos x}\right)\mathrm{d}x = 0.$$

Checks

Simpson's rule gives 0.2721982613 for both the tan integral and the $1 + x^2$ integral, matching $\frac{1}{8}\pi \ln 2 = 0.2721982613\ldots$; the third integral comes out as 0 to machine precision. A rough check by hand: the integrand $\ln(1 + \tan\theta)$ rises from 0 to $\ln 2$, so I should be roughly half of $\frac{1}{4}\pi \ln 2$. That is exactly the answer.

Common slips

- Forgetting that $\mathrm{d}\theta = -\mathrm{d}\phi$ *and* that the limits swap. The two sign changes cancel; keeping only one gives $I = -\frac{1}{4}\pi \ln 2 - I$ (and a negative answer to a positive integral).
- Writing $\tan(\frac{1}{4}\pi - \phi) = 1 - \tan\phi$, which forgets the denominator in the addition formula.
- Spending a long time on the third integral without noticing the $x \mapsto \frac{1}{2}\pi - x$ symmetry.

Beyond STEP

The trick of substituting a reflection and adding the result to the original, so that $\int_a^b f(x)\,\mathrm{d}x = \int_a^b f(a + b - x)\,\mathrm{d}x$, is called the King property in some texts. It also gives the classic $\int_0^{\frac{1}{2}\pi} \ln\sin x\,\mathrm{d}x = -\frac{1}{2}\pi \ln 2$. At university, results like these sit inside the theory of definite integrals with parameters and contour integration, where whole families of integrals are evaluated without finding an antiderivative. Needham's *Visual Complex Analysis* shows how complex methods produce such results, and Siklos's *Advanced Problems in Mathematics* has more STEP-style practice with these symmetry tricks.

WORKED SOLUTION

STEP I 1994, Question 9

Mechanics • Projectile Motion

STEP I 1994, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s1-q9

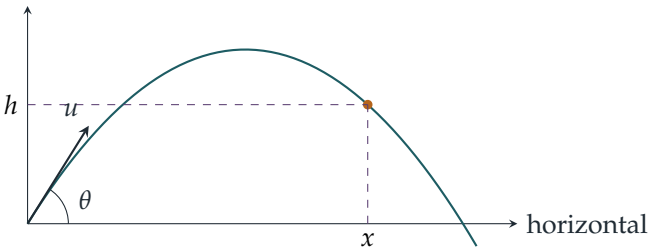
The key idea

The firing angle θ is the thing we don't know, so eliminate it: the horizontal and vertical equations each contain $u\cos\theta$ or $u\sin\theta$, and squaring and adding removes θ completely. The resulting equation is a quadratic in t^2, and a quadratic only has real roots when its discriminant is non-negative. That condition is exactly the bound on the range.

Solution

The equation in t

Model the cannon-ball as a particle moving under gravity alone, fired from the origin at angle θ above the horizontal. After time t it is at horizontal distance x and height h, so its distance from the cannon is $\sqrt{x^2+h^2}$, as required.



Using constant acceleration g downwards and no horizontal force,

$$x = ut\cos\theta,$$
$$h = ut\sin\theta - \tfrac{1}{2}gt^2.$$

Isolate the θ terms:

$$ut\cos\theta = x, \qquad ut\sin\theta = h + \tfrac{1}{2}gt^2.$$

Squaring and adding, and using $\cos^2\theta + \sin^2\theta = 1$,

$$\begin{aligned} u^2t^2 &= x^2 + \left(h + \tfrac{1}{2}gt^2\right)^2 \\ &= x^2 + h^2 + ght^2 + \tfrac{1}{4}g^2t^4, \end{aligned}$$

which rearranges to

$$\tfrac{1}{4}g^2t^4 - (u^2 - gh)t^2 + h^2 + x^2 = 0,$$

as required.

The bound on the range

Now fix u and h, and ask which horizontal distances x are possible at height h. If the ball is at (x, h) at some time t, then $T = t^2$ is a real root of

$$\tfrac{1}{4}g^2T^2 - (u^2 - gh)T + (h^2 + x^2) = 0.$$

Ask yourself The equation involves x, h, u and t, but we want a statement about x alone. What must be true of a quadratic if it has a real root?

A real root needs a non-negative discriminant:

$$\begin{aligned} (u^2 - gh)^2 - 4 \cdot \tfrac{1}{4}g^2(h^2 + x^2) &\geq 0 \\ u^4 - 2u^2gh + g^2h^2 - g^2h^2 - g^2x^2 &\geq 0 \\ g^2x^2 &\leq u^2(u^2 - 2gh). \end{aligned}$$

Since $u^2 > 2gh$ the right-hand side is positive, and taking non-negative square roots gives

$$x \leq \frac{u\sqrt{u^2 - 2gh}}{g}.$$

So the horizontal range at height h can never exceed this value, which is what we had to show. (If $u^2 < 2gh$ the inequality can't hold for any x: the height h is above the greatest height $u^2/2g$ the ball can reach.)

The bound is attained

Equality means the discriminant is zero, so there is a double root

$$t^2 = T = \frac{u^2 - gh}{2 \cdot \frac{1}{4}g^2} = \frac{2(u^2 - gh)}{g^2}.$$

This is positive (because $u^2 > 2gh \geq gh$ if $h \geq 0$, and trivially if $h < 0$), so it gives a genuine time $t > 0$. We still have to check that some angle θ produces this time with $x = u\sqrt{u^2 - 2gh}/g$. From the two component equations we need

$$\sin\theta = \frac{h + \frac{1}{2}gt^2}{ut}, \qquad \cos\theta = \frac{x}{ut}.$$

With the value of t^2 above, $h + \frac{1}{2}gt^2 = h + (u^2 - gh)/g = u^2/g$, so

$$\sin\theta = \frac{u}{gt}, \qquad \cos\theta = \frac{\sqrt{u^2 - 2gh}}{gt}.$$

Ask yourself We have chosen values for $\sin\theta$ and $\cos\theta$ separately. How do we know a single angle has both?

Both values are positive, and

$$\sin^2\theta + \cos^2\theta = \frac{u^2 + u^2 - 2gh}{g^2t^2} = \frac{2(u^2 - gh)}{g^2t^2} = 1$$

by the value of t^2. So there is exactly one angle θ between 0 and $\frac{1}{2}\pi$ that works, namely

$$\tan\theta = \frac{u}{\sqrt{u^2 - 2gh}}.$$

Firing at this angle, the ball is at height h at time t with horizontal distance exactly $u\sqrt{u^2 - 2gh}/g$, so the bound is always attained when $u^2 > 2gh$.

Notice that $\tan\theta > 1$ when $h > 0$: to reach furthest at a given height above the cannon, we fire more steeply than 45°. Also the vertical velocity at that moment is $u\sin\theta - gt = (u^2 - g^2t^2)/(gt) = (2gh - u^2)/(gt) < 0$, so the ball reaches its furthest point at height h on the way down.

Checks

- With $h = 0$ the bound is u^2/g and $\tan\theta = 1$: the familiar maximum range on level ground, at 45°.
- With $u = 20\,\mathrm{m\,s^{-1}}$, $g = 9.8\,\mathrm{m\,s^{-2}}$ and $h = 10\,\mathrm{m}$ the bound is about 29.15 m, at $\theta \approx 54.5°$.
- We confirmed numerically that the quartic holds along random trajectories, and that scanning thousands of firing angles never beats the bound but gets within 10^{-4} of it. The angle above hits it exactly.

Common slips

- Trying to find the best θ by differentiating x with respect to θ. You can, but it's messy. The discriminant does the optimisation for you.
- Forgetting the final step: a zero discriminant gives a value of t, but you must still show that a real angle produces it, by checking $\sin^2\theta + \cos^2\theta = 1$.

Beyond STEP

The points the ball can reach at speed u fill a region bounded by the *envelope* of all the trajectories, the "parabola of safety" $y = \frac{u^2}{2g} - \frac{gx^2}{2u^2}$. Rearranging the bound here shows that (x, h) is reachable exactly when it lies on or under this parabola. Envelopes of families of curves come from the same "double root" idea, and they come up again in the calculus of variations and in optics (caustics). For more on projectiles and the mechanics behind them, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP I 1994, Question 10

Mechanics • Circular Motion

STEP I 1994, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s1-q10

The key idea

This is a conical pendulum with a stretchy string. The particle has no vertical acceleration, so the vertical component of the tension balances the weight, while the horizontal component supplies the centripetal force. The new feature is that the radius of the circle depends on the extension, which depends on the tension, so Hooke's law ties everything together. The inequality then comes from asking which values of $\cos\theta$ actually describe a circle.

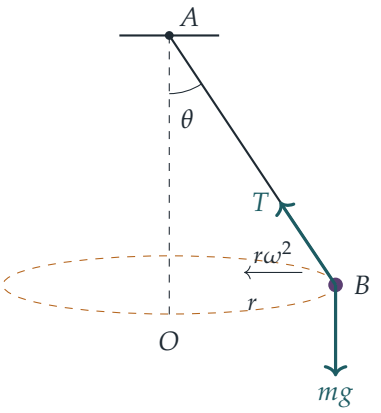
Solution

Finding $\cos\theta$

Model B as a particle and AB as a light elastic string obeying Hooke's law. Let the extension be x and the tension T, so

$$T = \frac{\lambda x}{l}, \qquad AB = l + x.$$

The particle moves in a horizontal circle of radius $r = (l + x)\sin\theta$ about the point O vertically below A. The only forces on it are its weight and the tension along BA.



Resolving vertically (no vertical acceleration):

$$T\cos\theta = mg. \tag{1}$$

Resolving horizontally, towards the centre O (acceleration $r\omega^2$):

$$T\sin\theta = m\omega^2(l+x)\sin\theta.$$

For a genuine circle $\sin\theta \neq 0$, so we can divide by it:

$$T = m\omega^2(l+x). \tag{2}$$

Now Hooke's law gives $\lambda x/l = m\omega^2(l+x)$, that is,

$$x\left(\lambda - m\omega^2 l\right) = m\omega^2 l^2.$$

Since $x > 0$ (from (1), $T > 0$, so the string is stretched) and $m\omega^2 l^2 > 0$, we have $\lambda - m\omega^2 l > 0$ and

$$x = \frac{m\omega^2 l^2}{\lambda - m\omega^2 l}, \qquad l + x = \frac{l\lambda}{\lambda - m\omega^2 l}.$$

Then (2) gives

$$T = \frac{m\omega^2 l\lambda}{\lambda - m\omega^2 l},$$

and (1) gives

$$\boxed{\cos\theta = \frac{mg}{T} = \frac{g\left(\lambda - m\omega^2 l\right)}{l\lambda\,\omega^2}}$$

When the motion is possible

Ask yourself What does each restriction on $\cos\theta$ say physically?

For the particle to move in a horizontal circle below A, with the string taut, we need $0 < \theta < \frac{1}{2}\pi$, i.e. $0 < \cos\theta < 1$. Conversely, if ω gives $0 < \cos\theta < 1$ from the formula,

then working back through the steps (with $\lambda - m\omega^2 l > 0$) produces a positive extension and tension satisfying (1) and (2), so the motion does exist.

Upper bound. $\cos\theta > 0$ holds exactly when $\lambda - m\omega^2 l > 0$, since the other factors are positive. This is

$$\omega^2 < \frac{\lambda}{ml}.$$

Physically: if ω is too large, no finite extension of the string can supply the required centripetal force, since the extra length needs extra force faster than the string provides it.

Lower bound. $\cos\theta < 1$ means

$$g\lambda - gm\omega^2 l < l\lambda\omega^2 \iff g\lambda < \omega^2 l(\lambda + mg) \iff \omega^2 > \frac{g\lambda}{l(\lambda + mg)},$$

where we divided by $l\lambda\omega^2 > 0$ first and by $l(\lambda + mg) > 0$ last, so the inequality directions are kept. Physically: if ω is too small, the formula would demand $\cos\theta \geq 1$; the only option is for the particle to hang at rest on the axis ($\theta = 0$), which is not circular motion.

Putting the two together, the motion is possible only if

$$\frac{g\lambda}{l(\lambda + mg)} < \omega^2 < \frac{\lambda}{ml},$$

as required.

Checks

- At the lower bound, $\cos\theta = 1$: the string is then vertical with tension mg and length $l(\lambda + mg)/\lambda$, which is exactly the equilibrium length of the hanging particle. Note $\omega^2 = g/(\text{that length})$, the familiar conical-pendulum threshold $\omega^2 > g/L$.
- As $\lambda \to \infty$ (inextensible string) the result becomes $\cos\theta = g/(l\omega^2)$, the standard conical pendulum, and the range becomes $\omega^2 > g/l$.
- Numerically, with $g = 9.8$ and random m, l, λ and ω, solving Hooke's law and the horizontal equation directly by bisection gives the same $\cos\theta$ as the formula, and both resolved equations balance; across 20 000 random cases the motion exists exactly when the inequality holds.

Common slips

- Using radius $l\sin\theta$ instead of $(l + x)\sin\theta$: the string is stretched.
- Forgetting to justify dividing by $\sin\theta$, and so missing why $\theta = 0$ (and hence $\cos\theta = 1$) is excluded.
- Only proving $\cos\theta < 1$ and forgetting that $\cos\theta > 0$ (equivalently, a positive extension) gives the upper bound.

Beyond STEP

The upper bound is a simple example of a *runaway* or instability: as $\omega^2 \to \lambda/(ml)$ the required extension blows up. The same effect appears for rotating elastic rods and in the stability of spinning bodies, and the lower bound is a bifurcation, where the hanging equilibrium gives way to circular motion. For more on rotating-frame mechanics, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP I 1994, Question 11

Mechanics • Connected Particles on Incline

STEP I 1994, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

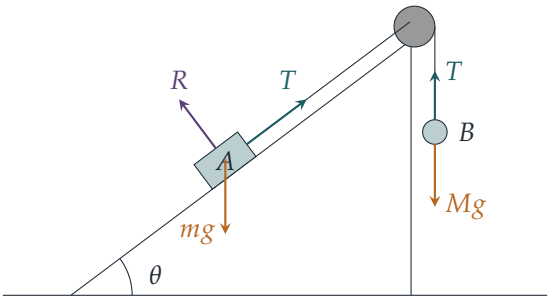
step.maths.org/questions/94-s1-q11

The key idea

Start with the geometry. The string is exactly as long as the track, so when the wagon is at the bottom the ball hangs right at the pulley. So the ball falls through the height of the incline, $d \sin\theta$, and the wagon climbs the same distance up the slope. While the string is taut the system moves with one constant acceleration. Once the ball lands, the string goes slack and the wagon just coasts up a smooth slope. So part (ii) asks whether its leftover speed carries it the rest of the way to the pulley.

Solution

We model the wagon and ball as particles, the track as smooth, and the string as light and inextensible over a light frictionless pulley. So the tension T is the same on both sides, and the wagon and ball have the same speed while the string is taut.



Part (i)

Let a be the acceleration of the ball downwards, which is also the acceleration of the wagon up the slope. Newton's second law for the ball (vertically) and for the wagon

(along the slope) gives

$$\begin{aligned} Mg - T &= Ma, \\ T - mg\sin\theta &= ma. \end{aligned}$$

The normal reaction R is perpendicular to the slope and does not enter. Adding the two equations eliminates T:

$$a = \frac{(M - m\sin\theta)\, g}{M + m}.$$

The system starts from rest, so the ball moves at all only if $a > 0$. (If $a \leq 0$ nothing happens: with $a = 0$ the system stays in equilibrium, and the string can't push, so $a < 0$ is impossible.) So the ball falls to the ground if and only if

$$M > m\sin\theta.$$

When the wagon is at the bottom of the track the full length d of the string lies along the slope, so the ball starts level with the pulley, at height $d\sin\theta$ above the ground. It falls this distance from rest with constant acceleration a, so by $v^2 = u^2 + 2as$,

$$v^2 = 2 \cdot \frac{(M - m\sin\theta)g}{M + m} \cdot d\sin\theta = \frac{2g(M - m\sin\theta)\, d\sin\theta}{M + m},$$

as required.

Part (ii)

When the ball hits the ground, the wagon has moved $d\sin\theta$ up the track and is moving at speed v. That leaves a distance

$$d - d\sin\theta = d(1 - \sin\theta)$$

to the pulley.

Ask yourself What happens to the string once the ball has landed?

The ball stops, but the wagon is still moving up the slope, towards the pulley. So the string goes slack and $T = 0$. The only force along the slope is now the component of gravity, so the wagon decelerates at $g\sin\theta$. It comes to rest after a further distance s given by $0 = v^2 - 2g\sin\theta\, s$:

$$s = \frac{v^2}{2g\sin\theta} = \frac{(M - m\sin\theta)\, d}{M + m}.$$

(The wagon then rolls back down, and nothing more happens near the pulley.)

The wagon doesn't collide with the pulley if it stops before reaching it, that is, if $s < d(1 - \sin\theta)$. Since $d > 0$ and $M + m > 0$, this is equivalent to

$$\begin{aligned} M - m\sin\theta &< (M + m)(1 - \sin\theta) \\ &= M + m - M\sin\theta - m\sin\theta, \end{aligned}$$

which simplifies to $M\sin\theta < m$.

Combining this with part (i), the ball reaches the ground and the wagon stops short of the pulley exactly when

$$m\sin\theta < M < \frac{m}{\sin\theta}.$$

If $M\sin\theta = m$, the wagon just reaches the pulley with zero speed. That's a touch, not a collision.

Checks

- The furthest the wagon gets up the track is $d\sin\theta + s = \dfrac{Md(1+\sin\theta)}{M+m}$. This is less than d exactly when $M\sin\theta < m$, which agrees with the condition above. As $m \to 0$ it tends to $d(1+\sin\theta) > d$: a weightless wagon always hits the pulley, as you'd expect.
- At $\theta = 90°$ the condition $M < m$ contradicts $M > m\sin\theta = m$ from part (i). That's right: on a vertical track the wagon starts at the bottom directly under the pulley, the ball starts at the top, and the wagon rises exactly as far as the ball falls, which is all the way to the pulley.
- Numerically, with $m = 2$, $M = 1.5$, $\theta = 30°$, $d = 3$ and $g = 9.8$, the formula gives $v^2 = 4.2$ and a furthest point of $\frac{27}{14} \approx 1.93 < 3$. Stepping the motion forward in small time steps gives the same values. Random cases also confirm that the collision happens exactly when $M\sin\theta > m$.

Common slips

- Taking the ball's drop to be d rather than $d\sin\theta$. The ball starts at the pulley, which is at height $d\sin\theta$.
- Assuming the string stays taut after the ball lands, or forgetting the coasting phase altogether and comparing only $d\sin\theta$ with d.
- Using g instead of $g\sin\theta$ for the wagon's deceleration once the string is slack.

Beyond STEP

The question splits the motion into phases with different constraints: string taut, then string slack. This is the simplest example of a *unilateral constraint*, which can pull but not push. Such constraints come up throughout mechanics, from chains falling off tables to contact problems in robotics. Energy methods give a quick alternative here: in the coasting phase, the wagon's kinetic energy $\frac{1}{2}mv^2$ turns into potential energy $mgs\sin\theta$. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops both viewpoints carefully.

WORKED SOLUTION

STEP I 1994, Question 13

Probability and Statistics • Indicator Variables

STEP I 1994, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s1-q13

The key idea

Writing a count as a sum of indicator variables (each 1 or 0) turns a hard expectation into an easy one, because expectation is linear *whether or not* the pieces are independent. Here the draws are certainly not independent, but symmetry tells us each X_i has the same expectation. Then we compute $\mathrm{E}(X)$ a second way, straight from the distribution of X, and setting the two answers equal gives a binomial-coefficient identity almost for free.

Solution

Throughout, $1 \le n \le M$, and we use the convention that $\binom{a}{b} = 0$ when $b > a$ or $b < 0$.

Part (i)

Each X_i is 1 if the ith token removed is red and 0 if it is not. So in the sum $X_1 + X_2 + \cdots + X_n$ every red token removed contributes exactly 1 and every non-red token contributes 0. The sum therefore counts the red tokens among the n removed, which is X.

Part (ii)

Since X_i takes only the values 0 and 1,

$$\mathrm{E}(X_i) = 0 \cdot \mathrm{P}(X_i = 0) + 1 \cdot \mathrm{P}(X_i = 1) = \mathrm{P}(i\text{th token removed is red}).$$

Ask yourself Doesn't the chance that the ith token is red depend on what was drawn before it?

Conditionally, yes; unconditionally, no. Imagine numbering the tokens and drawing them all out one by one: every ordering of the M tokens is equally likely. So the token in position i is equally likely to be any of the M tokens, just as the token in position 1

is. Stopping after n draws doesn't change what happens at position $i \le n$. Since m of the M tokens are red,

$$\mathrm{E}(X_i) = \mathrm{P}(X_i = 1) = \frac{m}{M} \qquad \text{for each } i = 1, \ldots, n.$$

(For $i = 2$ you can confirm this the long way: $\frac{m}{M} \cdot \frac{m-1}{M-1} + \frac{M-m}{M} \cdot \frac{m}{M-1} = \frac{m(M-1)}{M(M-1)} = \frac{m}{M}$.)

Part (iii)

By part (i) and linearity of expectation (which needs no independence),

$$\mathrm{E}(X) = \mathrm{E}(X_1) + \cdots + \mathrm{E}(X_n) = n \cdot \frac{m}{M} = \frac{mn}{M},$$

as required.

Part (iv)

The set of n tokens removed is equally likely to be any of the $\binom{M}{n}$ subsets of size n. To have exactly k red tokens we choose k of the m red tokens and $n-k$ of the $M-m$ others, so

$$\mathrm{P}(X = k) = \frac{\binom{m}{k}\binom{M-m}{n-k}}{\binom{M}{n}}, \qquad k = 0, 1, \ldots, n.$$

The convention on binomial coefficients makes this correctly 0 when $k > m$ (not enough red tokens) or $n - k > M - m$ (not enough others). This is the *hypergeometric* distribution.

Part (v)

Compute $\mathrm{E}(X)$ directly from part (iv); the $k = 0$ term is zero, so

$$\mathrm{E}(X) = \sum_{k=1}^{n} k\,\mathrm{P}(X = k) = \frac{1}{\binom{M}{n}} \sum_{k=1}^{n} k \binom{m}{k}\binom{M-m}{n-k}.$$

Equating this with part (iii),

$$\sum_{k=1}^{n} k \binom{m}{k}\binom{M-m}{n-k} = \frac{mn}{M}\binom{M}{n}.$$

Finally,

$$\frac{n}{M}\binom{M}{n} = \frac{n}{M} \cdot \frac{M!}{n!\,(M-n)!} = \frac{(M-1)!}{(n-1)!\,(M-n)!} = \binom{M-1}{n-1},$$

so the right-hand side is $m\binom{M-1}{n-1}$, which is the required identity.

Checks

- Take $M = 5$, $m = 2$, $n = 3$. Part (iv) gives $\mathrm{P}(X = 0, 1, 2) = \frac{1}{10}, \frac{6}{10}, \frac{3}{10}$, which sum to 1, and $\mathrm{E}(X) = \frac{6+6}{10} = \frac{6}{5} = \frac{mn}{M}$. The identity reads $1 \cdot 2 \cdot 3 + 2 \cdot 1 \cdot 3 = 12 = 2\binom{4}{2}$.

- If $n = M$ (take everything), $X = m$ for certain, and indeed $\mathrm{E}(X) = mM/M = m$.
- We checked $\mathrm{E}(X_i)$, $\mathrm{E}(X)$ and $\mathrm{P}(X = k)$ by listing every ordering of the tokens for all bags with $M \leq 6$, and the identity in (v) for every $M < 30$.

Common slips

- Claiming $\mathrm{E}(X) = \sum \mathrm{E}(X_i)$ needs independence. It doesn't: linearity of expectation always holds.
- Treating the draws as independent and writing a binomial $\mathrm{B}(n, \frac{m}{M})$ in (iv). This gets the mean right but the probabilities wrong, because the draws are without replacement.

Beyond STEP

The indicator method is one of the most useful tools in probability: it gives expected numbers of fixed points of a random permutation, of runs in coin tosses, of isolated vertices in a random graph, and it extends to variances via $\mathrm{Var}(\sum X_i) = \sum \mathrm{Var}(X_i) + \sum_{i \neq j} \mathrm{Cov}(X_i, X_j)$. Proving combinatorial identities by computing one quantity in two ways is just as general. Blitzstein and Hwang, *Introduction to Probability*, uses indicators throughout and is an excellent place to see both ideas in action.

WORKED SOLUTION

STEP I 1994, Question 14

Probability and Statistics • Exponential Order Statistics

STEP I 1994, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s1-q14

The key idea

Each T_i is exponentially distributed, so part (i) is a pair of standard integrals by parts. For the last and first essays, the trick is to turn a statement about a maximum or a minimum into a statement about *every* T_i: the last essay is in by time u exactly when all of them are, and the first one is still outstanding at time t exactly when all of them are. Independence then turns "all" into a product, and differentiating the distribution function gives the density.

Solution

Part (i)

Write X for any one of the T_i, with density $f(t) = \lambda \mathrm{e}^{-\lambda t}$ for $t \geq 0$. Integrating by parts, with $u = t$ and $\mathrm{d}v = \lambda \mathrm{e}^{-\lambda t}\,\mathrm{d}t$ (so $v = -\mathrm{e}^{-\lambda t}$),

$$\mathrm{E}(X) = \int_0^\infty t\,\lambda \mathrm{e}^{-\lambda t}\,\mathrm{d}t = \Big[-t\mathrm{e}^{-\lambda t}\Big]_0^\infty + \int_0^\infty \mathrm{e}^{-\lambda t}\,\mathrm{d}t = 0 + \frac{1}{\lambda} = \frac{1}{\lambda},$$

since $t\mathrm{e}^{-\lambda t} \to 0$ as $t \to \infty$ (exponentials beat powers). In the same way,

$$\mathrm{E}(X^2) = \int_0^\infty t^2 \lambda \mathrm{e}^{-\lambda t}\,\mathrm{d}t = \Big[-t^2\mathrm{e}^{-\lambda t}\Big]_0^\infty + \int_0^\infty 2t\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t = \frac{2}{\lambda}\,\mathrm{E}(X) = \frac{2}{\lambda^2},$$

where we recognised $\int_0^\infty t\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t$ as $\frac{1}{\lambda}\mathrm{E}(X)$. Therefore

$$\mathrm{E}(T_i) = \frac{1}{\lambda}, \qquad \mathrm{Var}(T_i) = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}.$$

Part (ii)

First we need the distribution function of a single T_i: for $u \geq 0$,

$$\mathrm{P}(T_i \leq u) = \int_0^u \lambda \mathrm{e}^{-\lambda t}\,\mathrm{d}t = 1 - \mathrm{e}^{-\lambda u}.$$

Now $U = \max(T_1, \ldots, T_n)$, and the largest of the T_i is at most u if and only if every one of them is at most u:

$$\{U \le u\} = \{T_1 \le u\} \cap \{T_2 \le u\} \cap \cdots \cap \{T_n \le u\}.$$

Ask yourself Why is it legitimate to multiply the probabilities here?

The events $\{T_i \le u\}$ concern different T_i, which are independent by assumption, so the probability of the intersection is the product of the probabilities. Hence, for $u \ge 0$,

$$\mathrm{P}(U \le u) = \prod_{i=1}^{n} \mathrm{P}(T_i \le u) = \left(1 - \mathrm{e}^{-\lambda u}\right)^n,$$

as required. The density of U is the derivative of this distribution function; by the chain rule,

$$f_U(u) = n\left(1 - \mathrm{e}^{-\lambda u}\right)^{n-1} \cdot \lambda \mathrm{e}^{-\lambda u} = n\lambda\, \mathrm{e}^{-\lambda u}\left(1 - \mathrm{e}^{-\lambda u}\right)^{n-1} \qquad (u \ge 0),$$

and $f_U(u) = 0$ for $u < 0$.

Part (iii)

Now $T = \min(T_1, \ldots, T_n)$. Here the useful event is the complement: the first essay arrives after time t if and only if *none* has arrived by then, that is, every $T_i > t$. With $\mathrm{P}(T_i > t) = \mathrm{e}^{-\lambda t}$ and independence again,

$$\mathrm{P}(T > t) = \prod_{i=1}^{n} \mathrm{P}(T_i > t) = \left(\mathrm{e}^{-\lambda t}\right)^n = \mathrm{e}^{-n\lambda t} \qquad (t \ge 0).$$

So $\mathrm{P}(T \le t) = 1 - \mathrm{e}^{-n\lambda t}$, and differentiating,

$$f_T(t) = n\lambda\, \mathrm{e}^{-n\lambda t} \qquad (t \ge 0),$$

with $f_T(t) = 0$ for $t < 0$.

Part (iv)

The density in (iii) is exactly the density of part (i) with λ replaced by $n\lambda$: T is exponential with parameter $n\lambda$. So we can simply write down, from part (i),

$$\mathrm{E}(T) = \frac{1}{n\lambda}, \qquad \mathrm{Var}(T) = \frac{1}{n^2\lambda^2}.$$

This makes sense: with n students, the first essay should arrive about n times sooner than any particular one.

Checks

- With $n = 1$ there is only one essay, so $T = U = T_1$; indeed both densities reduce to $\lambda \mathrm{e}^{-\lambda t}$.

- $\mathrm{P}(U \le u)$ rises from 0 at $u = 0$ to 1 as $u \to \infty$, as a distribution function must.
- We simulated 2×10^5 classes with $n = 5$, $\lambda = 0.7$. For example, the proportion with $U \le 2$ was 0.2425 against $(1 - \mathrm{e}^{-1.4})^5 \approx 0.2427$, and the mean of T was 0.2860 against $1/(5 \times 0.7) \approx 0.2857$. Simpson's rule confirmed that f_U integrates to 1.

Common slips

- Working with $\mathrm{P}(T \le t)$ directly for the minimum: "at least one $T_i \le t$" is not an intersection, so it does not factorise. Go via $\mathrm{P}(T > t)$.
- Forgetting to state that independence is what justifies the product.
- Writing $\mathrm{Var}(T_i) = \mathrm{E}(T_i^2) = 2/\lambda^2$ and forgetting to subtract $(\mathrm{E}\,T_i)^2$.

Beyond STEP

T and U are the smallest and largest *order statistics* of the sample. The minimum of independent exponentials being exponential with the rates added is the basis of Poisson processes and queueing theory. A lovely further fact, from the memoryless property, is that the gaps between successive essays are independent exponentials with rates $n\lambda, (n-1)\lambda, \ldots, \lambda$, so $\mathrm{E}(U) = \frac{1}{\lambda}\left(1 + \frac{1}{2} + \cdots + \frac{1}{n}\right)$. Both of these are developed in Grimmett and Welsh, *Probability: An Introduction*, and in Blitzstein and Hwang, *Introduction to Probability*.

WORKED SOLUTION

STEP II 1994, Question 1

Pure • Divisibility and Decimal Digits

STEP II 1994, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s2-q1

The key idea

The last few digits of a product depend only on the last few digits of the factors, so everything here is arithmetic modulo a power of 10. A number ending in 1, 3, 7 or 9 shares no factor with 10, and that makes it *invertible* modulo 10, and in fact modulo any power of 10. The middle part strips the factors of 2 and 5 out of n and turns them into zeros. The final part uses invertibility modulo 10^{10} to force the last ten digits of the product to be any block we like.

Solution

Part 1: a multiple ending in 1

The last digit of $n \times m$ is the last digit of (last digit of n)$\times$(last digit of m), because the other digits contribute multiples of 10. Since n doesn't end in 5 or an even digit, it ends in 1, 3, 7 or 9, and

$$1 \times 1 = 1, \qquad 3 \times 7 = 21, \qquad 7 \times 3 = 21, \qquad 9 \times 9 = 81.$$

So take $m = 1$, 7, 3 or 9 respectively. Then $n \times m$ ends in 1, as required.

Part 2: any n

Write

$$n = 2^a \, 5^b \, k,$$

where $a, b \geq 0$ and k is divisible by neither 2 nor 5. Then k doesn't end in 5 or an even digit, so by Part 1 there is m' with km' ending in 1. Now let

$$m = 2^b \, 5^a \, m'.$$

Then

$$n \times m = 2^{a+b} \, 5^{a+b} \, km' = 10^{a+b} \, (km').$$

Multiplying by 10^{a+b} just appends $a+b$ zeros to km'. So $n \times m$ ends in 1 if $a+b=0$, and in 1 followed by $a+b$ zeros otherwise. This is what we needed.

For example, $n = 40 = 2^3 \cdot 5$ has $k=1$, so $m = 2 \cdot 5^3 = 250$ and $n \times m = 10\,000$.

Part 3: all ten digits

Write $n = 10^r q$ with $r \geq 0$ and q ending in 1. Appending zeros never removes digits, so it is enough to find m with $q \times m$ containing every digit; then $n \times m = 10^r(qm)$ contains them too.

Let $N = 10^{10}$ and $X = 1234567890$, which contains every digit. We aim for

$$qm \equiv X \pmod{N},$$

that is, qm ends in the ten digits 1234567890.

> **Ask yourself** Why must some multiple of q leave remainder X on division by 10^{10}?

Consider the N numbers $q \times 0,\ q \times 1,\ \ldots,\ q \times (N-1)$ and their remainders modulo N. Suppose two were equal, say $qs \equiv qt \pmod{N}$ with $0 \leq t < s \leq N-1$. Then 10^{10} divides $q(s-t)$. But q ends in 1, so it is odd and not a multiple of 5, and it has no prime factor in common with $10^{10} = 2^{10}5^{10}$. Hence $2^{10}5^{10}$ divides $s-t$, which is impossible because $0 < s-t < N$. So the N remainders are all different. There are only N possible remainders, $0, 1, \ldots, N-1$, so every one of them occurs. In particular, some m with $0 \leq m \leq N-1$ has $qm \equiv X \pmod{N}$.

This m isn't 0, since $X \not\equiv 0$. So qm is a positive multiple with $qm \geq X \geq 10^9$. Its last ten digits are therefore exactly $1, 2, 3, 4, 5, 6, 7, 8, 9, 0$, with the leading 1 of X genuinely present. Thus $q \times m$, and hence $n \times m$, contains all the digits $0, 1, \ldots, 9$.

For instance, with $n = 21$ we can take $m = 7\,201\,646\,090$, and then $21 \times m = 151\,234\,567\,890$.

> **Ask yourself** Where was the hypothesis on n used, and why was it needed?

We used it only to know that q has no factor 2 or 5. Without that, the remainders qm mod 10^{10} can't take every value. For example, every multiple of 2 is even, so it can't end in $\ldots 1$. The hypothesis is what lets us dictate the *end* of the product. (Part 2 shows that any n has a multiple of this form, so combined with Part 3, *every* n has a multiple containing all ten digits.)

Checks

- Part 1 by hand: $13 \times 7 = 91$ and $29 \times 9 = 261$.
- A computer check confirmed the Part 1 recipe for every $n \leq 20\,000$ not ending in 5 or an even digit. It confirmed the Part 2 construction for every $n \leq 20\,000$ and $20\,000$

random n up to 10^{15}. It also computed the Part 3 multiplier for over 10 000 values of n of the required form, including ones ending in several zeros, and in every case $n \times m$ contained all ten digits. Examples: $21 \times 7\,201\,646\,090 = 151\,234\,567\,890$ and $11 \times 4\,657\,687\,990 = 51\,234\,567\,890$.

Common slips

- In Part 3, forgetting to check that the block 1234567890 really appears in full. The leading 1 needs $qm \geq 10^9$, which holds because $qm \equiv X$ and $qm > 0$.
- In Part 2, multiplying by 10^{a+b} instead of $2^b 5^a$. That adds zeros, but it leaves the 2s and 5s already in n where they are.

Beyond STEP

The pigeonhole argument in Part 3 proves a general fact: if $\gcd(q, N) = 1$ then q has a multiplicative inverse modulo N. The invertible residues form a group, written $(\mathbb{Z}/N\mathbb{Z})^\times$, and this is the starting point of elementary number theory and of RSA cryptography. A different trick shows that the final part holds for any n at all: some multiple of n lies between $X \cdot 10^k$ and $X \cdot 10^k + 10^k$ once $10^k \geq n$, and such a multiple *begins* with 1234567890. Davenport's *The Higher Arithmetic* is a lovely introduction to congruences and residues.

WORKED SOLUTION

STEP II 1994, Question 2

Pure • Legendre Polynomials

STEP II 1994, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s2-q2

The key idea

A factor $(x-a)^m$ survives, slightly weakened, every time we differentiate, so a repeated root of order m is also a root of the first $m-1$ derivatives. The polynomial $(x^2-1)^n = (x-1)^n(x+1)^n$ has repeated roots of order n at both ends of $[-1,1]$, so when we integrate $x^m \mathrm{P}_n(x)$ by parts, each boundary term vanishes. Each integration by parts moves one derivative off $(x^2-1)^n$ and onto x^m, and x^m runs out of derivatives before $(x^2-1)^n$ does.

Solution

Repeated roots and derivatives

By the product rule,

$$\begin{aligned}\mathrm{P}'(x) &= m(x-a)^{m-1}\mathrm{Q}(x) + (x-a)^m\mathrm{Q}'(x)\\ &= (x-a)^{m-1}\big[m\mathrm{Q}(x) + (x-a)\mathrm{Q}'(x)\big] = (x-a)^{m-1}\mathrm{R}(x),\end{aligned}$$

where $\mathrm{R}(x) = m\mathrm{Q}(x) + (x-a)\mathrm{Q}'(x)$ is a polynomial, because sums and products of polynomials are polynomials. This is the required form.

The result says that a polynomial with a factor $(x-a)^k$, $k \geq 1$, has a derivative with a factor $(x-a)^{k-1}$. Applying it repeatedly, starting from $k = m$, gives

$$\mathrm{P}^{(r)}(x) = (x-a)^{m-r}\mathrm{R}_r(x) \qquad (0 \leq r \leq m)$$

for some polynomials R_r. Each step is allowed, since we only differentiate $\mathrm{P}^{(r)}$ to get $\mathrm{P}^{(r+1)}$ when $r \leq m-1$, and then the exponent $m-r$ is at least 1. If $1 \leq r \leq m-1$ then $m-r \geq 1$, so $(x-a)$ is still a factor of $\mathrm{P}^{(r)}(x)$, and therefore $\mathrm{P}^{(r)}(a) = 0$. (This also holds for $r = 0$, and in general fails for $r = m$.)

P_n is a polynomial of degree n

Write $F(x) = (x^2-1)^n$. By the binomial theorem, $F(x)$ is x^{2n} plus terms of degree at most $2n-2$. Differentiating x^k n times gives either 0 (if $k < n$) or a multiple of x^{k-n}, so $P_n = F^{(n)}$ is a polynomial of degree at most n. Its x^n term comes only from x^{2n}:

$$\frac{d^n}{dx^n} x^{2n} = 2n(2n-1)\cdots(n+1)\,x^n = \frac{(2n)!}{n!}\,x^n.$$

This coefficient is non-zero, so P_n has degree exactly n.

The vanishing integrals

Since $F(x) = (x-1)^n(x+1)^n$, the first part with $a = 1$, $m = n$ and $Q(x) = (x+1)^n$ shows that $F^{(r)}(1) = 0$ for $1 \le r \le n-1$. The same argument with $a = -1$ gives $F^{(r)}(-1) = 0$. Also $F(\pm 1) = 0$ directly. So

$$F^{(r)}(\pm 1) = 0 \qquad \text{for } 0 \le r \le n-1. \tag{$*$}$$

Now take $j \ge 1$ and $1 \le k \le n$. Integrating by parts, with x^j differentiated and $F^{(k)}$ integrated, gives

$$\begin{aligned}\int_{-1}^{1} x^j F^{(k)}(x)\,dx &= \Big[x^j F^{(k-1)}(x)\Big]_{-1}^{1} - j\int_{-1}^{1} x^{j-1}F^{(k-1)}(x)\,dx \\ &= -j\int_{-1}^{1} x^{j-1}F^{(k-1)}(x)\,dx,\end{aligned}$$

because $k-1 \le n-1$, so the boundary term vanishes by $(*)$.

Ask yourself How many times can we use this before something goes wrong?

Each use lowers the power of x by one and the order of the derivative by one. We start at x^m and $F^{(n)}$, so we may apply it m times. The step needs $k \ge 1$, and the smallest k we use is $n-m+1 \ge 1$. After m steps,

$$\int_{-1}^{1} x^m P_n(x)\,dx = (-1)^m m! \int_{-1}^{1} F^{(n-m)}(x)\,dx.$$

(For $m = 0$ this is just the original integral.)

If $0 \le m \le n-1$ then $n-m \ge 1$, so the last integrand is itself a derivative and

$$\int_{-1}^{1} x^m P_n(x)\,dx = (-1)^m m!\Big[F^{(n-m-1)}(x)\Big]_{-1}^{1} = 0,$$

by $(*)$, since $0 \le n-m-1 \le n-1$. This is the required result.

The case $m = n$

The same reduction with $m = n$ is still valid (the last step uses $k = 1$) and gives

$$\int_{-1}^{1} x^n P_n(x)\,dx = (-1)^n n! \int_{-1}^{1} (x^2-1)^n\,dx = n!\int_{-1}^{1} (1-x^2)^n\,dx,$$

since $(x^2-1)^n = (-1)^n(1-x^2)^n$. The integrand is even, and the substitution $x = \sin t$, $\mathrm{d}x = \cos t\,\mathrm{d}t$, takes $[0,1]$ to $[0, \frac{\pi}{2}]$ with $1 - x^2 = \cos^2 t$. So, using the formula in the hint,

$$\int_{-1}^{1} (1-x^2)^n \,\mathrm{d}x = 2\int_0^{\pi/2} \cos^{2n+1} t \,\mathrm{d}t = \frac{2^{2n+1}(n!)^2}{(2n+1)!}.$$

Therefore

$$\boxed{\int_{-1}^{1} x^n \mathrm{P}_n(x) \,\mathrm{d}x = \frac{2^{2n+1}(n!)^3}{(2n+1)!}.}$$

Checks

For $n = 1$, $\mathrm{P}_1(x) = 2x$, so $\int_{-1}^{1} x \cdot 2x \,\mathrm{d}x = \frac{4}{3}$, and the formula gives $\frac{8\cdot 1}{6} = \frac{4}{3}$. For $n = 2$, $\mathrm{P}_2(x) = 12x^2 - 4$. Then $\int_{-1}^{1}(12x^2-4)\,\mathrm{d}x = 8 - 8 = 0$ and $\int_{-1}^{1} x(12x^2-4)\,\mathrm{d}x = 0$ (an odd integrand), as predicted. Also $\int_{-1}^{1} x^2(12x^2-4)\,\mathrm{d}x = \frac{24}{5} - \frac{8}{3} = \frac{32}{15}$, which matches $\frac{2^5\cdot 8}{120} = \frac{32}{15}$. We checked all of this exactly by computer for $n \le 10$.

Common slips

- Forgetting that the boundary terms need $\mathrm{F}^{(r)}(\pm 1) = 0$ for $r = 0$ as well as $1 \le r \le n-1$. The final step for $m = n-1$ uses $r = 0$.
- Losing the sign: $(x^2-1)^n = (-1)^n(1-x^2)^n$ cancels the $(-1)^n$ from the integrations by parts, so the answer is positive. That is reasonable, since $x^n\mathrm{P}_n(x)$ has positive leading coefficient.

Beyond STEP

Up to a constant, P_n is the Legendre polynomial of degree n, and this formula for it is Rodrigues' formula. The first result says P_n is orthogonal to every polynomial of lower degree with respect to the inner product $\int_{-1}^{1} fg\,\mathrm{d}x$. This is the property behind Legendre series and Gaussian quadrature. Legendre polynomials also turn up in the solution of Laplace's equation in spherical coordinates. Boyce and DiPrima, *Elementary Differential Equations*, meets them through Legendre's differential equation.

WORKED SOLUTION

STEP II 1994, Question 3

Pure • Functional Equations

STEP II 1994, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s2-q3

The key idea

We know nothing about f except one identity, valid for *all* x and y, and one value, $f(0) = 1$. So the only tool is substitution: choose x and y so that most arguments become 0 or a, where we know (or are about to know) the values. The identity should remind you of $\cos(x-y) = \cos x \cos y + \sin x \sin y$, with $f(a-x)$ playing the part of $\pm \sin x$; that analogy tells you both what to expect and what the examples will be.

Solution

Throughout, write the given identity as

$$f(x-y) = f(x)\,f(y) - f(a-x)\,f(a+y) \qquad \text{for all } x, y. \tag{$*$}$$

Every step below is a substitution into $(*)$, so we make no assumption of continuity, differentiability or anything else.

Showing $f(a) = 0$

Put $x = y = 0$ in $(*)$:

$$f(0) = f(0)^2 - f(a)\,f(a), \qquad \text{so} \qquad 1 = 1 - f(a)^2.$$

Hence $f(a)^2 = 0$, and since $f(a)$ is a real number, $f(a) = 0$.

Part (i)

Put $x = 0$ and $y = t$ in $(*)$:

$$f(-t) = f(0)\,f(t) - f(a)\,f(a+t) = 1 \cdot f(t) - 0 = f(t),$$

using $f(0) = 1$ and $f(a) = 0$. So f is even.

Part (ii)

Put $x = y = a$ in $(*)$:

$$\mathrm{f}(0) = \mathrm{f}(a)^2 - \mathrm{f}(0)\,\mathrm{f}(2a), \qquad \text{so} \qquad 1 = 0 - \mathrm{f}(2a),$$

giving $\mathrm{f}(2a) = -1$.

Part (iii)

> **Ask yourself** Which single substitution makes the first product on the right vanish but keeps t free?

Setting $x = a$ kills the term $\mathrm{f}(x)\mathrm{f}(y)$ (because $\mathrm{f}(a) = 0$) and turns $\mathrm{f}(a - x)$ into $\mathrm{f}(0) = 1$, while y is still free. With $x = a$, $(*)$ becomes

$$\mathrm{f}(a - y) = 0 \cdot \mathrm{f}(y) - \mathrm{f}(0)\,\mathrm{f}(a + y) = -\mathrm{f}(a + y) \qquad \text{for all } y.$$

So f is "odd about a". Now choose $y = a - t$: then $a - y = t$ and $a + y = 2a - t$, so

$$\mathrm{f}(t) = -\mathrm{f}(2a - t),$$

which is $\mathrm{f}(2a - t) = -\mathrm{f}(t)$ for all t, as required. (Putting $t = 0$ recovers part (ii), a useful consistency check.)

Part (iv)

Apply (iii) twice, with (i) in between. First, (iii) with t replaced by $-t$ gives

$$\mathrm{f}(2a + t) = -\mathrm{f}(-t) = -\mathrm{f}(t), \tag{1}$$

using (i). So shifting the argument by $2a$ changes the sign. Shifting by $2a$ again, i.e. applying (1) with t replaced by $2a + t$,

$$\mathrm{f}(4a + t) = -\mathrm{f}(2a + t) = -\big(-\mathrm{f}(t)\big) = \mathrm{f}(t).$$

So f is periodic with period $4a$ (when $a \neq 0$), exactly like cosine with period $4 \cdot \frac{\pi}{2} = 2\pi$.

Examples

With $a = \pi/2$. Take $\mathrm{f}(x) = \cos x$. Then $\mathrm{f}(0) = 1$, and since $\cos(\frac{\pi}{2} - x) = \sin x$ and $\cos(\frac{\pi}{2} + y) = -\sin y$,

$$\mathrm{f}(x)\mathrm{f}(y) - \mathrm{f}(\tfrac{\pi}{2} - x)\mathrm{f}(\tfrac{\pi}{2} + y) = \cos x \cos y + \sin x \sin y = \cos(x - y) = \mathrm{f}(x - y).$$

With $a = -2$. Try $\mathrm{f}(x) = \cos kx$. The results above say we need $\mathrm{f}(a) = 0$, i.e. $\cos(2k) = 0$, and period $4|a| = 8$; so take $k = \frac{\pi}{4}$, i.e. $\mathrm{f}(x) = \cos\frac{\pi x}{4}$. To verify: $\mathrm{f}(0) = 1$, and with $u = \frac{\pi x}{4}$, $v = \frac{\pi y}{4}$,

$$\mathrm{f}(-2 - x) = \cos\big(-\tfrac{\pi}{2} - u\big) = \cos\big(\tfrac{\pi}{2} + u\big) = -\sin u, \qquad \mathrm{f}(-2 + y) = \cos\big(v - \tfrac{\pi}{2}\big) = \sin v,$$

so

$$\begin{aligned}\mathrm{f}(x)\mathrm{f}(y) - \mathrm{f}(-2-x)\mathrm{f}(-2+y) &= \cos u \cos v + \sin u \sin v \\ &= \cos(u-v) = \cos\tfrac{\pi(x-y)}{4} = \mathrm{f}(x-y).\end{aligned}$$

Both functions are clearly non-constant. (More generally, $\cos kx$ works for any a with $\cos ka = 0$, so $\cos\frac{3\pi x}{4}$ would also do for $a = -2$.)

Checks

Part (iii) at $t = 0$ gives $\mathrm{f}(2a) = -\mathrm{f}(0) = -1$, agreeing with (ii); and (iii) at $t = a$ gives $\mathrm{f}(a) = -\mathrm{f}(a)$, agreeing with $\mathrm{f}(a) = 0$. Numerically, for $\mathrm{f}(x) = \cos\frac{\pi x}{4}$ with $a = -2$, at $x = 1$, $y = 3$ the left-hand side is $\mathrm{f}(-2) = 0$ and the right-hand side is $\mathrm{f}(1)\mathrm{f}(3) - \mathrm{f}(-3)\mathrm{f}(1) = \cos\frac{\pi}{4}\cos\frac{3\pi}{4} - \cos\frac{3\pi}{4}\cos\frac{\pi}{4} = 0$. A computer testing both examples at 20 000 random pairs (x, y) found the identity holding to within 10^{-13}.

Common slips

- Getting the sign of the half-period shift wrong: (iii) and (i) give $\mathrm{f}(t+2a) = -\mathrm{f}(t)$, so the period is $4a$, not $2a$.
- Assuming f is differentiable (or is cosine) when proving (i)–(iv). The question forbids it; only substitutions are allowed.
- For $a = -2$, offering $\cos\frac{\pi x}{4}$ without checking that it satisfies the identity: matching $\mathrm{f}(a) = 0$ and the period is a good guide, not a proof.

Beyond STEP

Determining a function from an identity it satisfies is the subject of *functional equations*. Here one identity alone forces evenness, a "half-period" sign change and periodicity. In analysis courses one meets the classic results: Cauchy's equation $\mathrm{f}(x+y) = \mathrm{f}(x) + \mathrm{f}(y)$ forces $\mathrm{f}(x) = cx$ once we add continuity, and without such a condition there are wild solutions. In the same way the cosine subtraction formula characterises $\cos kx$ among continuous functions. For a careful treatment of how continuity enters such arguments, see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP II 1994, Question 4

Pure • Areas of Ellipses

STEP II 1994, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s2-q4

The key idea

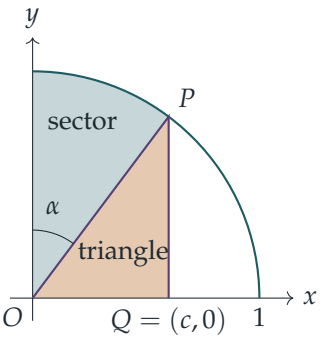
The first integral is an area under a quarter circle, and that area splits cleanly into a right-angled triangle and a circular sector, so no substitution is needed. An ellipse is a circle stretched by different factors in the two directions, and stretching multiplies areas by a constant, which gives the second result at once. For the final part, symmetry does almost all the work: the two ellipses are reflections of each other in the line $y = x$, so we only need one-eighth of the intersection, and that piece is the second result minus a triangle.

Solution

The integral of $(1 - x^2)^{1/2}$

The region in question is bounded by the quarter circle $y = (1 - x^2)^{1/2}$, the axes and the line $x = c$. (The curve has to be read as the boundary: the region is the set of points with $0 \leq y \leq (1 - x^2)^{1/2}$ and $0 \leq x \leq c$.) Its area is exactly the integral we want.

Let $P = \left(c, (1 - c^2)^{1/2}\right)$ be the point of the circle above $x = c$, let $Q = (c, 0)$, and let $\alpha = \sin^{-1} c$, so that $0 < \alpha \leq \frac{\pi}{2}$. Joining O to P cuts the region into two pieces.



- The triangle OQP has a right angle at Q, base c and height $(1-c^2)^{1/2}$, so its area is $\frac{1}{2}c(1-c^2)^{1/2}$.
- The rest is the sector of the unit circle between OP and the y-axis. The angle between OP and the y-axis is α, because the horizontal distance of P from the y-axis is $c = 1 \cdot \sin\alpha$. A sector of radius 1 and angle α has area $\frac{1}{2}\alpha = \frac{1}{2}\sin^{-1} c$.

Adding the two pieces,

$$\int_0^c (1-x^2)^{\frac{1}{2}}\,\mathrm{d}x = \tfrac{1}{2}\left[c(1-c^2)^{\frac{1}{2}} + \sin^{-1} c\right], \qquad 0 < c \le 1,$$

as required. (When $c = 1$ the triangle is flat and we get the quarter-circle area $\frac{\pi}{4}$.)

The area under the ellipse

Again the region lies under the upper half of the curve, $y = b\left(1 - x^2/a^2\right)^{1/2}$, for $0 \le x \le c$, so its area is

$$\int_0^c b\left(1 - \frac{x^2}{a^2}\right)^{\frac{1}{2}}\mathrm{d}x.$$

Substituting $x = au$, so that $\mathrm{d}x = a\,\mathrm{d}u$ and the limits become 0 and c/a (with $0 < c/a \le 1$, since $0 < c \le a$), this is

$$ab\int_0^{c/a} (1-u^2)^{\frac{1}{2}}\,\mathrm{d}u = \frac{ab}{2}\left[\frac{c}{a}\left(1 - \frac{c^2}{a^2}\right)^{\frac{1}{2}} + \sin^{-1}\left(\frac{c}{a}\right)\right],$$

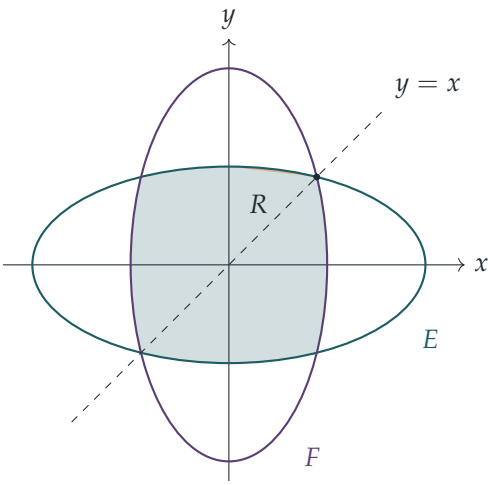
by the first result with c replaced by c/a. This is the required area.

Geometrically: the ellipse is the unit circle stretched by a horizontally and b vertically, and that stretch multiplies every area by ab.

The intersection of the two ellipses

Both E and F are unchanged by $x \mapsto -x$ and by $y \mapsto -y$, so $E \cap F$ is too, and its area is four times the area of its part in the quadrant $x, y \ge 0$. Also, swapping x and y turns E into F and F into E, so $E \cap F$ is symmetric in the line $y = x$. So the quarter in the first quadrant is made of two mirror-image halves, and

$$\text{area}(E \cap F) = 8 \times \text{area of } R, \qquad R = \{(x,y) \in E \cap F :\ 0 \le x \le y\}.$$



Ask yourself In R, which of the two ellipses actually forms the boundary?

For any point,

$$\left(\frac{x^2}{b^2}+\frac{y^2}{a^2}\right)-\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}\right)=(x^2-y^2)\left(\frac{1}{b^2}-\frac{1}{a^2}\right).$$

When $0 \le x \le y$ we have $x^2 - y^2 \le 0$, and since $0 < b \le a$ we have $\frac{1}{b^2} - \frac{1}{a^2} \ge 0$. So the difference is ≤ 0: on that side of $y = x$, any point in E is automatically in F. Hence

$$R = \{(x, y) \in E : \ 0 \le x \le y\},$$

and only E matters. (This is where $b \le a$ is used; if $b > a$ the roles of the ellipses swap.)

The boundary of E meets $y = x$ in the first quadrant where $x^2\left(\frac{1}{a^2}+\frac{1}{b^2}\right) = 1$, that is at $x = y = x_0$ with

$$x_0 = \frac{ab}{\sqrt{a^2+b^2}}.$$

For $0 \le x \le x_0$ the upper half of E lies on or above $y = x$ (it starts at height $b > 0$, and meets the line only at x_0), while for $x > x_0$ it lies below the line. So R is the region between $y = x$ and the curve $y = b(1 - x^2/a^2)^{1/2}$ for $0 \le x \le x_0$:

$$\text{area}(R) = \underbrace{\int_0^{x_0} b\left(1-\frac{x^2}{a^2}\right)^{\frac{1}{2}}\mathrm{d}x}_{\text{previous result with } c=x_0} - \underbrace{\tfrac{1}{2}x_0^2}_{\text{triangle under } y=x}.$$

With $c = x_0$ (and $0 < x_0 < a$, as needed),

$$\frac{c}{a}=\frac{b}{\sqrt{a^2+b^2}}, \qquad \left(1-\frac{c^2}{a^2}\right)^{\frac{1}{2}}=\frac{a}{\sqrt{a^2+b^2}}, \qquad \frac{c}{a}\left(1-\frac{c^2}{a^2}\right)^{\frac{1}{2}}=\frac{ab}{a^2+b^2}.$$

So

$$\begin{aligned}\text{area}(E \cap F) &= 8\,\text{area}(R)\\ &= 4ab\left[\frac{ab}{a^2+b^2} + \sin^{-1}\Big(\frac{b}{\sqrt{a^2+b^2}}\Big)\right] - \frac{4a^2b^2}{a^2+b^2}\\ &= 4ab\sin^{-1}\Big(\frac{b}{\sqrt{a^2+b^2}}\Big),\end{aligned}$$

since the two algebraic terms cancel exactly. This is the required area.

Ask yourself Why do the algebraic terms cancel?

The term $\frac{ab}{2}\cdot\frac{c}{a}(1-c^2/a^2)^{1/2} = \frac{1}{2}x_0y_0$ is the triangle O, $(x_0,0)$, (x_0,y_0) from the circle picture, stretched. Here $y_0 = x_0$, so it is exactly the triangle under $y = x$ that we subtract. What is left is a stretched sector, which is why the answer is a pure inverse sine.

Checks

- If $a = b$ the ellipses are the same circle, and the formula gives $4a^2\sin^{-1}\frac{1}{\sqrt{2}} = 4a^2\cdot\frac{\pi}{4} = \pi a^2$, as it should.
- As $b \to 0$ with a fixed, the area behaves like $4ab\cdot\frac{b}{a} = 4b^2$: the intersection is nearly the square of side $2b$ where E's thin horizontal strip crosses F's thin vertical one.
- For $a = 2$, $b = 1$ the formula gives $8\sin^{-1}\frac{1}{\sqrt{5}} \approx 3.70918$. Simpson's rule on the lower of the two boundaries gives 3.70918, and a Monte Carlo estimate with 2×10^5 points gives 3.707. The first two results were also checked by quadrature for many random a, b, c.

Common slips

- Getting the sector angle wrong: the sector is between OP and the y-axis, with angle $\sin^{-1}c$, not $\cos^{-1}c$.
- Not checking which ellipse bounds the region. In the half $0 \le x \le y$ it is E only because $b \le a$; with the other choice the answer would come out with a in place of b inside the inverse sine.

Beyond STEP

A linear map $(x,y)\mapsto(ax,by)$ multiplies every area by ab, and in general a linear map multiplies areas by the absolute value of its determinant. This is the two-dimensional case of the Jacobian factor in the change-of-variables formula for multiple integrals,

which you meet in first-year calculus. Strang's *Introduction to Linear Algebra* explains the determinant as an area (and volume) scale factor.

WORKED SOLUTION

STEP II 1994, Question 5

Pure • Counting Real Roots

STEP II 1994, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s2-q5

The key idea

Counting roots of $f(x) = c$ means asking how many times the horizontal line $y = c$ meets the graph of f, so what matters is the shape of f: where it is smallest, and where it is increasing or decreasing. Every expression here is built from two brackets placed symmetrically about a centre point. Shifting the variable so that the centre becomes the origin makes the odd-power terms cancel (for fourth powers) or the even-power terms cancel (for cubes), and the shape becomes obvious.

Solution

Part (i)

Expand by the binomial theorem. In $(x+1)^4$ and $(x-1)^4$ the terms with odd powers of x have opposite signs, so they cancel in the sum:

$$f(x) = (x-1)^4 + (x+1)^4 = 2\left(x^4 + 6x^2 + 1\right) = 2x^4 + 12x^2 + 2.$$

So f is even, and $f(x) = 2 + 2x^4 + 12x^2 \geq 2$, with equality only at $x = 0$. For $x \geq 0$, $f'(x) = 8x^3 + 24x > 0$ when $x > 0$, so f is strictly increasing on $[0, \infty)$, and $f(x) \to \infty$ as $x \to \infty$. Since f is continuous it takes every value in $[2, \infty)$ exactly once on $[0, \infty)$; by symmetry it does the same on $(-\infty, 0]$.

- If $c > 2$: there is exactly one root $x = a > 0$, and then $x = -a$ is a second; the two are distinct and there are no others. So there are **exactly two** real roots.
- If $c = 2$: $f(x) = 2$ forces $2x^4 + 12x^2 = 0$, so $x = 0$ is the **only** root.
- If $c < 2$: $f(x) \geq 2 > c$ for all x, so there are **no** roots.

This is exactly what was required.

(Alternatively, $2x^4 + 12x^2 + 2 - c = 0$ is a quadratic in $u = x^2$ with roots $u = -3 \pm \sqrt{8 + c/2}$; only the $+$ root can be non-negative, and it is positive, zero or negative according as $c > 2$, $c = 2$ or $c < 2$.)

Part (ii)

The brackets $x-3$ and $x-1$ are centred on $x=2$, so put $y=x-2$. Then

$$(x-3)^4+(x-1)^4=(y-1)^4+(y+1)^4,$$

which is the expression from part (i) in the variable y. Since $x \mapsto y = x-2$ is a one-to-one correspondence between real numbers, the equation has the same number of real roots as in part (i):

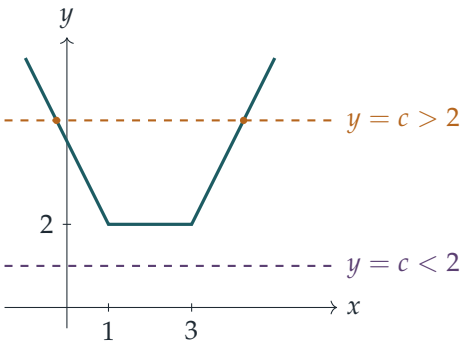
$$\text{two roots if } c>2, \qquad \text{one root } (x=2) \text{ if } c=2, \qquad \text{no roots if } c<2.$$

Part (iii)

Let $g(x)=|x-3|+|x-1|$ and remove the modulus signs on each of the three intervals cut out by $x=1$ and $x=3$:

$$g(x)=\begin{cases}(3-x)+(1-x)=4-2x, & x<1,\\ (3-x)+(x-1)=2, & 1\le x\le 3,\\ (x-3)+(x-1)=2x-4, & x>3.\end{cases}$$

So the graph is a flat-bottomed V: constant at 2 between 1 and 3, strictly decreasing to the left and strictly increasing to the right, with $g>2$ outside $[1,3]$.



- If $c<2$: $g(x)\ge 2>c$ everywhere, so there are **no** roots.
- If $c=2$: every x with $1\le x\le 3$ is a root, and nothing outside, so there are **infinitely many** roots.
- If $c>2$: there is no root in $[1,3]$; on $x<1$, $4-2x=c$ gives $x=2-\frac{c}{2}$, which does lie below 1 since $c>2$; on $x>3$, $2x-4=c$ gives $x=2+\frac{c}{2}>3$. So there are **exactly two** roots, $x=2\pm\frac{c}{2}$.

? **Ask yourself** Why is the answer for $c=2$ so different from part (ii)?

With fourth powers the sum has a single sharp minimum at the centre point, because both terms are strictly convex. With moduli, moving x between 1 and 3 makes one distance grow by exactly as much as the other shrinks, so the sum stays at 2 (the distance between 1 and 3) along the whole interval.

Part (iv)

Again put $y = x - 2$. For cubes it is the even powers of y that cancel:

$$(x-3)^3 + (x-1)^3 = (y-1)^3 + (y+1)^3 = 2y^3 + 6y.$$

Let $h(y) = 2y^3 + 6y$. Then

$$h'(y) = 6y^2 + 6 \geq 6 > 0$$

for every real y, so h is strictly increasing on the whole real line, and hence takes each value at most once. Also $h(y) \to \infty$ as $y \to \infty$ and $h(y) \to -\infty$ as $y \to -\infty$, so, since h is continuous, by the intermediate value theorem it takes every real value at least once.

Hence for **every** real c the equation has **exactly one** real root.

Checks

- At the centre point $x = 2$, both fourth-power expressions equal $1 + 1 = 2$, matching the $c = 2$ cases, and $(2-3)^3 + (2-1)^3 = 0$.
- In part (iv), $c = 8$ should give $x = 3$: indeed $0^3 + 2^3 = 8$, and $h(1) = 2 + 6 = 8$ with $y = 1$.
- In part (iii) with $c = 4$, the roots $x = 0$ and $x = 4$ give $3 + 1 = 4$ and $1 + 3 = 4$.

We confirmed numerically that the expansions hold at random points, and counted sign changes of each function minus c on a fine grid for a range of values of c (for example $c = 1.9$, 2.5, 100 in (i) and (ii), and c from -1000 to 5000 in (iv)): the counts were 0, 2, 2 and always 1, as claimed.

Common slips

- Answering "one root" or "two roots" for $c = 2$ in part (iii): the whole interval $1 \leq x \leq 3$ consists of roots.
- In part (iv), arguing that a cubic "has one or three real roots" without deciding which. The fact that $h'(y) > 0$ everywhere settles it.
- Forgetting to check that $x = 2 - \frac{c}{2}$ really lies in the region $x < 1$ where the formula $4 - 2x$ was used.

Beyond STEP

The arguments here rest on two theorems that a first analysis course proves carefully: the intermediate value theorem (a continuous function on an interval takes every value in between) and the fact that a function with positive derivative is strictly increasing, which follows from the mean value theorem. Part (iii) is also a first taste of convexity and of the ℓ^1 distance: the sum of distances to two points is minimised along the

whole segment between them. Abbott's *Understanding Analysis* develops these ideas rigorously.

WORKED SOLUTION

STEP II 1994, Question 6

Pure • Telescoping Trigonometric Series

STEP II 1994, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s2-q6

The key idea

Everything rests on one small identity: $\tan x = \cot x - 2\cot 2x$. It turns each term of the sum into a difference of two consecutive terms of the form $\frac{1}{2^r}\cot\frac{\theta}{2^r}$, so the sum telescopes, and that same identity is exactly what makes the inductive step work. The infinite sum then needs only the limit $\frac{\sin\varphi}{\varphi} \to 1$.

Solution

A useful identity

For any x with $\sin x$, $\cos x$ and $\sin 2x$ all non-zero,

$$\cot x - \tan x = \frac{\cos x}{\sin x} - \frac{\sin x}{\cos x} = \frac{\cos^2 x - \sin^2 x}{\sin x \cos x} = \frac{\cos 2x}{\frac{1}{2}\sin 2x} = 2\cot 2x,$$

so

$$\tan x = \cot x - 2\cot 2x. \qquad (*)$$

We shall only use $(*)$ with $x = \theta/2^r$ for $r \geq 1$. Since $0 < \theta < \pi$, such an x lies in $(0, \frac{\pi}{2})$, so $\sin x$, $\cos x$ and $\sin 2x$ are all positive and every tan and cot we write down is defined.

The finite sum, by induction

Write S_n for the left-hand side, $\sum_{r=1}^{n} \frac{1}{2^r}\tan\frac{\theta}{2^r}$, and let $P(n)$ be the statement

$$S_n = \frac{1}{2^n}\cot\frac{\theta}{2^n} - \cot\theta.$$

Base case $n = 1$. Put $x = \theta/2$ in $(*)$ and halve:

$$S_1 = \tfrac{1}{2}\tan\tfrac{\theta}{2} = \tfrac{1}{2}\cot\tfrac{\theta}{2} - \cot\theta,$$

which is $P(1)$.

Inductive step. Suppose $P(k)$ holds for some $k \geq 1$. Put $x = \theta/2^{k+1}$ in $(*)$, so that $2x = \theta/2^k$, and divide by 2^{k+1}:

$$\frac{1}{2^{k+1}} \tan \frac{\theta}{2^{k+1}} = \frac{1}{2^{k+1}} \cot \frac{\theta}{2^{k+1}} - \frac{1}{2^k} \cot \frac{\theta}{2^k}.$$

Adding this to $P(k)$,

$$\begin{aligned} S_{k+1} &= S_k + \frac{1}{2^{k+1}} \tan \frac{\theta}{2^{k+1}} \\ &= \left(\frac{1}{2^k} \cot \frac{\theta}{2^k} - \cot \theta \right) + \left(\frac{1}{2^{k+1}} \cot \frac{\theta}{2^{k+1}} - \frac{1}{2^k} \cot \frac{\theta}{2^k} \right) \\ &= \frac{1}{2^{k+1}} \cot \frac{\theta}{2^{k+1}} - \cot \theta, \end{aligned}$$

which is $P(k+1)$. By induction $P(n)$ holds for every $n \geq 1$ and every $\theta \in (0, \pi)$, as required.

Ask yourself How would you spot the identity $(*)$ if you hadn't seen it before?

Work backwards from the induction. Going from $n = k$ to $n = k+1$, the right-hand side changes by $\frac{1}{2^{k+1}} \cot \frac{\theta}{2^{k+1}} - \frac{1}{2^k} \cot \frac{\theta}{2^k}$, and the left-hand side by $\frac{1}{2^{k+1}} \tan \frac{\theta}{2^{k+1}}$. Multiplying by 2^{k+1} and writing $x = \theta/2^{k+1}$, the two changes agree precisely when $\tan x = \cot x - 2 \cot 2x$. So the question tells you which identity to prove; you then check it with the double-angle formulae.

The infinite sum

The terms are all positive, and we have an exact formula for the partial sums, so it suffices to find the limit of S_n as $n \to \infty$. Let $\varphi_n = \theta/2^n$, so $\varphi_n \to 0^+$ and $\frac{1}{2^n} = \frac{\varphi_n}{\theta}$. Then

$$\frac{1}{2^n} \cot \frac{\theta}{2^n} = \frac{\varphi_n}{\theta} \cdot \frac{\cos \varphi_n}{\sin \varphi_n} = \frac{1}{\theta} \cdot \frac{\varphi_n}{\sin \varphi_n} \cdot \cos \varphi_n \longrightarrow \frac{1}{\theta} \cdot 1 \cdot 1 = \frac{1}{\theta},$$

using the standard limit $\frac{\sin \varphi}{\varphi} \to 1$ as $\varphi \to 0$ and $\cos \varphi \to 1$. Since $\cot \theta$ does not depend on n,

$$\sum_{r=1}^{\infty} \frac{1}{2^r} \tan \frac{\theta}{2^r} = \lim_{n \to \infty} S_n = \frac{1}{\theta} - \cot \theta,$$

which is the required result. In particular the series converges for every $\theta \in (0, \pi)$.

Ask yourself Why is it legitimate to say "the infinite sum equals the limit of S_n"?

That is the definition of the sum of an infinite series: $\sum_{r=1}^{\infty} a_r$ means $\lim_{n \to \infty} \sum_{r=1}^{n} a_r$, when that limit exists. We have shown the limit exists by computing it, so nothing more is needed.

Checks

- $\theta = \frac{\pi}{2}$: the right-hand side is $\frac{2}{\pi} - 0 \approx 0.63662$. Summing the first 60 terms of the series numerically also gives 0.6366198.
- $\theta = 1$: both the series and $1 - \cot 1$ give 0.3579074.
- Small θ: $\tan\frac{\theta}{2^r} \approx \frac{\theta}{2^r}$, so the series is about $\theta\sum 4^{-r} = \frac{\theta}{3}$. This agrees with the expansion $\frac{1}{\theta} - \cot\theta = \frac{\theta}{3} + O(\theta^3)$.
- As $\theta \to \pi^-$, $\cot\theta \to -\infty$ and the first term $\frac{1}{2}\tan\frac{\theta}{2} \to +\infty$, so both sides blow up together, as they should.

Common slips

- Getting the factor of 2 in $(*)$ wrong, for example writing $\tan x = \cot x - \cot 2x$. Test it at $x = \frac{\pi}{4}$: $1 = 1 - 2\cdot 0$.
- Claiming $\frac{1}{2^n}\cot\frac{\theta}{2^n} \to 0$ because $\frac{1}{2^n} \to 0$. The cotangent grows like $2^n/\theta$, so the product tends to $\frac{1}{\theta}$, not 0.
- Forgetting to say why the tangents and cotangents are defined: it is the condition $0 < \theta < \pi$ that keeps every angle $\theta/2^r$ inside $(0, \frac{\pi}{2})$.

Beyond STEP

The function $\frac{1}{\theta} - \cot\theta$ belongs to a famous family: differentiating the partial-fraction expansion $\cot\theta = \frac{1}{\theta} + \sum_{k\geq 1}\frac{2\theta}{\theta^2 - k^2\pi^2}$ gives the Basel sum $\sum 1/k^2 = \pi^2/6$, and Euler obtained such results from the infinite product for $\sin\theta$. Taking logarithms and differentiating in Viète's product $\frac{\sin\theta}{\theta} = \prod_{r\geq 1}\cos\frac{\theta}{2^r}$ gives exactly this question's series, which is another way to see the result. Making such manipulations of infinite sums and products rigorous is a first-year analysis topic; see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP II 1994, Question 7

Pure • Circles of Apollonius

STEP II 1994, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s2-q7

The key idea

The angle of elevation of the top of a pole of height h seen from distance r has tangent h/r, so "equal angles of elevation" just says that the soldier's distances from the two feet are in the fixed ratio $h_1 : h_2$. Squaring that condition gives an equation of exactly the form in the first part, so the locus is a circle (a circle of Apollonius). For the last part, put the poles at general positions and write each centre as a weighted combination of two feet; the identity then drops out, and its coefficients add up to zero, which is what forces collinearity.

Solution

The circle

Since $a > 0$ we may divide by a and complete the square:

$$x^2 + y^2 + \frac{2g}{a}x + \frac{2f}{a}y + \frac{c}{a} = 0 \iff \left(x + \frac{g}{a}\right)^2 + \left(y + \frac{f}{a}\right)^2 = \frac{g^2 + f^2 - ac}{a^2}.$$

The right-hand side is positive because $f^2 + g^2 > ac$, so this is a circle with

$$\text{centre } \left(-\frac{g}{a}, -\frac{f}{a}\right), \qquad \text{radius } \frac{\sqrt{g^2 + f^2 - ac}}{a}.$$

(Without the condition the "circle" would be a single point or empty.)

Two flagpoles

Put the foot of the pole of height h_1 at the origin and the foot of the other at $(d, 0)$. If the soldier is at (x, y), at distances r_1 and r_2 from the two feet, the angles of elevation θ_1, θ_2 satisfy $\tan\theta_i = h_i/r_i$. Both angles lie in $(0, \frac{\pi}{2})$, where tan is one-to-one, so

$$\theta_1 = \theta_2 \iff \frac{h_1}{r_1} = \frac{h_2}{r_2} \iff h_2^2 r_1^2 = h_1^2 r_2^2,$$

the last step being reversible because all the quantities are positive. (The soldier can't stand at a foot: $r_1 = 0$ would force $r_2 = 0$ too.) So the path is

$$h_2^2(x^2+y^2) = h_1^2((x-d)^2+y^2) \iff (h_2^2-h_1^2)(x^2+y^2) + 2h_1^2 d\,x - h_1^2 d^2 = 0.$$

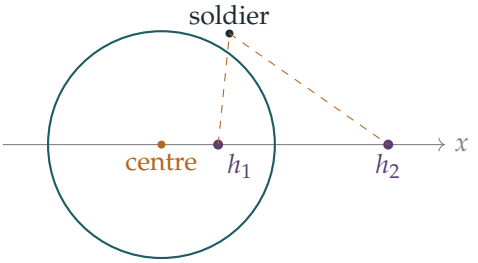
Label the poles so that $h_2 > h_1$ (if they're different, one is taller). Then this is the first part's equation with

$$a = h_2^2 - h_1^2 > 0, \quad g = h_1^2 d, \quad f = 0, \quad c = -h_1^2 d^2,$$

and

$$g^2 + f^2 - ac = h_1^4 d^2 + (h_2^2 - h_1^2)h_1^2 d^2 = h_1^2 h_2^2 d^2 > 0.$$

So the soldier marches round a circle with centre $\left(-\dfrac{h_1^2 d}{h_2^2 - h_1^2},\, 0\right)$ and radius $\dfrac{h_1 h_2 d}{h_2^2 - h_1^2}$. The centre lies on the line through the feet, beyond the shorter pole. The circle goes round the shorter pole, which makes sense: to make the short pole look as tall as the tall one, you have to stand nearer to it.



(Drawn for $h_1 = 1$, $h_2 = 2$, $d = 3$: the centre is $(-1, 0)$ and the radius is 2.)

If $h_1 = h_2$, the $x^2 + y^2$ terms cancel and the equation becomes $2h_1^2 d\,x = h_1^2 d^2$, that is $x = \frac{1}{2}d$. The soldier walks along the perpendicular bisector of the line joining the feet, a straight line rather than a circle. This is the limit of the circles above: as $h_2 \to h_1$ the radius tends to infinity.

Three flagpoles

Now we can't put all three feet on an axis, so let the pole of height h_i stand at $\mathbf{p}_i = (p_i, q_i)$. Repeating the argument above with $r_i^2 = (x-p_i)^2 + (y-q_i)^2$, the soldier assigned to poles i and j walks along

$$(h_j^2 - h_i^2)(x^2+y^2) - 2(h_j^2 p_i - h_i^2 p_j)x - 2(h_j^2 q_i - h_i^2 q_j)y + \text{const} = 0.$$

This is a circle (the value of $g^2 + f^2 - ac$ doesn't change when we move the origin, so it is still $h_i^2 h_j^2 |\mathbf{p}_i - \mathbf{p}_j|^2 > 0$, after multiplying through by -1 if $h_j < h_i$). Using the first part, its centre $\mathbf{c}_{ij} = (x_{ij}, y_{ij})$ satisfies

$$(h_j^2 - h_i^2)\,\mathbf{c}_{ij} = h_j^2 \mathbf{p}_i - h_i^2 \mathbf{p}_j.$$

This expression doesn't change if we swap i and j (both sides change sign), so it doesn't matter which way round we label $\mathbf{c}_{31}$.

Ask yourself How do we turn three of these formulae into the given identity without solving for anything?

Each term in the identity already has a factor $\pm(h_j^2 - h_i^2)$ next to x_{ij}, so we can substitute the formula above directly, using vectors so that the x- and y-coordinates are done together:

$$\begin{aligned} h_3^2(h_1^2 - h_2^2)\,\mathbf{c}_{12} &= -h_3^2(h_2^2\mathbf{p}_1 - h_1^2\mathbf{p}_2) = -h_2^2h_3^2\,\mathbf{p}_1 + h_1^2h_3^2\,\mathbf{p}_2, \\ h_1^2(h_2^2 - h_3^2)\,\mathbf{c}_{23} &= -h_1^2(h_3^2\mathbf{p}_2 - h_2^2\mathbf{p}_3) = -h_1^2h_3^2\,\mathbf{p}_2 + h_1^2h_2^2\,\mathbf{p}_3, \\ h_2^2(h_3^2 - h_1^2)\,\mathbf{c}_{31} &= h_2^2(h_3^2\mathbf{p}_1 - h_1^2\mathbf{p}_3) = h_2^2h_3^2\,\mathbf{p}_1 - h_1^2h_2^2\,\mathbf{p}_3. \end{aligned}$$

Adding, everything cancels in pairs, so

$$h_3^2(h_1^2 - h_2^2)\,\mathbf{c}_{12} + h_1^2(h_2^2 - h_3^2)\,\mathbf{c}_{23} + h_2^2(h_3^2 - h_1^2)\,\mathbf{c}_{31} = \mathbf{0}.$$

The x-component is the required relation between the x_{ij}, and the y-component is the same relation between the y_{ij}.

Ask yourself Why does a linear relation between three points make them collinear?

Call the three coefficients $A = h_3^2(h_1^2 - h_2^2)$, $B = h_1^2(h_2^2 - h_3^2)$, $C = h_2^2(h_3^2 - h_1^2)$. The heights are positive and all different, so none of A, B, C is zero. The key extra fact is that they add up to zero:

$$A + B + C = h_1^2h_3^2 - h_2^2h_3^2 + h_1^2h_2^2 - h_1^2h_3^2 + h_2^2h_3^2 - h_1^2h_2^2 = 0.$$

So $C = -A - B$, and the relation becomes

$$A(\mathbf{c}_{12} - \mathbf{c}_{31}) + B(\mathbf{c}_{23} - \mathbf{c}_{31}) = \mathbf{0}, \qquad \text{so} \qquad \mathbf{c}_{12} - \mathbf{c}_{31} = -\frac{B}{A}\,(\mathbf{c}_{23} - \mathbf{c}_{31}).$$

The vectors from $\mathbf{c}_{31}$ to the other two centres are parallel (or one is zero, if two centres coincide), so the three centres lie on a straight line. Without $A + B + C = 0$ a linear relation would prove nothing: any three points satisfy some relation of that kind.

Checks

- With $h_1 = 1$, $h_2 = 2$, $d = 3$ our formulae give centre $(-1, 0)$ and radius 2. The point $(1, 0)$ is at distances 1 and 2 from the feet, and $(-3, 0)$ is at distances 3 and 6: both in ratio $1 : 2$, as they should be.
- A numerical check put random points on each circle and confirmed that the two angles of elevation agreed. For 2000 random triples of poles in random positions, it also confirmed the identity and the collinearity of the three centres.
- Another reason to expect collinearity: if a point lies on two of the circles, then $r_1 : r_2 = h_1 : h_2$ and $r_2 : r_3 = h_2 : h_3$, so $r_1 : r_3 = h_1 : h_3$ and it lies on the third. When the circles meet, they all pass through the same two points, so their centres all lie on the perpendicular bisector of those points.

Common slips

- Setting $h_1/r_1 = h_2/r_2$ without saying why equal angles give equal tangents (both angles are acute).
- Forgetting to check $g^2 + f^2 > ac$: the first part only gives a circle under that condition.
- Deducing collinearity from the identity alone, without using $A + B + C = 0$ and $A, B, C \neq 0$.

Beyond STEP

The set of points whose distances from two fixed points are in a fixed ratio $k \neq 1$ is a *circle of Apollonius*. Taking all values of k gives a *coaxal system* of circles, a family in which any two circles share the same radical axis. Families like this are where inversive geometry begins, and in the complex plane they are the circles $|z - a| = k|z - b|$, which Möbius transformations map to one another. Coxeter and Greitzer, *Geometry Revisited*, covers coaxal circles and inversion; Needham, *Visual Complex Analysis*, shows the complex-number side.

WORKED SOLUTION

STEP II 1994, Question 8

Pure • Optimising a Quadratic Model

STEP II 1994, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s2-q8

The key idea

Follow the money through one delivery cycle. Stock falls at a steady rate r until it runs out, so $S(t)$ is a simple ramp, and the storage bill is b times the area under that ramp. The profit per cycle is then a quadratic in X. Maximising it over the allowed range $0 < X \leq rT$ gives the condition, and the hypothesis $T \geq p/b$ is exactly what puts the vertex of the parabola inside that range.

Solution

Why $X \leq rT$

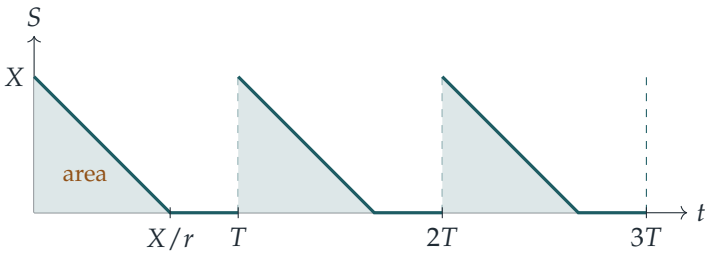
In T hours she can sell at most rT units. If $X > rT$, at least $X - rT$ units are still on the shelf when the next lorry arrives. Each cycle then adds at least $X - rT$ more to this unsold surplus, so after n cycles it is at least $n(X - rT)$. That grows without bound, and so does the storage bill of b per unit per hour. She pays for these units (and for storing them) but they bring in no extra sales, because sales are capped at rT per cycle anyway. So ordering more than rT per delivery only adds cost, and she should take $X \leq rT$.

The stock level

With $X \leq rT$ each cycle starts with exactly X units. In the continuous model she sells at rate r while there's stock, so

$$S(t) = \begin{cases} X - rt, & 0 \leq t \leq X/r, \\ 0, & X/r \leq t \leq T. \end{cases}$$

The graph is a straight line falling from X to zero at $t = X/r$, then flat at zero until the next delivery at $t = T$ (where it jumps back up to X). This gives a sawtooth over several cycles.



Profit per cycle

Over one period of T hours:

- *Sales.* All X units are sold (since $X \le rT$), at $£(p+q)$ each, giving $£(p+q)X$.
- *Delivery.* $£(a+qX)$.
- *Storage.* Storing $S(t)$ units for a short time δt costs $bS(t)\,\delta t$, so the total is

$$b\int_0^T S(t)\,\mathrm{d}t = b\int_0^{X/r}(X-rt)\,\mathrm{d}t = b\cdot\tfrac{1}{2}\cdot X\cdot\frac{X}{r} = \frac{bX^2}{2r},$$

  the area of the shaded triangle times b.
- *Running costs.* $£cT$.

The qX terms cancel, so the profit per cycle is

$$P(X) = pX - \frac{bX^2}{2r} - a - cT.$$

So p is the real profit margin on each unit: q is simply passed on to the customer.

When can the business be profitable?

With T fixed, she controls X in the range $0 < X \le rT$. Completing the square,

$$P(X) = \frac{p^2 r}{2b} - a - cT - \frac{b}{2r}\left(X - \frac{pr}{b}\right)^2.$$

Ask yourself Is the best X actually allowed?

The parabola peaks at $X = pr/b$, and we need $pr/b \le rT$, which is $T \ge p/b$. That's precisely the hypothesis. So she can choose $X = pr/b$, and then

$$P = \frac{p^2 r}{2b} - (a + cT).$$

This is positive exactly when $p^2 r > 2b(a+cT)$, that is, when

$$p^2 > 2\,\frac{(a+cT)b}{r}.$$

So if this inequality holds, the order size $X = pr/b$ makes the business profitable, as required. (In fact pr/b is the best possible choice, so the condition is also necessary when $T \geq p/b$.)

If instead $T < p/b$, the vertex lies beyond rT. Then P is increasing on the whole allowed range and the best she can do is $X = rT$. That's why the hypothesis on T is needed for this particular condition.

Checks

- Take $a = 10$, $b = 1$, $r = 10$, $p = 4$, $c = 1$, $T = 5$. Then $T \geq p/b = 4$, and $p^2 = 16 > 2(15)(1)/10 = 3$. The best order is $X = 40 \leq rT = 50$, with profit $160 - 10 - 80 - 5 = 65 = \frac{16 \cdot 10}{2} - 15$.
- The computer stepped $S(t)$ numerically through a cycle for 30 random sets of parameters and matched $P(X)$ to within 10^{-7}. A grid search over X for 2000 random cases with $T \geq p/b$ confirmed the maximum $\frac{p^2 r}{2b} - a - cT$.
- Units: $p^2 r/b$ is $(£/\text{unit})^2 \times (\text{units/h})/(£/\text{unit/h}) = £$, as are a and cT.

Common slips

- Charging storage on all X units for all T hours (bXT) instead of on the stock actually held, $b \int S \, dt$.
- Forgetting to check that $X = pr/b$ satisfies $X \leq rT$. That check is the only place the condition $T \geq p/b$ is used.

Beyond STEP

This is the classic *economic order quantity* model of operations research. Balancing a fixed cost per order against a holding cost per unit-time gives a square-root rule. Here, if she can also choose T, the profit *rate* P/T is maximised by trading off a against storage. Constrained optimisation of this kind (here the vertex versus the boundary $X = rT$) grows into linear and nonlinear programming and the Karush–Kuhn–Tucker conditions. For the habit of building and testing such models, Siklos, *Advanced Problems in Mathematics*, is a good companion.

WORKED SOLUTION

STEP II 1994, Question 9

Mechanics • Equilibrium of Rigid Bodies

STEP II 1994, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s2-q9

The key idea

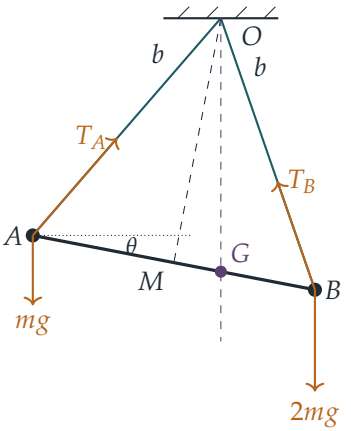
Treat the rod, the two strings and the two particles as one rigid body. The only external forces on it are the two weights and the support at O, so for equilibrium the combined weight, acting at the centre of mass G, must act along a line through O: G hangs vertically below O. That turns the angle into pure geometry of an isosceles triangle. For the tension, take moments about B, which removes both unknown forces acting at B.

Solution

Model. The rod and strings are light, the strings inextensible, and A, B are particles. Since $OA = OB = b$ and $AB = 2a$, triangle OAB is isosceles. Let M be the midpoint of AB; then $OM \perp AB$ and, by Pythagoras,

$$OM = h = \sqrt{b^2 - a^2}.$$

The heavier particle B hangs lower, and the rod makes angle θ with the horizontal.



The angle

Consider the whole system: rod, strings and particles together. The external forces on it are the weights mg at A and $2mg$ at B, and the force from the support at O. Taking moments about O, the support force has no moment, so the total moment of the weights about O must vanish. The two weights are parallel, so they combine into a single force $3mg$ through the centre of mass G of the particles. Its moment about O is zero only if G lies on the vertical through O. So G is directly below O.

Now find G. It lies on AB with $AG : GB = 2m : m = 2 : 1$, so $AG = \frac{2}{3} \cdot 2a = \frac{4a}{3}$, and therefore

$$MG = AG - AM = \tfrac{4a}{3} - a = \tfrac{a}{3},$$

with G on the B side of M.

Ask yourself Why is the angle MOG the same as θ?

OG is vertical and OM is perpendicular to the rod. Rotating the vertical onto OM is the same rotation that takes the horizontal onto the rod, so the angle between OG and OM equals the angle between the rod and the horizontal, namely θ.

Triangle OMG has a right angle at M, so

$$\tan\theta = \frac{MG}{OM} = \frac{a/3}{\sqrt{b^2 - a^2}} = \frac{a}{3\sqrt{b^2 - a^2}},$$

as required.

The tension in AO

Again take the rod and particles together. The forces on this system are T_A along AO, T_B along BO, and the two weights. (The force in the rod is internal.) Taking moments about B kills T_B and the weight $2mg$.

The perpendicular distance from B to the line OA is twice the area of triangle OAB divided by OA. That area is $\frac{1}{2} \cdot 2a \cdot h = ah$, so the distance is $2ah/b$. The weight mg at A has moment arm equal to the horizontal distance between A and B, which is $2a\cos\theta$. The two moments have opposite senses, and balancing them gives

$$T_A \cdot \frac{2ah}{b} = mg \cdot 2a\cos\theta \quad \Longrightarrow \quad T_A = \frac{mgb\cos\theta}{h}.$$

From $\tan\theta = \dfrac{a}{3h}$, with θ acute, we get $\cos\theta = \dfrac{3h}{\sqrt{9h^2 + a^2}}$, and $9h^2 + a^2 = 9b^2 - 8a^2$. Hence

$$T_A = \frac{mgb}{h} \cdot \frac{3h}{\sqrt{9b^2 - 8a^2}} = \frac{3mgb}{\sqrt{9b^2 - 8a^2}}.$$

(The same argument with moments about A gives $T_B = 2T_A$: each string tension is proportional to the mass it supports.)

Checks

- If $a \to 0$ the strings become vertical and $T_A \to 3mgb/(3b) = mg$, which is exactly what particle A needs.
- Vertical resolution for the whole system should give $3mg$. With $a = 1$, $b = 2$, $m = 1$, $g = 9.8$ we get $T_A \approx 11.11\,\text{N}$ and $T_B \approx 22.22\,\text{N}$. Resolving these along the string directions gives a total vertical component of $29.4\,\text{N} = 3mg$.
- A numerical check places the rod at the predicted angle for 2000 random (a, b, m) and solves the force balance at each particle separately. The two particles agree on the rod force, which is a thrust, and the tension matches the formula to about 10^{-15}.

Common slips

- Putting G a distance $\frac{2a}{3}$ from M, or on the wrong side. The centre of mass is nearer the heavier particle, at $\frac{a}{3}$ from the midpoint.
- Taking the moment arm of T_A about B to be $2a$ or b. You need the perpendicular distance from B to the line OA.
- Leaving the answer in terms of θ, when it is asked for in terms of m, g, a and b.

Beyond STEP

The principle that a body hanging freely from one point settles with its centre of mass below that point is the simplest case of a deeper fact: stable equilibria minimise potential energy, and here the potential energy is just $3mg$ times the height of G. At university this becomes the virtual work principle and Lagrangian mechanics, where equilibria are found by differentiating an energy rather than by resolving forces. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body statics and energy methods carefully.

WORKED SOLUTION

STEP II 1994, Question 10

Mechanics • Elastic Strings and SHM

STEP II 1994, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s2-q10

The key idea

Work with the *extension* of the rope rather than with the trailer's position. Because the truck's velocity is constant, the extension obeys exactly the same equation as the trailer's acceleration, up to sign, and that equation is simple harmonic motion. So the rope stays taut for half an oscillation, and then lets go of a trailer that is moving faster than the truck. After that it's a plain constant-velocity catch-up problem.

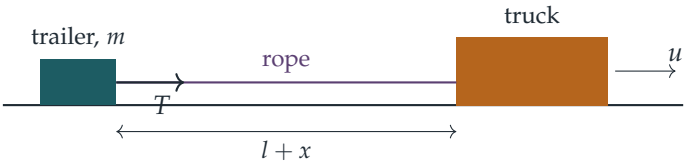
Solution

Setting up

Model the trailer as a particle, the rope as light and obeying Hooke's law while taut (and exerting no force while slack), and ignore friction. Measure time t from the moment the rope first becomes taut, and distances along the direction of motion. At $t = 0$ the truck is moving at speed u (and keeps that speed), the trailer is at rest, and the truck is a distance l ahead of it.

Let the trailer be at position $y(t)$, with $y(0) = 0$, so the truck is at $l + ut$. The extension of the rope is

$$x = (l + ut) - y - l = ut - y.$$



While the rope is taut

While $x > 0$ the tension is $T = \lambda x/l$, and it is the only horizontal force on the trailer, pulling it forwards. So

$$m\ddot{y} = \frac{\lambda x}{l}.$$

Since $x = ut - y$ and u is constant, $\ddot{x} = -\ddot{y}$, and hence

$$\ddot{x} = -\omega^2 x, \qquad \text{where } \omega^2 = \frac{\lambda}{ml}.$$

This is simple harmonic motion. The initial conditions are $x(0) = 0$ (the rope has just become taut) and $\dot{x}(0) = u - \dot{y}(0) = u$, so

$$x = \frac{u}{\omega}\sin\omega t, \qquad \dot{x} = u\cos\omega t.$$

The trailer's velocity is $\dot{y} = u - \dot{x} = u(1 - \cos\omega t)$.

> **Ask yourself** The SHM formula makes x negative after a while. Is it still valid then?

No. The equation $\ddot{x} = -\omega^2 x$ used the tension $\lambda x/l$, which only applies when $x > 0$; a rope can't push. The formula holds for $0 \le t \le \pi/\omega$, where $\sin\omega t \ge 0$, and the rope goes slack again at

$$t_1 = \frac{\pi}{\omega} = \pi\left(\frac{lm}{\lambda}\right)^{1/2}.$$

At that moment the extension is zero, so the trailer is again a distance l behind the truck, and its velocity is

$$\dot{y}(t_1) = u(1 - \cos\pi) = 2u.$$

After the rope goes slack

For $t > t_1$ the trailer moves at $2u$ and the truck at u, so the gap between them, which is l at $t = t_1$, shrinks at the steady rate $2u - u = u$. The gap is then less than l, so the rope stays slack, there is still no horizontal force on the trailer, and its speed really does stay at $2u$. The gap closes to zero after a further time l/u.

So the trailer collides with the truck at time

$$t_1 + \frac{l}{u} = \pi\left(\frac{lm}{\lambda}\right)^{1/2} + \frac{l}{u}$$

after the rope first becomes taut, as required.

Checks

- Units: λ is a force, so lm/λ has units of $\mathrm{m}\cdot\mathrm{kg}/\mathrm{N} = \mathrm{s}^2$, and both terms are times.

- Change of frame: the rope behaves like a spring in an elastic collision between the trailer and a truck of effectively infinite mass. In the truck's frame the trailer approaches at u and leaves at u in the opposite direction, so it ends with ground speed $2u$, just as found.
- A numerical integration of the trailer's equation of motion (with the force switched off whenever the rope is slack) for random values of m, l, λ, u reproduced the formula; for example m, l, λ, u chosen so that $\pi/\omega \approx 2.9944$ and $l/u \approx 2.3017$ gave a collision time of 5.2962 against the predicted 5.2961.

Common slips

- Applying the SHM solution for all time, forgetting that the rope goes slack when the extension returns to zero.
- Writing the equation of motion for the trailer's position y directly, which gives $m\ddot{y} = \lambda(ut - y)/l$, a forced equation. It's solvable, but the substitution $x = ut - y$ makes it unforced at once.
- Forgetting the final l/u: when the rope goes slack the trailer is still a full natural length behind the truck.

Beyond STEP

Changing to the truck's frame turns the whole episode into an elastic "collision" between the trailer and a spring attached to an immovable wall: the trailer comes in at speed u and leaves at speed u, and the contact lasts exactly half a period, whatever the speed. This is the simplest example of how Galilean invariance and energy conservation constrain collisions, an idea that runs through first-year mechanics and leads on to scattering theory. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats both oscillations and collisions in this spirit.

WORKED SOLUTION

STEP II 1994, Question 11

Mechanics • Projectiles and Explosions

STEP II 1994, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s2-q11

The key idea

The explosion happens at a known height, the top of the vertical flight, and at that moment the shell is at rest. So the two fragments start from the same point with equal and opposite velocities. If we describe the direction by one angle θ, each landing time comes from a quadratic, and the gap between the landing points becomes a single function of θ. The two cases arise because the maximum of that function lies either at an end of the allowed range or at an interior turning point.

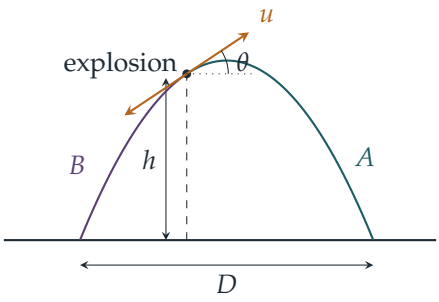
Solution

Setting up

We model the shell and the fragments as particles moving under gravity alone. The shell rises with initial speed v, so it comes to rest at height

$$h = \frac{v^2}{2g}, \qquad \text{that is,} \qquad 2gh = v^2.$$

Just after the explosion, call the fragments A and B. Their velocities are equal and opposite, so both lie in one vertical plane through the explosion point. Take x horizontal in that plane and y vertically upwards, with the explosion point at $(0, h)$. We can label the fragments so that A does not move downwards. Then A has velocity $(u\cos\theta,\ u\sin\theta)$ and B has velocity $(-u\cos\theta,\ -u\sin\theta)$, with $0 \le \theta \le \frac{\pi}{2}$. Write $s = \sin\theta$ and $c = \cos\theta$, so $c \ge 0$.



Landing times

Fragment A is at height $h + ust - \frac{1}{2}gt^2$ at time t after the explosion. It lands when

$$\tfrac{1}{2}gt^2 - us\,t - h = 0.$$

The product of the roots is $-2h/g < 0$, so there is exactly one positive root:

$$t_A = \frac{us + \sqrt{u^2s^2 + 2gh}}{g} = \frac{us + \sqrt{u^2s^2 + v^2}}{g}.$$

For B the vertical velocity is $-us$, so we replace s by $-s$:

$$t_B = \frac{-us + \sqrt{u^2s^2 + v^2}}{g}.$$

This is positive, since $\sqrt{u^2s^2 + v^2} > us$.

The distance between the landing points

Horizontally, A travels $uc\,t_A$ in the positive x-direction and B travels $uc\,t_B$ in the negative direction, so they land on opposite sides of the launch point. The distance between the landing points is therefore

$$D = uc\,(t_A + t_B) = \frac{2uc\sqrt{u^2s^2 + v^2}}{g}.$$

Since $D \geq 0$, maximising D is the same as maximising D^2. Put $x = s^2 \in [0,1]$, so that $c^2 = 1 - x$. Then

$$\frac{g^2D^2}{4u^2} = (1-x)(v^2 + u^2x) = f(x), \qquad \text{say,}$$

and

$$f(x) = v^2 + (u^2 - v^2)x - u^2x^2.$$

This is a quadratic in x with negative leading coefficient, and its vertex is at

$$x^* = \frac{u^2 - v^2}{2u^2}.$$

Ask yourself The vertex of f is a maximum, so why isn't the answer always $f(x^*)$?

Only $x \in [0,1]$ is allowed, because $x = \sin^2\theta$. Note $x^* < \frac{1}{2}$ always, so the vertex never lies beyond the top end. Whether the vertex is allowed depends only on its sign.

Case $u \le v$. Here $x^* \le 0$. On $[0,1]$ we are to the right of the vertex of a downward parabola, so f is decreasing there, and the maximum is at $x = 0$, where $f(0) = v^2$. This corresponds to $\theta = 0$, a horizontal explosion. Hence

$$D_{\max} = \frac{2u}{g}\sqrt{v^2} = \frac{2uv}{g}.$$

Case $v \le u$. Here $0 \le x^* < \frac{1}{2}$, so the vertex is allowed and gives the maximum:

$$\begin{aligned} f(x^*) &= v^2 + \frac{(u^2-v^2)^2}{2u^2} - \frac{(u^2-v^2)^2}{4u^2} = v^2 + \frac{(u^2-v^2)^2}{4u^2} \\ &= \frac{4u^2v^2 + u^4 - 2u^2v^2 + v^4}{4u^2} = \frac{(u^2+v^2)^2}{4u^2}. \end{aligned}$$

Therefore

$$D_{\max} = \frac{2u}{g}\cdot\frac{u^2+v^2}{2u} = \frac{u^2+v^2}{g}.$$

This happens when $\sin^2\theta = \dfrac{u^2-v^2}{2u^2}$, so the fragments fly off at an angle to the horizontal.

When $u = v$ both formulas give $2u^2/g$, so the two answers agree on the boundary between the cases, as they must. This completes the proof.

Checks

- Units: uv/g and $(u^2+v^2)/g$ are both (speed)^2/(acceleration), which is a length.
- Taking $g = 9.8$, $u = 30$ and $v = 10$, the best angle is $\theta \approx 41.8^\circ$, and $D = 1000/9.8 \approx 102.04\,\text{m}$ from both the general formula for D and $(u^2+v^2)/g$. A horizontal explosion gives only $2\cdot 30\cdot 10/9.8 \approx 61.2\,\text{m}$.
- Stepping the two trajectories numerically and scanning θ over a fine grid reproduces both maxima for hundreds of random pairs (u,v).
- With $v = 0$ the shell explodes at ground level, and the answer becomes u^2/g. That is the usual maximum range of a projectile launched at speed u at 45°, as it should be: B goes into the ground at once, and $x^* = \frac{1}{2}$ gives $\theta = 45^\circ$.

Common slips

- Assuming the best explosion is horizontal. That is only true when $u \le v$.

- Using the same time of flight for both fragments. The one thrown upwards stays in the air longer.
- Forgetting the constraint $0 \leq \sin^2\theta \leq 1$ when maximising, and so missing the case split.

Beyond STEP

The tidy answer has a neat explanation. Relative to the centre of mass, which simply falls straight down, the fragments move apart in a straight line at relative speed $2u$. But they hit the ground at different times, so the problem becomes one of optimising under a constraint, with a maximum that switches from the boundary to the interior as a parameter changes. This kind of switch reappears in the Lagrange multiplier and Karush–Kuhn–Tucker conditions of first-year optimisation. For conservation of momentum, centre-of-mass frames and projectile problems taken further, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP II 1994, Question 12

Probability and Statistics • Conditional Probability

STEP II 1994, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s2-q12

The key idea

Once the first throw has fixed a "point" s, every later throw is the same independent experiment with three outcomes: roll s (win), roll 7 (lose), or roll anything else (carry on). So, given the point, the throw on which the game ends is geometric, and the whole question is a matter of splitting into cases by the point and adding up with the law of total probability. The final part is the same calculation with a different conditioning event.

Solution

Let S be the sum of two fair dice. Counting the 36 equally likely outcomes,

$$\mathrm{P}(S = s) = \frac{6 - |s - 7|}{36}, \qquad s = 2, 3, \ldots, 12,$$

so the sums $2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12$ have probabilities $\frac{1}{36} \times (1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1)$.

Winning on the first throw

She wins at once if the first sum is 7 or 11:

$$\mathrm{P}(\text{win on throw 1}) = \frac{6}{36} + \frac{2}{36} = \frac{2}{9}.$$

Winning on the nth throw, given more than one throw

She throws more than once exactly when the first sum is one of the points $4, 5, 6, 8, 9, 10$, which has probability

$$\mathrm{P}(M) = \frac{3 + 4 + 5 + 5 + 4 + 3}{36} = \frac{2}{3},$$

where M is the event "she throws more than once".

Fix a point s and write $p_s = \mathrm{P}(S = s)$. Each later throw is independent of the earlier ones and, on it,

$$\mathrm{P}(\text{sum } s) = p_s, \qquad \mathrm{P}(\text{sum } 7) = \tfrac{1}{6}, \qquad \mathrm{P}(\text{neither}) = r_s = 1 - p_s - \tfrac{1}{6}.$$

(Since $s \neq 7$ these two events are disjoint.) She wins on throw $n \geq 2$ exactly when throws $2, \ldots, n-1$ are all "neither" and throw n shows s. So, by independence,

$$\mathrm{P}(\text{win on throw } n \mid \text{point } s) = r_s^{n-2}\, p_s.$$

Points come in pairs with equal probability (4 and 10, 5 and 9, 6 and 8). Write $p_s = k/36$ with $k = 3, 4, 5$; then $r_s = (30-k)/36$, which is $\frac{3}{4}$, $\frac{13}{18}$, $\frac{25}{36}$ respectively. By the law of total probability, for $n \geq 2$,

$$\mathrm{P}(\text{win on throw } n) = \sum_s p_s \cdot r_s^{n-2} p_s = 2\sum_{k=3}^{5} \Big(\frac{k}{36}\Big)^2 \Big(\frac{30-k}{36}\Big)^{n-2}.$$

Winning on throw $n \geq 2$ implies M, so dividing by $\mathrm{P}(M) = \frac{2}{3}$ multiplies each coefficient by $\frac{3}{2}$:

$$2 \cdot \frac{k^2}{1296} \cdot \frac{3}{2} = \frac{k^2}{432} \quad \text{gives} \quad \frac{9}{432} = \frac{1}{48}, \quad \frac{16}{432} = \frac{1}{27}, \quad \frac{25}{432}.$$

Hence

$$\mathrm{P}(\text{win on throw } n \mid M) = \frac{1}{48}\Big(\frac{3}{4}\Big)^{n-2} + \frac{1}{27}\Big(\frac{13}{18}\Big)^{n-2} + \frac{25}{432}\Big(\frac{25}{36}\Big)^{n-2},$$

as required.

Ask yourself Why is it wrong to work with a single "average" point and one geometric distribution?

Because the continuation probability r_s depends on the point. The mixture of three geometric distributions is not itself geometric, which is why the answer has three separate powers rather than one.

Given more than m throws

Let M_m be the event "she throws more than m times", with $m > 1$. This happens exactly when the first throw gives a point s and throws $2, \ldots, m$ are all "neither", so

$$\mathrm{P}(M_m) = \sum_s p_s\, r_s^{m-1} = \frac{2}{36}\sum_{k=3}^{5} k\Big(\frac{30-k}{36}\Big)^{m-1}.$$

If $n \leq m$, winning on throw n is incompatible with M_m, so the conditional probability is 0. If $n > m$, winning on throw n implies M_m, so

$$\mathrm{P}(\text{win on throw } n \mid M_m) = \frac{\mathrm{P}(\text{win on throw } n)}{\mathrm{P}(M_m)} = \frac{9\big(\frac{3}{4}\big)^{n-2} + 16\big(\frac{13}{18}\big)^{n-2} + 25\big(\frac{25}{36}\big)^{n-2}}{36\Big[3\big(\frac{3}{4}\big)^{m-1} + 4\big(\frac{13}{18}\big)^{m-1} + 5\big(\frac{25}{36}\big)^{m-1}\Big]},$$

where the factors $\frac{2}{1296}$ and $\frac{2}{36}$ have been cancelled. Equivalently, multiplying top and bottom by $\frac{1}{432}$,

$$\frac{\frac{1}{48}\left(\frac{3}{4}\right)^{n-2}+\frac{1}{27}\left(\frac{13}{18}\right)^{n-2}+\frac{25}{432}\left(\frac{25}{36}\right)^{n-2}}{\frac{1}{4}\left(\frac{3}{4}\right)^{m-1}+\frac{1}{3}\left(\frac{13}{18}\right)^{m-1}+\frac{5}{12}\left(\frac{25}{36}\right)^{m-1}}, \qquad n > m.$$

Note that the denominator is $\mathrm{P}(M_m \mid M)$, as it should be: the numerator is the earlier answer, conditioned on M, and $M_m \subseteq M$.

Checks

- With $m = 1$ the denominator becomes $\frac{1}{4}+\frac{1}{3}+\frac{5}{12} = 1$, recovering the previous part.
- For $n = 2$ the conditional formula gives $\frac{1}{48}+\frac{1}{27}+\frac{25}{432} = \frac{50}{432} = \frac{25}{216} \approx 0.1157$; directly, $\frac{3}{2}\cdot\frac{2(9+16+25)}{1296}$ gives the same.
- For $m = 2$, $n = 3$ the answer is $\frac{107}{930} \approx 0.1151$. We checked all the formulae against exact step-by-step propagation of the game with fractions, and a simulation of 4×10^5 games agreed to within sampling error.

Common slips

- Forgetting to divide by $\mathrm{P}(M) = \frac{2}{3}$, which gives coefficients $\frac{1}{72}$, $\frac{2}{81}$, $\frac{25}{648}$.
- Using $\mathrm{P}(M_m) = \left(\text{something}\right)^{m-1}$ with a single ratio: the conditioning event is a mixture over points too.
- Counting the "neither" throws as $n-1$ rather than $n-2$: throw 1 sets the point and throw n is the win.

Beyond STEP

The game is a small Markov chain with absorbing states "win" and "lose", and the time to absorption here is a mixture of geometric distributions. Conditioning on survival past m throws shifts the mixture weights towards the points that last longest (here 4 and 10, the least likely to be repeated), a simple example of how a mixture of memoryless distributions is not itself memoryless. Grimmett and Welsh, *Probability: An Introduction*, covers conditional probability and Markov chains at this level.

WORKED SOLUTION

STEP II 1994, Question 13

Probability and Statistics • Coupon Collector Problem

STEP II 1994, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s2-q13

This question has a full guided treatment in *Beyond STEP*, Section 9.11, *Collecting the Set*: https://beyondstep.keithmcnulty.org/ch09/coupon-collector.html

The key idea

This is the classic "coupon collector" problem, and despite the database tag it is pure probability. Split the long wait for a full set into n shorter waits: the wait for the first new card, then the second, and so on. Each short wait is geometric, so its mean is easy, and linearity of expectation adds them up. The total is n times a harmonic sum, and the integral inequality is there to tell us how big that sum is.

Solution

Waiting for a new card

We model each packet as containing one of the n cards, each equally likely with probability $\frac{1}{n}$, independently of every other packet.

Suppose I already hold r different cards, where $0 \leq r \leq n-1$. A packet gives a new card with probability

$$p = \frac{n-r}{n} > 0,$$

and an old card with probability $q = 1 - p = \frac{r}{n}$. Let X be the number of packets I open up to and including the first new card. Because the packets are independent, $X = k$ means $k-1$ old cards followed by a new one, so

$$\mathrm{P}(X = k) = q^{k-1}p, \qquad k = 1, 2, 3, \ldots$$

Hence

$$\mathrm{E}(X) = \sum_{k=1}^{\infty} kpq^{k-1} = p\,(1-q)^{-2} = \frac{p}{p^2} = \frac{1}{p},$$

using $\sum_{k\geq 1} kq^{k-1} = (1-q)^{-2}$ for $|q| < 1$ (differentiate the geometric series, or expand $(1-q)^{-2}$ binomially). Here $0 \leq q < 1$ because $r \leq n-1$. So

$$\mathrm{E}(X) = \frac{n}{n-r},$$

as required.

(A slicker route: the first packet either gives a new card, and we stop, or with probability q it puts us back where we started, so $\mathrm{E}(X) = 1 + q\,\mathrm{E}(X)$, giving $\mathrm{E}(X) = 1/(1-q) = 1/p$. This assumes $\mathrm{E}(X)$ is finite, which the series above confirms.)

The integral inequality

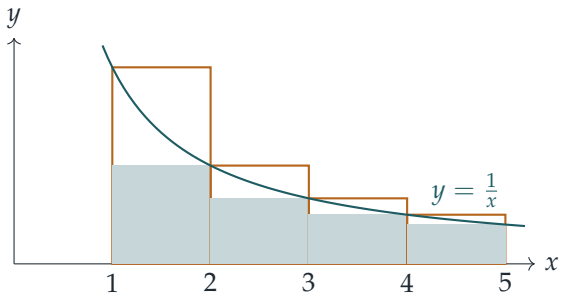
Let $r \geq 1$. For $r \leq x \leq r+1$ we have

$$\frac{1}{r+1} \leq \frac{1}{x} \leq \frac{1}{r},$$

because $1/x$ is decreasing for $x > 0$. Integrating over an interval of length 1 keeps these inequalities, so

$$\frac{1}{r+1} \leq \int_r^{r+1} \frac{1}{x}\,\mathrm{d}x \leq \frac{1}{r}.$$

In the picture, the area under $y = 1/x$ between $x = r$ and $x = r+1$ lies between the short rectangle of height $\frac{1}{r+1}$ and the tall rectangle of height $\frac{1}{r}$.



(The vertical scale is stretched for clarity.) The shaded rectangles sit under the curve; the outlined ones sit above it.

Now add the inequality for $r = 1, 2, \ldots, n-1$, where $n \geq 2$. The integrals join up into one:

$$\sum_{r=1}^{n-1} \frac{1}{r+1} \leq \int_1^n \frac{1}{x}\,\mathrm{d}x \leq \sum_{r=1}^{n-1} \frac{1}{r}.$$

The middle term is $\ln n - \ln 1 = \ln n$, and in the left-hand sum we relabel $r+1$ as r, which runs from 2 to n. So

$$\sum_{r=2}^{n} \frac{1}{r} \leq \ln n \leq \sum_{r=1}^{n-1} \frac{1}{r},$$

as required.

Collecting the whole set

Let X_r be the number of packets opened while I hold exactly r different cards, ending with the packet that gives card number $r+1$. The total number of packets needed for the full set is

$$T = X_0 + X_1 + \cdots + X_{n-1}.$$

Ask yourself The X_r depend on what happened earlier. Can we still just add their expectations?

Yes. Linearity of expectation, $\mathrm{E}(A+B) = \mathrm{E}(A) + \mathrm{E}(B)$, needs no independence at all. (In fact, given the history so far, X_r is geometric with $p = \frac{n-r}{n}$ regardless, so the first part applies to each X_r.) Therefore

$$\mathrm{E}(T) = \sum_{r=0}^{n-1} \frac{n}{n-r} = n\left(\frac{1}{n} + \frac{1}{n-1} + \cdots + \frac{1}{1}\right) = n\sum_{k=1}^{n}\frac{1}{k},$$

writing $k = n - r$.

Write $H_n = \sum_{k=1}^{n} \frac{1}{k}$. The two halves of the inequality give

$$H_n = 1 + \sum_{k=2}^{n}\frac{1}{k} \le 1 + \ln n, \qquad H_n = \sum_{k=1}^{n-1}\frac{1}{k} + \frac{1}{n} \ge \ln n + \frac{1}{n}.$$

So

$$n\ln n + 1 \le \mathrm{E}(T) \le n\ln n + n.$$

The gap between the bounds is less than n, which is tiny next to $n\ln n$ when n is large (their ratio $1/\ln n \to 0$). So for large n I should expect to eat roughly

$$n\ln n \text{ packets.}$$

For example, 100 mathematicians would mean about 460 packets of Cruncho.

Checks

- $r = 0$: the first packet is always new, and $n/(n-0) = 1$. $r = n-1$: one specific card is missing, which turns up with probability $\frac{1}{n}$, so we expect n packets, matching $n/1$.
- $n = 2$: $\mathrm{E}(T) = 2(1 + \frac{1}{2}) = 3$, and the bounds give $2\ln 2 + 1 \approx 2.39$ and $2\ln 2 + 2 \approx 3.39$.
- A simulation of 10^5 complete collections with $n = 20$ gave an average of 71.9 packets, against $20H_{20} \approx 71.95$; the bounds are 60.9 and 79.9.

Common slips

- Summing the integral inequality from $r = 0$: the bound $\frac{1}{r}$ fails there, and $\int_0^1 \frac{1}{x}\,\mathrm{d}x$ diverges. The sum must start at $r = 1$.

- Thinking the X_r must be independent before adding expectations. Linearity needs no independence.
- Answering $\ln n$ or n: the expected number of packets grows like $n \ln n$, faster than n.

Beyond STEP

The difference $H_n - \ln n$ decreases to a limit, the Euler–Mascheroni constant $\gamma \approx 0.5772$, so a sharper answer is $\mathrm{E}(T) \approx n(\ln n + \gamma)$; for $n = 100$ that gives about 519 packets rather than 460. Comparing sums with integrals like this is the integral test for series, met early in a first analysis course. The coupon collector problem itself reappears in the analysis of algorithms and in the study of how many random shuffles mix a deck. Grimmett and Welsh, *Probability: An Introduction*, is a good place to see geometric waiting times and linearity of expectation used together.

WORKED SOLUTION

STEP II 1994, Question 14

Probability and Statistics • Geometrical Probability

STEP II 1994, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s2-q14

The key idea

"Uniform over the disc" means probability is proportional to area. So the chance of landing within distance r of the centre is the area of the smaller disc divided by the area of the board, and that gives the distribution function of R straight away. Part (ii) then reduces to a single probability p per throw, a geometric-style "at least one success" calculation, and the minimisation of a function of an integer, which we do by looking at consecutive differences rather than by calculus.

Solution

Part (i)

Model each dart as a point uniformly distributed over a disc of radius a, so the probability that it lands in any region of the board is that region's area divided by πa^2.

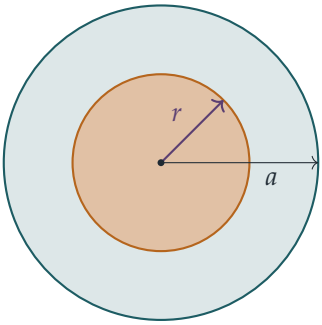
For $0 \leq r \leq a$, the event $R \leq r$ means the dart lands in the concentric disc of radius r, so

$$\mathrm{P}(R \leq r) = \frac{\pi r^2}{\pi a^2} = \frac{r^2}{a^2}.$$

Differentiating, R has probability density function

$$f(r) = \frac{2r}{a^2} \qquad (0 \leq r \leq a),$$

and $f(r) = 0$ otherwise.



Note that R is *not* uniform on $[0, a]$: there is more area (a longer thin ring) at larger radii, which is why the density grows like r.

Now

$$\mathrm{E}(R) = \int_0^a r \cdot \frac{2r}{a^2}\,\mathrm{d}r = \frac{2}{a^2} \cdot \frac{a^3}{3} = \frac{2a}{3},$$
$$\mathrm{E}(R^2) = \int_0^a r^2 \cdot \frac{2r}{a^2}\,\mathrm{d}r = \frac{2}{a^2} \cdot \frac{a^4}{4} = \frac{a^2}{2},$$

so

$$\mathrm{Var}(R) = \mathrm{E}(R^2) - \big(\mathrm{E}(R)\big)^2 = \frac{a^2}{2} - \frac{4a^2}{9} = \frac{a^2}{18},$$

as required.

Part (ii)

(We read "with $a/\sqrt{10}$" as "within $a/\sqrt{10}$".) By part (i), the probability that a single throw lands within $a/\sqrt{10}$ of the centre is

$$p = \frac{(a/\sqrt{10})^2}{a^2} = \frac{1}{10}.$$

We assume the throws are independent (each is a fresh, uniformly random point on the board). Then the probability that none of the m throws is within $a/\sqrt{10}$ is $(0.9)^m$, so

$$\mathrm{P}(\text{he wins back £12}) = 1 - (0.9)^m.$$

His loss is the $£(12 + m)$ he pays minus anything he wins back, so his expected loss, in pounds, is

$$L(m) = 12 + m - 12\big(1 - (0.9)^m\big) = m + 12(0.9)^m.$$

Each extra throw costs £1 for certain but makes the £12 more likely; we need to find where these balance.

> **Ask yourself** How do we minimise a function defined only on whole numbers?

Compare neighbouring values. For $m \geq 1$,

$$L(m+1) - L(m) = 1 + 12(0.9)^m(0.9 - 1) = 1 - 1.2\,(0.9)^m.$$

Since $(0.9)^m$ decreases as m increases, this difference is increasing in m, so L decreases while the difference is negative and increases once it is positive. The difference is negative exactly when $(0.9)^m > \frac{1}{1.2} = \frac{5}{6}$. Now

$$(0.9)^1 = 0.9 > \tfrac{5}{6}, \qquad (0.9)^2 = 0.81 < \tfrac{5}{6},$$

so $L(2) - L(1) < 0$, while $L(m+1) - L(m) > 0$ for every $m \geq 2$. Hence

$$L(1) > L(2) < L(3) < L(4) < \cdots,$$

and indeed $L(1) = 11.80$, $L(2) = 11.72$, $L(3) = 11.748$. (He is determined to play, so $m = 0$ is not allowed; in any case $L(0) = 12$ is larger.)

He should choose $m = 2$ throws, with an expected loss of £11.72. Since the minimum is strict, $m = 2$ is the only such value.

Checks

- A simulation of 4×10^5 uniform points in the unit disc gave mean 0.667, mean square 0.500 and variance 0.0556 $\approx \frac{1}{18}$, matching $\frac{2}{3}$, $\frac{1}{2}$ and $\frac{1}{18}$; about 10% of the points fell within $1/\sqrt{10}$ of the centre.
- Exactly, $L(1) = \frac{59}{5}$, $L(2) = \frac{293}{25}$, $L(3) = \frac{2937}{250}$, and a search over $1 \leq m \leq 99$ confirms $m = 2$ is the minimum.
- Sense check: the variance $a^2/18$ is far less than that of a uniform variable on $[0, a]$, namely $a^2/12$, because R is concentrated towards the rim.

Common slips

- Taking R to be uniform on $[0, a]$, which gives variance $a^2/12$.
- Forgetting the fixed £12 is only part of the cost: the loss is $12 + m$ minus the expected winnings, not just m.
- Setting the derivative of $m + 12(0.9)^m$ to zero gives $m \approx 2.3$, which is not an answer; you must compare the integer values on either side.

Beyond STEP

Finding the distribution of R from a uniform point in the plane is the simplest example of transforming a joint density into polar coordinates, where the Jacobian factor r in $\mathrm{d}A = r\,\mathrm{d}r\,\mathrm{d}\theta$ is exactly what makes the density of R proportional to r. Change of variables for joint densities, and optimisation over a discrete choice, are standard first-year probability topics. Grimmett and Welsh, *Probability: An Introduction*, covers both.

WORKED SOLUTION

STEP III 1994, Question 1

Pure • Reduction Formulae

STEP III 1994, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s3-q1

The key idea

For the first integral, write $\operatorname{sech} t$ as $\dfrac{\cosh t}{\cosh^2 t} = \dfrac{\cosh t}{1+\sinh^2 t}$: the numerator is the derivative of $\sinh t$, so this is an arctan in disguise. For the reduction formula, the natural move is to split off a factor $\operatorname{sech}^2 t$, which is the derivative of $\tanh t$, and integrate by parts. The identity $\tanh^2 t = 1 - \operatorname{sech}^2 t$ then brings I_n back on the right-hand side, and we collect it.

Solution

The integral of $\operatorname{sech} t$

Multiplying top and bottom by $\cosh t$ and using $\cosh^2 t = 1 + \sinh^2 t$,

$$\int_0^x \operatorname{sech} t \, \mathrm{d}t = \int_0^x \frac{\cosh t}{1+\sinh^2 t} \, \mathrm{d}t.$$

With $s = \sinh t$, $\mathrm{d}s = \cosh t \, \mathrm{d}t$, and the limits become 0 and $\sinh x$, so

$$\int_0^x \operatorname{sech} t \, \mathrm{d}t = \int_0^{\sinh x} \frac{\mathrm{d}s}{1+s^2} = \arctan(\sinh x).$$

An equally good route is $\operatorname{sech} t = \dfrac{2\mathrm{e}^t}{\mathrm{e}^{2t}+1}$ with $u = \mathrm{e}^t$, which gives the answer in the form $2\arctan(\mathrm{e}^x) - \frac{\pi}{2}$. The two agree: both vanish at $x = 0$ and have derivative $\operatorname{sech} x$. (This function of x is called the *Gudermannian*.)

The reduction formula

Take $n \geq 2$ and write $\operatorname{sech}^n t = \operatorname{sech}^{n-2} t \cdot \operatorname{sech}^2 t$. Integrate by parts with

$$u = \operatorname{sech}^{n-2} t, \qquad \frac{\mathrm{d}v}{\mathrm{d}t} = \operatorname{sech}^2 t,$$

so that $v = \tanh t$ and, by the chain rule (using $\frac{\mathrm{d}}{\mathrm{d}t}\operatorname{sech} t = -\operatorname{sech} t \tanh t$),

$$\frac{\mathrm{d}u}{\mathrm{d}t} = (n-2)\operatorname{sech}^{n-3} t \cdot (-\operatorname{sech} t \tanh t) = -(n-2)\operatorname{sech}^{n-2} t \tanh t.$$

Then

$$\begin{aligned} I_n &= \left[\operatorname{sech}^{n-2} t \tanh t\right]_0^x + (n-2)\int_0^x \operatorname{sech}^{n-2} t \tanh^2 t \,\mathrm{d}t \\ &= \operatorname{sech}^{n-2} x \tanh x + (n-2)\int_0^x \operatorname{sech}^{n-2} t \left(1 - \operatorname{sech}^2 t\right) \mathrm{d}t \\ &= \operatorname{sech}^{n-2} x \tanh x + (n-2) I_{n-2} - (n-2) I_n, \end{aligned}$$

where the boundary term at 0 vanishes because $\tanh 0 = 0$.

Ask yourself We seem to have gone in a circle: I_n has reappeared on the right. Is that a problem?

Not at all; it is exactly what we want. The I_n on the right has coefficient $-(n-2)$, not $+1$, so moving it to the left gives $(n-1)I_n$ and we can divide by $n-1 \neq 0$. The reduction formula is

$$\boxed{(n-1) I_n = \operatorname{sech}^{n-2} x \tanh x + (n-2)\, I_{n-2} \qquad (n \geq 2).}$$

Finding I_5

The odd chain starts from $I_1 = \arctan(\sinh x)$. Putting $n = 3$:

$$2I_3 = \operatorname{sech} x \tanh x + I_1, \qquad \text{so} \qquad I_3 = \tfrac{1}{2}\operatorname{sech} x \tanh x + \tfrac{1}{2}\arctan(\sinh x).$$

Putting $n = 5$:

$$4I_5 = \operatorname{sech}^3 x \tanh x + 3I_3,$$

so

$$I_5 = \tfrac{1}{4}\operatorname{sech}^3 x \tanh x + \tfrac{3}{8}\operatorname{sech} x \tanh x + \tfrac{3}{8}\arctan(\sinh x).$$

Finding I_6

The even chain starts from $I_0 = \int_0^x 1 \,\mathrm{d}t = x$, or more directly from $I_2 = \int_0^x \operatorname{sech}^2 t \,\mathrm{d}t = \tanh x$ (the formula with $n = 2$ confirms this: $I_2 = \tanh x + 0$). Putting $n = 4$:

$$3I_4 = \operatorname{sech}^2 x \tanh x + 2I_2, \qquad \text{so} \qquad I_4 = \tfrac{1}{3}\operatorname{sech}^2 x \tanh x + \tfrac{2}{3}\tanh x.$$

Putting $n = 6$:

$$5I_6 = \operatorname{sech}^4 x \tanh x + 4I_4,$$

so

$$I_6 = \tfrac{1}{5}\operatorname{sech}^4 x \tanh x + \tfrac{4}{15}\operatorname{sech}^2 x \tanh x + \tfrac{8}{15}\tanh x.$$

The "otherwise" for even powers

Even powers have a neater direct route. With $u = \tanh t$, $\mathrm{d}u = \operatorname{sech}^2 t \,\mathrm{d}t$ and $\operatorname{sech}^2 t = 1 - u^2$, so

$$I_6 = \int_0^x (\operatorname{sech}^2 t)^2 \operatorname{sech}^2 t \,\mathrm{d}t = \int_0^{\tanh x} (1-u^2)^2 \,\mathrm{d}u = \tanh x - \tfrac{2}{3}\tanh^3 x + \tfrac{1}{5}\tanh^5 x.$$

Replacing $\operatorname{sech}^2 x$ by $1 - \tanh^2 x$ in our earlier answer and expanding gives exactly this, so the two forms agree.

Checks

- As $x \to \infty$, $\tanh x \to 1$, $\operatorname{sech} x \to 0$ and $\arctan(\sinh x) \to \frac{\pi}{2}$, so $I_5 \to \frac{3\pi}{16} \approx 0.58905$ and $I_6 \to \frac{8}{15}$. Simpson's rule on $[0, 30]$ gives 0.589049 and 0.533333.
- At $x = 1$, Simpson's rule gives $I_5 \approx 0.561566$ and $I_6 \approx 0.518343$, matching the formulae; we also checked the reduction formula for $2 \le n \le 8$ at 200 random values of x.
- Both I_5 and I_6 are odd functions of x, as they must be, since sech is even.

Common slips

- Getting the sign of $\frac{\mathrm{d}}{\mathrm{d}t}\operatorname{sech} t$ wrong: it is $-\operatorname{sech} t \tanh t$, and this minus sign is what makes the boundary term positive.
- Dividing by n instead of $n - 1$ after collecting the two copies of I_n.
- Leaving the answer as an indefinite integral: with limits 0 and x there is no arbitrary constant, and every term should vanish at $x = 0$.

Beyond STEP

Reduction formulae like this one let you evaluate a whole family of integrals by recursion, and the same idea underlies Wallis's formula for $\int_0^{\pi/2} \sin^n t \,\mathrm{d}t$, which leads to a famous infinite product for π. The limits $\int_0^\infty \operatorname{sech}^n t \,\mathrm{d}t$ found here are values of the Beta function, a central object in first-year analysis and probability. Siklos, *Advanced Problems in Mathematics*, has more reduction-formula practice, and Abbott, *Understanding Analysis*, makes the improper integrals rigorous.

WORKED SOLUTION

STEP III 1994, Question 2

Pure • Palindromic Quartic Equations

STEP III 1994, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s3-q2

The key idea

The quartic in (i) is *palindromic*: its coefficients read the same forwards and backwards. Dividing by x^2 pairs x^2 with x^{-2} and x with x^{-1}, and both pairs can be written in terms of $y = x + x^{-1}$, so the quartic collapses to a quadratic in y. In (ii) there is no hint, so we look for a twisted version of the same symmetry: the coefficients 1, 1, -10, -4, 16 pair up once we allow a scaling factor, which suggests the substitution $y = x - 4/x$.

Solution

Part (i)

The constant term is $1 \neq 0$, so $x = 0$ is not a root and we may divide by x^2:

$$x^2 + 10x + 26 + \frac{10}{x} + \frac{1}{x^2} = 0, \qquad \text{that is} \qquad \left(x^2 + \frac{1}{x^2}\right) + 10\left(x + \frac{1}{x}\right) + 26 = 0.$$

With $y = x + x^{-1}$ we have $y^2 = x^2 + 2 + x^{-2}$, so $x^2 + x^{-2} = y^2 - 2$. The equation becomes

$$y^2 - 2 + 10y + 26 = 0 \iff y^2 + 10y + 24 = 0 \iff (y+4)(y+6) = 0,$$

so $y = -4$ or $y = -6$. Now we return to x, multiplying $x + x^{-1} = y$ through by x (again fine, as $x \neq 0$):

$$x + \frac{1}{x} = -4 \iff x^2 + 4x + 1 = 0 \iff x = -2 \pm \sqrt{3},$$
$$x + \frac{1}{x} = -6 \iff x^2 + 6x + 1 = 0 \iff x = -3 \pm 2\sqrt{2}.$$

These are four distinct numbers, and a quartic has at most four roots, so the solutions are

$$x = -2 \pm \sqrt{3}, \qquad x = -3 \pm 2\sqrt{2}.$$

Notice that the roots come in reciprocal pairs: $(-2+\sqrt{3})(-2-\sqrt{3}) = 1$ and $(-3+2\sqrt{2})(-3-2\sqrt{2}) = 1$. That is exactly what a palindromic polynomial must do, since replacing x by $1/x$ and multiplying by x^4 gives back the same equation.

Part (ii)

Ask yourself The coefficients 1, 1, -10, -4, 16 are not symmetric. Is there a substitution like the one in (i) that still works?

Compare the outer pairs of coefficients. The constant term is $16 = 4^2$ times the leading coefficient, and the coefficient of x is -4 times the coefficient of x^3. So the terms pair up as

$$x^4 + 16 = x^2\left(x^2 + \frac{16}{x^2}\right), \qquad x^3 - 4x = x^2\left(x - \frac{4}{x}\right).$$

This is the palindromic pattern with $1/x$ replaced by $-4/x$. Since $x = 0$ is not a root (the constant term is 16), dividing by x^2 gives

$$\left(x^2 + \frac{16}{x^2}\right) + \left(x - \frac{4}{x}\right) - 10 = 0.$$

Let $y = x - 4/x$. Then $y^2 = x^2 - 8 + 16/x^2$, so $x^2 + 16/x^2 = y^2 + 8$, and

$$y^2 + 8 + y - 10 = 0 \iff y^2 + y - 2 = 0 \iff (y-1)(y+2) = 0,$$

giving $y = 1$ or $y = -2$. Multiplying $x - 4/x = y$ by x:

$$x - \frac{4}{x} = 1 \iff x^2 - x - 4 = 0 \iff x = \frac{1 \pm \sqrt{17}}{2},$$
$$x - \frac{4}{x} = -2 \iff x^2 + 2x - 4 = 0 \iff x = -1 \pm \sqrt{5}.$$

Again these are four distinct roots, so they are all of them:

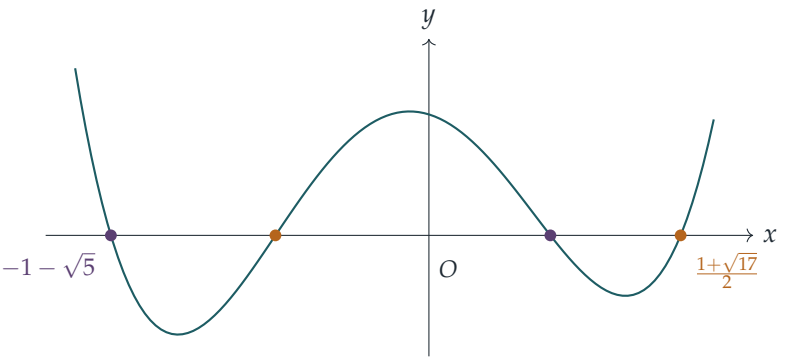
$$x = \frac{1 \pm \sqrt{17}}{2}, \qquad x = -1 \pm \sqrt{5}.$$

Here the roots pair up with product -4 rather than 1: the symmetry is now $x \mapsto -4/x$.

As a check, the working amounts to the factorisation

$$x^4 + x^3 - 10x^2 - 4x + 16 = (x^2 - x - 4)(x^2 + 2x - 4),$$

which you can confirm by expanding: the x^3 terms give $2 - 1 = 1$, the x^2 terms give $-4 - 2 - 4 = -10$, and the x terms give $4 - 8 = -4$.



The graph of the quartic in (ii) crosses the axis four times: the amber roots come from $x^2 - x - 4 = 0$ and the plum roots from $x^2 + 2x - 4 = 0$.

Checks

- Products of roots: in (i) all four roots multiply to $1 \cdot 1 = 1$, the constant term; in (ii) they multiply to $(-4)(-4) = 16$, again the constant term.
- Sums of roots: in (i) $(-4) + (-6) = -10$, minus the x^3 coefficient; in (ii) $1 + (-2) = -1$, likewise.
- Numerically, $x = -2 + \sqrt{3} \approx -0.268$ and $x = -3 - 2\sqrt{2} \approx -5.828$ both make the quartic in (i) vanish to within 10^{-9}, and $x = \frac{1}{2}(1 + \sqrt{17}) \approx 2.562$ and $x = -1 - \sqrt{5} \approx -3.236$ do the same in (ii).

Common slips

- Forgetting the cross term: $x^2 + x^{-2} = y^2 - 2$, not y^2; and in (ii) $x^2 + 16x^{-2} = y^2 + 8$, with a plus sign because the cross term of $(x - 4/x)^2$ is -8.
- Trying $y = x + 4/x$ in (ii). That gives $x^2 + 16/x^2 = y^2 - 8$ but cannot produce $x^3 - 4x$, whose signs demand $x - 4/x$.
- Stopping at the values of y. The question asks for x, and each value of y gives two values of x.

Beyond STEP

The trick works because the polynomials are invariant under a substitution: $x \mapsto 1/x$ in (i) and $x \mapsto -4/x$ in (ii). The quantity y is unchanged by that substitution, and any polynomial expression that is unchanged can be written in terms of y. This is a first glimpse of Galois theory, where the symmetries among the roots of a polynomial decide how its equation can be solved. Allenby's *Rings, Fields and Groups* is a good place to see these ideas developed.

WORKED SOLUTION

STEP III 1994, Question 3

Pure • Spheres, Planes and Linking

STEP III 1994, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s3-q3

The key idea

Each circle lies in its own plane, and the two planes meet in the line L. So the only places where C_1 can pass through the plane of C_2 are its two points on L, and the same is true the other way round. Whether the circles are linked therefore comes down to a one-dimensional question: how the four points on L are arranged along the line.

Solution

Planes and spheres

Let a sphere have centre Q and radius r, and let d be the distance from Q to a plane Π. Let F be the foot of the perpendicular from Q to Π. For a point X of Π, Pythagoras gives $QX^2 = d^2 + FX^2$, so X is on the sphere exactly when $FX^2 = r^2 - d^2$. Hence:

- if $d > r$, the plane misses the sphere;
- if $d = r$, the plane touches the sphere at the single point F (it is a tangent plane);
- if $d < r$, the intersection is a circle in Π with centre F and radius $\sqrt{r^2 - d^2}$. When $d = 0$ this is a great circle, of radius r.

The data

Completing the square, S_2 is

$$x^2 + (y-3)^2 + (z-2)^2 = 3,$$

with centre $(0,3,2)$ and radius $\sqrt{3}$. The sphere S_1 has centre O and radius $\sqrt{7}$.

The distance from O to P_1 is $\frac{|-1|}{\sqrt{10}} = \frac{1}{\sqrt{10}} < \sqrt{7}$, so C_1 is a genuine circle, of radius $\sqrt{7 - \frac{1}{10}}$. The distance from $(0,3,2)$ to P_2 is $\frac{|0-3+1|}{\sqrt{2}} = \sqrt{2} < \sqrt{3}$, so C_2 is a circle of radius $\sqrt{3-2} = 1$. Its centre is the foot of the perpendicular, $(0,3,2) + (1,-1,0) = (1,2,2)$; check that $1 - 2 + 1 = 0$.

The line L

Subtracting the plane equations gives $2x - 2 = 0$, so $x = 1$, and then $y = x + 1 = 2$, while z is free. So

$$L = \{(1, 2, t) : t \in \mathbb{R}\},$$

the vertical line through $(1, 2, 0)$.

On S_1: $1 + 4 + t^2 = 7$, so $t = \pm\sqrt{2}$. Thus L meets S_1 at

$$(1, 2, \sqrt{2}) \quad \text{and} \quad (1, 2, -\sqrt{2}).$$

On S_2: $1 + 1 + (t - 2)^2 = 3$, so $t - 2 = \pm 1$. Thus L meets S_2 at

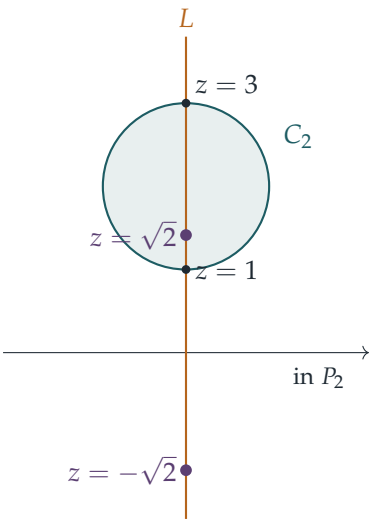
$$(1, 2, 1) \quad \text{and} \quad (1, 2, 3).$$

Since L lies in P_1, the points of L on S_1 are exactly the points of L on C_1; similarly the points of L on S_2 are the points of L on C_2. In particular $(1, 2, 1)$ and $(1, 2, 3)$ are the ends of a diameter of C_2, consistent with its centre $(1, 2, 2)$ and radius 1.

Are the circles linked?

First, the circles do not meet. Any common point lies in $P_1 \cap P_2 = L$, and the points of C_1 on L ($z = \pm\sqrt{2}$) differ from those of C_2 on L ($z = 1, 3$).

Let D_2 be the flat disc in P_2 bounded by C_2. It meets L in the diameter from $(1, 2, 1)$ to $(1, 2, 3)$, that is, in the points $(1, 2, t)$ with $1 \le t \le 3$.



The picture is the plane P_2, with C_2 and its disc shaded. The purple points are where C_1 passes through P_2.

Ask yourself Why does C_1 really pass *through* P_2 at these two points, rather than just touching it?

The line L divides the plane P_1 into two half-planes, and these lie on opposite sides of P_2, because P_1 crosses P_2 along L. The circle C_1 meets L in two distinct points, so L cuts C_1 into two arcs, one in each half-plane. Hence one arc of C_1 lies strictly on one side of P_2 and the other arc strictly on the other side, and C_1 crosses P_2 at $(1, 2, \sqrt{2})$ and at $(1, 2, -\sqrt{2})$, and nowhere else.

Now $1 < \sqrt{2} < 3$, so the crossing at $(1, 2, \sqrt{2})$ is inside the disc D_2. But $-\sqrt{2} < 1$, so the crossing at $(1, 2, -\sqrt{2})$ is outside it. Going once round C_1, we pass through the disc spanned by C_2 exactly once, and come back across P_2 outside the disc. In other words C_1 threads through C_2 once, like two links of a chain.

Along L the four points come in the order

$$-\sqrt{2} \;<\; 1 \;<\; \sqrt{2} \;<\; 3,$$

so the pairs from C_1 and from C_2 *interleave*. That is exactly what makes C_1 cross D_2 once. (If neither pair separated the other, C_1 would cross D_2 either twice or not at all, and the circles would not be linked.) Hence **the circles C_1 and C_2 are linked**.

One cannot pull them apart: any continuous movement taking them to two far-apart circles would have to change the number of times C_1 crosses the disc of C_2 from odd to even. The crossings come and go in pairs unless one circle passes through the other, which a separation without cutting does not allow.

Checks

Substitute each point: for example $(1, 2, 1)$ gives $3 - 2 - 1 = 0$, $1 - 2 + 1 = 0$ and $1 + 4 + 1 - 12 - 4 + 10 = 0$. Numerically, we parametrised both circles and evaluated Gauss's linking integral $\frac{1}{4\pi} \oint\oint \frac{(\mathbf{r}_1 - \mathbf{r}_2)\cdot(\mathrm{d}\mathbf{r}_1 \times \mathrm{d}\mathbf{r}_2)}{|\mathbf{r}_1 - \mathbf{r}_2|^3}$, getting 1.002 (an unlinked pair gives 0). We also found that the closest distance between the circles is about 0.09, so they come close but do not touch.

Common slips

- Checking only whether the circles intersect. They don't, but that alone says nothing about linking.
- Forgetting that the crossings of C_1 with P_2 must lie on L, and looking for them elsewhere.
- Getting the centre of S_2 wrong: the $-6y$ and $-4z$ terms give centre $(0, 3, 2)$, not $(0, -3, -2)$.

Beyond STEP

Whether two disjoint closed curves are linked is measured by an integer invariant, the *linking number*. It can be computed by counting signed crossings of one curve through a surface spanning the other, which is what we did, or by Gauss's double integral in the Checks. Linking numbers are a first step into knot theory and algebraic topology. Colin Adams's *The Knot Book* is a very readable introduction.

WORKED SOLUTION

STEP III 1994, Question 4

Pure • Singular Solutions of ODEs

STEP III 1994, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s3-q4

This question has a full guided treatment in *Beyond STEP*, Section 8.1, *When Solutions Aren't Unique*: https://beyondstep.keithmcnulty.org/ch08/non-uniqueness.html

The key idea

Take square roots: the equation says $\mathrm{d}y/\mathrm{d}x = \pm 2\sqrt{y}$, which separates at once. Better still, it is worth noticing that every parabola $y = (x-c)^2$ satisfies the equation, so the solutions are simply copies of $y = x^2$ slid along the x-axis. Once you see that picture, the later parts are about which copies pass through which points, and what all the copies have in common.

Solution

The general solution

Since $(\mathrm{d}y/\mathrm{d}x)^2 \geq 0$, any solution has $y \geq 0$. Where $y > 0$ we may write

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \pm 2\sqrt{y} \quad \Longrightarrow \quad \frac{1}{2\sqrt{y}}\frac{\mathrm{d}y}{\mathrm{d}x} = \pm 1 \quad \Longrightarrow \quad \sqrt{y} = \pm(x-c)$$

for a constant c, and squaring either sign gives

$$y = (x-c)^2.$$

It is worth checking directly that this whole parabola is a solution, including the left- and right-hand halves and the vertex: $y' = 2(x-c)$, so $(y')^2 = 4(x-c)^2 = 4y$ for every x.

Ask yourself The two signs gave the same formula. So where has the "$\pm$" gone?

It is hidden in the two halves of each parabola. On $y = (x-c)^2$ the gradient $2(x-c)$ is $-2\sqrt{y}$ to the left of the vertex and $+2\sqrt{y}$ to the right, so a single parabola uses both signs. The choice of sign reappears as a choice of c when we fit a point.

Solutions through (a, b^2)

We need $(a-c)^2 = b^2$, so $a - c = \pm b$, that is $c = a - b$ or $c = a + b$. The two solutions are

$$y = (x - a + b)^2 \qquad \text{and} \qquad y = (x - a - b)^2.$$

These are different because $b \neq 0$ makes $a - b \neq a + b$. (If $b = 0$ they would merge into one parabola with vertex at $(a, 0)$; the point would then also lie on the solution $y = 0$, which we meet again at the end.) Replacing b by $-b$ just swaps the two, as it should, since the point (a, b^2) is unchanged.

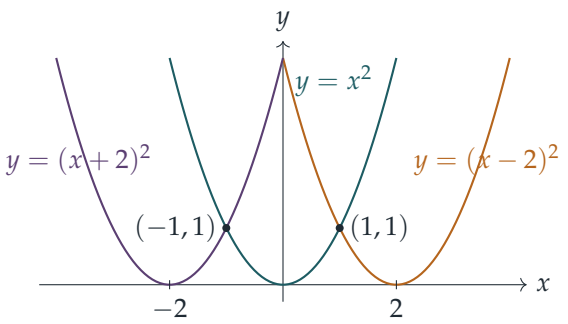
The points $(a_1, 1)$ and $(a_2, 1)$

Take $b = 1$: through $(a, 1)$ the solutions are $y = (x - a + 1)^2$ and $y = (x - a - 1)^2$. A parabola $y = (x - c)^2$ passes through the origin exactly when $c = 0$, so we need $a - 1 = 0$ or $a + 1 = 0$. Hence

$$a_1 = 1, \qquad a_2 = -1.$$

Through $(1, 1)$ the solutions are $y = x^2$ (from $c = a_1 - 1 = 0$) and $y = (x-2)^2$. Through $(-1, 1)$ the solutions are $y = (x+2)^2$ and $y = x^2$ (from $c = a_2 + 1 = 0$).

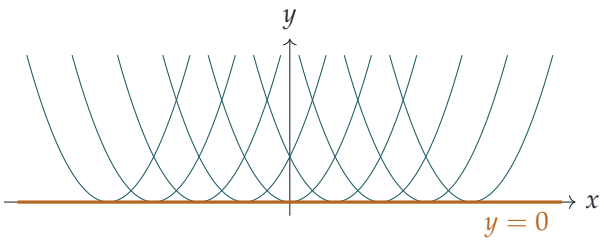
The solutions through the origin are, for both points, the same curve $y = x^2$, so their graphs coincide. This is no surprise: $(1, 1)$ and $(-1, 1)$ are the two points of $y = x^2$ at height 1, one on each side of the vertex. The other two solutions are $y = (x-2)^2$ and $y = (x+2)^2$, which are $y = x^2$ shifted 2 to the right and to the left.



At $(1, 1)$ the curve $y = x^2$ has gradient $+2$ and $y = (x-2)^2$ has gradient -2: the two solutions through a point always cross there with gradients $\pm 2b$, one for each square root.

The family and its common tangent

Varying a and b just varies $c = a \pm b$ over all real numbers, so the family is every horizontal translate of $y = x^2$.



Every parabola touches the x-axis at its vertex $(c, 0)$, where the gradient is 0. So the line $y = 0$ is a common tangent to the whole family (it is their *envelope*).

The relevance is that $y = 0$ is itself a solution, since then $y' = 0$ and $(y')^2 = 0 = 4y$. But it is not one of the parabolas for any c: it is a *singular solution*, which the separation of variables missed because we divided by $\sqrt{y}$. At each point $(c, 0)$ the solution $y = 0$ and the parabola $y = (x - c)^2$ have the same value and the same gradient, so solutions through points of the x-axis are not unique. In fact we can splice: for $p < q$,

$$y = \begin{cases} (x-p)^2, & x < p, \\ 0, & p \le x \le q, \\ (x-q)^2, & x > q \end{cases}$$

has a continuous derivative and satisfies the equation everywhere. So strictly the "two solutions" through (a, b^2) are the two that agree with the parabolas near that point; away from it, once a solution reaches the x-axis, it may run along the axis for a while before leaving.

Checks

- Substitute back: for $y = (x - a \pm b)^2$, at $x = a$ we get $y = b^2$, and $(y')^2 = 4y$ holds identically.
- The two gradients at (a, b^2) are $2(a - c) = \pm 2b$, whose squares are both $4b^2 = 4y$, as required.
- Numerically, a central-difference derivative confirmed $(y')^2 = 4y$ for random parabolas and for the spliced curve above with $p = -1$, $q = 1$, and that the solutions through $(1, 1)$ and $(-1, 1)$ have $c \in \{0, 2\}$ and $c \in \{-2, 0\}$, sharing only $c = 0$.

Common slips

- Writing $y' = 2\sqrt{y}$ only, and so getting just one half-parabola through each point.

- Forgetting $y = 0$: it is a genuine solution, and it is exactly the common tangent the last part is after.
- Thinking the two solutions through $(a_1, 1)$ and $(a_2, 1)$ that pass through the origin are different curves; both are $y = x^2$.

Beyond STEP

This equation is the standard example of where uniqueness of solutions breaks down. The Picard–Lindelöf theorem guarantees exactly one solution of $y' = f(x, y)$ through a point when f is Lipschitz in y, but $\pm 2\sqrt{y}$ is not Lipschitz near $y = 0$, and that is precisely where the parabolas and the singular solution meet. Envelopes of one-parameter families of curves, and the singular solutions they give for equations like Clairaut's, are a classical topic in first courses on ODEs. Boyce and DiPrima, *Elementary Differential Equations*, discusses existence and uniqueness with examples like this one.

WORKED SOLUTION

STEP III 1994, Question 5

Pure • Maclaurin Series

STEP III 1994, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s3-q5

The key idea

Differentiating $\sin^{-1} x$ once gives a square root, and differentiating again gives a relation between f'' and f' that has no square roots in it. Leibniz's theorem then turns that one equation into a recurrence for all the higher derivatives. At $x = 0$ most of the terms vanish, which gives a simple recurrence for the Maclaurin coefficients. For g, the laws of logarithms reduce everything to the familiar series for $\ln(1 \pm x)$.

Solution

The differential equation

For $-1 < x < 1$,

$$\mathrm{f}'(x) = \frac{1}{\sqrt{1-x^2}} = (1-x^2)^{-1/2}, \qquad \mathrm{f}''(x) = -\tfrac{1}{2}(1-x^2)^{-3/2} \cdot (-2x) = \frac{x}{(1-x^2)^{3/2}}.$$

Multiplying the second by $1 - x^2$ gives $(1-x^2)\mathrm{f}''(x) = x(1-x^2)^{-1/2} = x\mathrm{f}'(x)$, so

$$(1-x^2)\mathrm{f}''(x) - x\mathrm{f}'(x) = 0,$$

as required.

The recurrence

Differentiate this identity n times using Leibniz's theorem, $(uv)^{(n)} = \sum_r \binom{n}{r} u^{(r)} v^{(n-r)}$. In the first product we take $u = 1 - x^2$. Its derivatives are $-2x$, -2 and then 0, so only three terms survive:

$$\frac{\mathrm{d}^n}{\mathrm{d}x^n}\left[(1-x^2)\mathrm{f}''\right] = (1-x^2)\mathrm{f}^{(n+2)} + n(-2x)\mathrm{f}^{(n+1)} + \binom{n}{2}(-2)\mathrm{f}^{(n)}.$$

In the second product $u = x$, and only two terms survive:

$$\frac{\mathrm{d}^n}{\mathrm{d}x^n}\left[x\mathrm{f}'\right] = x\mathrm{f}^{(n+1)} + n\mathrm{f}^{(n)}.$$

Subtracting, and using $2\binom{n}{2} = n(n-1)$,

$$\begin{aligned} 0 &= (1-x^2)\mathrm{f}^{(n+2)} - (2n+1)x\mathrm{f}^{(n+1)} - \big(n(n-1)+n\big)\mathrm{f}^{(n)} \\ &= (1-x^2)\mathrm{f}^{(n+2)} - (2n+1)x\mathrm{f}^{(n+1)} - n^2\mathrm{f}^{(n)}, \end{aligned}$$

which is the required result. It holds for every $n \geq 1$, and for $n = 0$ it is just the first equation. (If you prefer, you can prove it by induction instead: differentiate the case n once and collect terms to get the case $n+1$. That is the same calculation done one step at a time.)

The Maclaurin series of f

Ask yourself Which value of x makes this recurrence easy to use?

Put $x = 0$. The middle term vanishes and $1 - x^2 = 1$, so

$$\mathrm{f}^{(n+2)}(0) = n^2\,\mathrm{f}^{(n)}(0) \qquad (n \geq 0).$$

We start from $\mathrm{f}(0) = \sin^{-1} 0 = 0$ and $\mathrm{f}'(0) = 1$.

Even derivatives. Since $\mathrm{f}(0) = 0$, the recurrence makes every even derivative zero at 0. This is what we expect, because $\sin^{-1}$ is an odd function.

Odd derivatives. With $n = 1, 3, 5, \ldots$,

$$\mathrm{f}'''(0) = 1^2, \quad \mathrm{f}^{(5)}(0) = 3^2 \cdot 1^2, \quad \mathrm{f}^{(7)}(0) = 5^2 \cdot 3^2 \cdot 1^2,$$

and in general

$$\mathrm{f}^{(2k+1)}(0) = \big(1 \cdot 3 \cdot 5 \cdots (2k-1)\big)^2 \qquad (k \geq 1).$$

Therefore

$$\sin^{-1} x = x + \frac{1^2}{3!}x^3 + \frac{1^2 \cdot 3^2}{5!}x^5 + \frac{1^2 \cdot 3^2 \cdot 5^2}{7!}x^7 + \cdots = x + \frac{x^3}{6} + \frac{3x^5}{40} + \frac{5x^7}{112} + \cdots.$$

The coefficient of x^{2k+1} simplifies neatly. Write $(2k+1)! = \big(1 \cdot 3 \cdots (2k-1)\big)\big(2 \cdot 4 \cdots 2k\big)(2k+1)$ and cancel one copy of the odd product:

$$a_{2k+1} = \frac{\big(1 \cdot 3 \cdots (2k-1)\big)^2}{(2k+1)!} = \frac{1 \cdot 3 \cdot 5 \cdots (2k-1)}{2 \cdot 4 \cdot 6 \cdots (2k)} \cdot \frac{1}{2k+1}.$$

The ratio test shows that this series converges for $|x| < 1$, since $a_{2k+3}/a_{2k+1} \to 1$. That it actually equals $\sin^{-1} x$ there can be confirmed by integrating the binomial series for $(1-x^2)^{-1/2}$ term by term, which gives exactly the same coefficients.

The function g

By the laws of logarithms, $\mathrm{g}(x) = \frac{1}{2}\big(\ln(1+x) - \ln(1-x)\big)$. For $|x| < 1$,

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots, \qquad \ln(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \cdots.$$

In the difference the even powers cancel and the odd powers double. So

$$g(x) = x + \frac{x^3}{3} + \frac{x^5}{5} + \cdots = \sum_{k=0}^{\infty} \frac{x^{2k+1}}{2k+1}.$$

(You may recognise g as $\tanh^{-1} x$.)

Comparing coefficients

? **Ask yourself** Both coefficients have a factor $\frac{1}{2k+1}$. What is left to compare?

For $n \geq 1$, the coefficient of x^{2n+1} is $\dfrac{1}{2n+1}$ in g and

$$\frac{1}{2} \cdot \frac{3}{4} \cdot \frac{5}{6} \cdots \frac{2n-1}{2n} \cdot \frac{1}{2n+1}$$

in f. Each of the n factors $\frac{2r-1}{2r}$ lies strictly between 0 and 1, and there is at least one of them because $n \geq 1$. So their product is strictly less than 1, and hence

$$\frac{1 \cdot 3 \cdots (2n-1)}{2 \cdot 4 \cdots (2n)} \cdot \frac{1}{2n+1} < \frac{1}{2n+1}.$$

The coefficient in g is therefore greater, as required. When $n = 0$ both coefficients equal 1, which is why the question excludes that case.

Checks

- At $x = -\frac{1}{2}$, the series summed up to x^{59} gives -0.5235987756, matching $\sin^{-1}(-\frac{1}{2}) = -\pi/6$.
- The coefficients $1, \frac{1}{6}, \frac{3}{40}, \frac{5}{112}$ agree with those generated directly from $f^{(n+2)}(0) = n^2 f^{(n)}(0)$. The comparison $\frac{1}{6} < \frac{1}{3}$, $\frac{3}{40} < \frac{1}{5}$, $\frac{5}{112} < \frac{1}{7}$ holds as claimed.
- Since every coefficient of f is at most the matching one of g, we should have $\sin^{-1} x < \tanh^{-1} x$ for $0 < x < 1$. At $x = 0.5$: $0.5236 < 0.5493$.

Common slips

- Forgetting the $\binom{n}{2}$ term in Leibniz's theorem, which gives n instead of n^2 in the final coefficient.
- Writing $f^{(2k+1)}(0) = 1 \cdot 3 \cdots (2k-1)$ without the square. Each step of the recurrence multiplies by n^2, not n.
- Dividing by $(2k)!$ or $(2k+1)$ alone instead of $(2k+1)!$ when forming the Maclaurin coefficient.

Beyond STEP

The ratio $\frac{1\cdot3\cdots(2n-1)}{2\cdot4\cdots(2n)}$ equals $\binom{2n}{n}/4^n$, and it behaves like $1/\sqrt{\pi n}$ for large n. This is a consequence of Wallis's product and Stirling's formula. The series for $\sin^{-1} x$ converges at $x = 1$ too, giving $\pi/2$, but proving that needs Abel's theorem on power series at the edge of their interval of convergence. When a function's derivatives satisfy a differential equation, a recurrence for the coefficients follows; this is the method of series solutions for ODEs, covered in Boyce and DiPrima, *Elementary Differential Equations*. For careful proofs of when a Taylor series really equals its function, see Abbott, *Understanding Analysis*.

WORKED SOLUTION

STEP III 1994, Question 6

Pure • Rotations in the Argand Diagram

STEP III 1994, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s3-q6

The key idea

A rotation about the point p through angle θ is the map $z \mapsto p + e^{i\theta}(z - p)$: shift p to the origin, rotate, shift back. Composing several such maps always gives something of the form $z \mapsto \lambda z + \mu$, and "$P_{\text{last}} = P_1$ for *every* P_1" means this map is the identity, so $\lambda = 1$ and $\mu = 0$. Everything else is reading those two equations geometrically.

Solution

Finding z_5

A positive quarter-turn is multiplication by i, so rotating P_1 about A gives

$$z_2 = a + \mathrm{i}(z_1 - a) = (1 - \mathrm{i})a + \mathrm{i}z_1.$$

In the same way $z_3 = (1 - \mathrm{i})b + \mathrm{i}z_2$, $z_4 = (1 - \mathrm{i})c + \mathrm{i}z_3$ and $z_5 = (1 - \mathrm{i})d + \mathrm{i}z_4$. Substituting back step by step,

$$\begin{aligned} z_5 &= (1 - \mathrm{i})\big(d + \mathrm{i}c + \mathrm{i}^2 b + \mathrm{i}^3 a\big) + \mathrm{i}^4 z_1 \\ &= (1 - \mathrm{i})\big(d - b + \mathrm{i}(c - a)\big) + z_1. \end{aligned}$$

When does P_5 return to P_1?

Four quarter-turns make a full turn, so the coefficient of z_1 is exactly 1, and

$$z_5 - z_1 = (1 - \mathrm{i})\big(d - b + \mathrm{i}(c - a)\big),$$

which does not depend on z_1 at all. The composite map is a pure translation. Since $1 - \mathrm{i} \neq 0$,

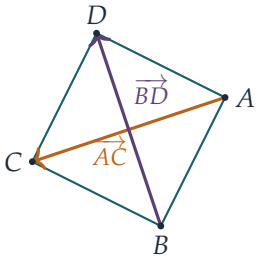
$$\begin{aligned} z_5 = z_1 \text{ for every } z_1 &\iff d - b + \mathrm{i}(c - a) = 0 \\ &\iff \mathrm{i}(c - a) = b - d \\ &\iff a - c = \mathrm{i}(b - d), \end{aligned}$$

where the last step multiplies both sides by $-\mathrm{i}$ (and $-\mathrm{i}\cdot\mathrm{i}=1$). Every step is reversible, so this is the required "if and only if". (If the condition fails, the translation is non-zero, and then $P_5 \neq P_1$ for *every* choice of P_1.)

Ask yourself What does $a-c=\mathrm{i}(b-d)$ say about the shape $ABCD$?

Multiply by -1: the condition is $c-a=\mathrm{i}(d-b)$. Now $c-a$ represents the vector $\overrightarrow{AC}$ and $d-b$ the vector $\overrightarrow{BD}$. Multiplying by i rotates a vector by $\pi/2$ without changing its length. So:

$\overrightarrow{AC}$ is $\overrightarrow{BD}$ turned through a positive right angle; that is, the diagonals AC and BD of the quadrilateral $ABCD$ are equal in length and perpendicular, with AC a quarter-turn anticlockwise from BD.



In the example drawn, $\overrightarrow{BD}=-1+3\mathrm{i}$ and $\overrightarrow{AC}=-3-\mathrm{i}=\mathrm{i}(-1+3\mathrm{i})$. The orientation matters: a square labelled *clockwise* satisfies the condition, but one labelled anticlockwise does not (there $\overrightarrow{AC}=-\mathrm{i}\,\overrightarrow{BD}$).

Three rotations through θ

Write $w=\mathrm{e}^{\mathrm{i}\theta}$. Exactly as before, each step is $z\mapsto(1-w)p+wz$, so

$$\begin{aligned} z_4 &= (1-w)c+w\big((1-w)b+w\big((1-w)a+wz_1\big)\big)\\ &= (1-w)\big(c+wb+w^2a\big)+w^3z_1. \end{aligned}$$

This is $\lambda z_1+\mu$ with $\lambda=w^3$ and $\mu=(1-w)(c+wb+w^2a)$. It equals z_1 for every z_1 if and only if $\lambda=1$ and $\mu=0$ (take $z_1=0$ to get $\mu=0$, then $z_1=1$ to get $\lambda=1$).

The angle. $w^3=1$ means 3θ is a multiple of 2π. We read $\theta\neq 0$ as "not a whole number of turns", so $w\neq 1$; hence w is a non-real cube root of unity and

$$\theta=\pm\tfrac{2\pi}{3}\pmod{2\pi}.$$

The points. Since $w\neq 1$, $\mu=0$ becomes $c+wb+w^2a=0$. Multiplying by w and using $w^3=1$ gives $a+w^2b+wc=0$, and since $1+w+w^2=0$ (as $w^3=1$, $w\neq 1$) we can rewrite $a=-(w+w^2)a$, so

$$w(c-a)+w^2(b-a)=0 \quad\Longrightarrow\quad c-a=-w\,(b-a)=\mathrm{e}^{\mathrm{i}(\theta+\pi)}(b-a)=\mathrm{e}^{\mathrm{i}(\theta-\pi)}(b-a).$$

Ask yourself What is $e^{i(\theta-\pi)}$ when $\theta = \pm 2\pi/3$?

For $\theta = 2\pi/3$ it is $e^{-i\pi/3}$; for $\theta = -2\pi/3$ it is $e^{i\pi/3}$ (since $e^{-5\pi i/3} = e^{i\pi/3}$). So $\overrightarrow{AC}$ is $\overrightarrow{AB}$ turned through $\mp\pi/3$ with the same length: ABC is **equilateral**, with A, B, C in **clockwise** order when $\theta = 2\pi/3$ and **anticlockwise** order when $\theta = -2\pi/3$. Conversely, each of these steps reverses, so these conditions also give $\mu = 0$.

Since A, B, C lie on the unit circle, that circle is the triangle's circumcircle, and for an equilateral triangle the circumcentre is the centre, so the vertices are equally spaced around $|z| = 1$. Both cases combine neatly into one statement:

$$\theta = \pm\tfrac{2\pi}{3} \pmod{2\pi}, \qquad b = a\,e^{-i\theta}, \quad c = b\,e^{-i\theta}.$$

In words, the three rotations are by a third of a turn, and A, B, C are equally spaced on the unit circle, stepping round in the *opposite* sense to the rotation.

Checks

- A computer applied the four quarter-turns to random points about random A, B, C, D and matched the formula for z_5 to 10^{-9}; with $c = a - i(b - d)$ every P_1 returned home, and nudging c by 0.1 broke it.
- For random a on the unit circle and $\theta = \pm 2\pi/3$, $b = ae^{-i\theta}$, $c = be^{-i\theta}$ gave $z_4 = z_1$ for random z_1; reversed labelling and other θ did not.
- By hand: rotate $z_1 = 0$ about the clockwise unit square $a = 0$, $b = i$, $c = 1 + i$, $d = 1$. You get $0 \to 0 \to 1 + i \to 1 + i \to 0$. Back home.

Common slips

- Writing a rotation about A as $z \mapsto iz$, forgetting to shift A to the origin first.
- Dropping orientation: the diagonals' sense of rotation, and the triangle's for each of $\theta = \pm 2\pi/3$, are part of the conditions.

Beyond STEP

The maps $z \mapsto \lambda z + \mu$ with $|\lambda| = 1$ are the orientation-preserving isometries of the plane, and they form a group. Angles add, so a composite of rotations is a rotation by the total angle, or a translation when the total is a whole number of turns: that is why $z_5 - z_1$ came out constant. Needham's *Visual Complex Analysis* develops this geometric view of complex numbers beautifully.

WORKED SOLUTION

STEP III 1994, Question 7

Pure • Group Isomorphisms

STEP III 1994, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s3-q7

The key idea

Two groups can only be isomorphic if they share every structural property that an isomorphism preserves. The two most useful such properties are the list of element orders and whether the group is commutative (abelian). Listing the orders already separates S_3 from Z_6. To prove that two groups *are* isomorphic, we write down an explicit bijection and check that it respects the operations. The final part comes down to commutativity: every group made of complex numbers is abelian, and S_3 isn't.

Solution

The elements of S_3 and Z_6

Label the three objects $1, 2, 3$ and use cycle notation, composing from right to left, so $\sigma\tau$ means "do τ first, then σ". The six permutations are listed below.

element of S_3	e	(12)	(13)	(23)	(123)	(132)
order	1	2	2	2	3	3

A transposition swaps two objects, and doing it twice restores everything, so it has order 2. A 3-cycle such as (123) sends $1 \to 2 \to 3 \to 1$, so it needs three applications to return each object home, and it has order 3.

In $Z_6 = \{0, 1, 2, 3, 4, 5\}$ the identity is 0, and the order of a is the least $n \geq 1$ with $na \equiv 0 \pmod 6$, which is $6/\gcd(a, 6)$.

element of Z_6	0	1	2	3	4	5
order	1	6	3	2	3	6

For example, $4 + 4 = 8 \equiv 2$ and $4 + 4 + 4 = 12 \equiv 0$, so 4 has order 3.

Are S_3 and Z_6 isomorphic?

No. An isomorphism θ preserves orders: if g has order n then $\theta(g)^k = \theta(g^k)$, and this is the identity exactly when g^k is, because θ is injective. So isomorphic groups have the same number of elements of each order. But Z_6 has two elements of order 6 and S_3 has none, so they aren't isomorphic. (They also differ in their elements of order 2: S_3 has three of them and Z_6 has only one.)

There is a second reason, which we'll reuse at the end. Z_6 is abelian, since addition is commutative. S_3 isn't:

$$(12)(13) = (132), \qquad (13)(12) = (123).$$

For the first product, (13) sends $1 \to 3$ and then (12) fixes 3, so $1 \to 3$. Similarly $3 \to 2$ and $2 \to 1$. An isomorphism carries commuting pairs to commuting pairs, so a non-abelian group can't be isomorphic to an abelian one.

C_6 is isomorphic with Z_6

Define $\phi : Z_6 \to C_6$ by $\phi(k) = \alpha^k$ for $k = 0, 1, \ldots, 5$.

- *Bijective.* The numbers $\alpha^k = e^{ik\pi/3}$ for $k = 0, \ldots, 5$ have arguments $0, \frac{\pi}{3}, \ldots, \frac{5\pi}{3}$. These are distinct angles in $[0, 2\pi)$, so the six values are distinct points on the unit circle. Hence ϕ is injective, and so it is a bijection between two sets of size 6.
- *Operations.* Let $a, b \in Z_6$, and write $a + b = 6q + r$ with $r = (a + b) \bmod 6$. Since $\alpha^6 = e^{2\pi i} = 1$,

$$\phi(a)\,\phi(b) = \alpha^a \alpha^b = \alpha^{a+b} = (\alpha^6)^q \alpha^r = \alpha^r = \phi\big((a + b) \bmod 6\big).$$

So ϕ turns addition mod 6 into complex multiplication, and it is an isomorphism. Hence $C_6 \cong Z_6$. As a sanity check, α and α^5 (the images of 1 and 5) have order 6, α^2 and α^4 have order 3, and $\alpha^3 = -1$ has order 2, which matches the table for Z_6.

A subgroup of $\mathbb{C}$ isomorphic with S_3?

> **Ask yourself** What property do *all* groups of complex numbers share?

They are all abelian. A subgroup H of $(\mathbb{C}, +)$ is abelian because $z + w = w + z$ for all complex z, w. A subgroup H of the non-zero complex numbers under multiplication is abelian because $zw = wz$. A subgroup uses the same operation as the big group, so it inherits commutativity.

Now suppose some such H had an isomorphism $\theta : S_3 \to H$. Apply it to the non-commuting pair above. Since H is abelian,

$$\theta\big((12)(13)\big) = \theta(12)\,\theta(13) = \theta(13)\,\theta(12) = \theta\big((13)(12)\big),$$

written multiplicatively (replace the products by sums for the additive case). Because θ is injective, this would give $(12)(13) = (13)(12)$, which is false. So **no** subgroup of the complex numbers, additive or multiplicative, is isomorphic with S_3.

Checks

- In each group the orders divide 6 (Lagrange's theorem): $1, 2, 3$ in S_3 and $1, 2, 3, 6$ in Z_6.
- A computer recomputed every order in S_3, Z_6 and C_6, confirmed $(12)(13) \neq (13)(12)$, and checked $\alpha^{(a+b) \bmod 6} = \alpha^a \alpha^b$ numerically for all 36 pairs (a, b), to within 10^{-12}.

Common slips

- Saying "both have six elements, so they're isomorphic". Having the same size is necessary but nowhere near sufficient.
- Saying the elements of order 3 in S_3 are (12) and (13). Transpositions have order 2; only the 3-cycles have order 3.
- In the last part, only ruling out *finite* multiplicative subgroups (the roots of unity). The commutativity argument covers every subgroup, finite or infinite, additive or multiplicative, at once.

Beyond STEP

This question is about invariants: properties, such as element orders and commutativity, that isomorphic groups must share. In fact every group of order 6 is isomorphic to either Z_6 or S_3. More generally, any finite subgroup of the non-zero complex numbers under multiplication is cyclic, so it is a group of roots of unity. You'll meet both results early in a first course on groups. Allenby's *Rings, Fields and Groups* is a friendly place to read about them.

WORKED SOLUTION

STEP III 1994, Question 8

Pure • Four-Square Identity

STEP III 1994, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s3-q8

This question has a full guided treatment in *Beyond STEP*, Section 7.8, *Four Squares*: https://beyondstep.keithmcnulty.org/ch07/four-squares.html

The key idea

A matrix of the form $\begin{pmatrix} z & w \\ -w^* & z^* \end{pmatrix}$ has determinant $zz^* + ww^* = |z|^2 + |w|^2$, which is a sum of *four* real squares. If we pack a, b, c, d into one such matrix and p, q, r, s into another, then $\det(AB) = \det A \det B$ gives the left-hand side. The work is to show that the product AB has the same special form, because then its determinant is also a sum of four squares, and those squares are the L_i^2.

Solution

Setting up

For complex numbers z and w write

$$M(z,w) = \begin{pmatrix} z & w \\ -w^* & z^* \end{pmatrix}, \qquad \det M(z,w) = zz^* - w(-w^*) = |z|^2 + |w|^2.$$

Choose

$$z_1 = a + b\mathrm{i}, \quad z_2 = c + d\mathrm{i}, \quad z_3 = p + q\mathrm{i}, \quad z_4 = r + s\mathrm{i}.$$

Then

$$\det M(z_1, z_2) = a^2 + b^2 + c^2 + d^2, \qquad \det M(z_3, z_4) = p^2 + q^2 + r^2 + s^2.$$

The product has the same form

Multiplying out,

$$M(z_1,z_2)\,M(z_3,z_4)=\begin{pmatrix} z_1z_3-z_2z_4^* & z_1z_4+z_2z_3^* \\ -z_2^*z_3-z_1^*z_4^* & -z_2^*z_4+z_1^*z_3^* \end{pmatrix}.$$

Put $w_1=z_1z_3-z_2z_4^*$ and $w_2=z_1z_4+z_2z_3^*$, the top row. Since the conjugate of a product is the product of the conjugates, and $(z^*)^*=z$,

$$w_1^*=z_1^*z_3^*-z_2^*z_4, \qquad -w_2^*=-z_1^*z_4^*-z_2^*z_3,$$

and these are exactly the bottom-right and bottom-left entries. So

$$M(z_1,z_2)\,M(z_3,z_4)=M(w_1,w_2).$$

Ask yourself Why does it matter that the product has this special shape?

For a general 2×2 complex matrix the determinant is not a sum of squares at all. It is only because the product is again of the form $M(w_1,w_2)$ that its determinant is $|w_1|^2+|w_2|^2$, which splits into four real squares.

Taking determinants

Using $\det(AB)=\det A\,\det B$,

$$(a^2+b^2+c^2+d^2)(p^2+q^2+r^2+s^2)=\det M(w_1,w_2)=|w_1|^2+|w_2|^2.$$

It remains to find the real and imaginary parts of w_1 and w_2:

$$\begin{aligned} w_1&=(a+b\mathrm{i})(p+q\mathrm{i})-(c+d\mathrm{i})(r-s\mathrm{i})\\ &=\big[(ap-bq)+(aq+bp)\mathrm{i}\big]-\big[(cr+ds)+(dr-cs)\mathrm{i}\big]\\ &=(ap-bq-cr-ds)+(aq+bp+cs-dr)\mathrm{i},\end{aligned}$$

$$\begin{aligned} w_2&=(a+b\mathrm{i})(r+s\mathrm{i})+(c+d\mathrm{i})(p-q\mathrm{i})\\ &=\big[(ar-bs)+(as+br)\mathrm{i}\big]+\big[(cp+dq)+(dp-cq)\mathrm{i}\big]\\ &=(ar-bs+cp+dq)+(as+br-cq+dp)\mathrm{i}.\end{aligned}$$

So we may take

$$L_1=ap-bq-cr-ds, \qquad L_2=aq+bp+cs-dr,$$
$$L_3=ar-bs+cp+dq, \qquad L_4=as+br-cq+dp,$$

and then $|w_1|^2+|w_2|^2=L_1^2+L_2^2+L_3^2+L_4^2$, giving

$$(a^2+b^2+c^2+d^2)(p^2+q^2+r^2+s^2)=L_1^2+L_2^2+L_3^2+L_4^2.$$

Each L_i is a sum of terms (one letter from a,b,c,d) $\times$ (one letter from p,q,r,s), so it is linear in each set of variables separately, as required. (The answer is not unique: changing any signs of the L_i, or choosing the z_j differently, gives other valid sets.)

Checks

- Take $(a,b,c,d) = (1,2,3,4)$ and $(p,q,r,s) = (5,6,7,8)$. The left-hand side is $30 \times 174 = 5220$. The formulae give $(L_1, L_2, L_3, L_4) = (-60, 12, 30, 24)$, and $3600 + 144 + 900 + 576 = 5220$.
- Put $c = d = r = s = 0$. Then $L_3 = L_4 = 0$ and the identity reduces to $(a^2+b^2)(p^2+q^2) = (ap-bq)^2 + (aq+bp)^2$, the two-square identity from $|z_1 z_3| = |z_1||z_3|$.
- A computer check confirmed the identity exactly for 20,000 random rational choices of the eight variables, and confirmed $M(z_1,z_2)M(z_3,z_4) = M(w_1,w_2)$ numerically.

Common slips

- Forgetting the conjugates in the bottom row when multiplying, which breaks the special form of the product.
- Sign errors in $(c+d\mathrm{i})(r-s\mathrm{i})$: the term $-ds\,\mathrm{i}^2$ becomes $+ds$.

Beyond STEP

This is Euler's four-square identity. The matrices $M(z,w)$ are a concrete model of Hamilton's quaternions: they are closed under multiplication, and the determinant is the squared "length", which is multiplicative. The identity is the key step in Lagrange's theorem that every positive integer is a sum of four squares, since it reduces the problem to primes. Davenport's *The Higher Arithmetic* covers sums of squares, and Needham's *Visual Complex Analysis* gives geometric intuition for the two-square case.

WORKED SOLUTION

STEP III 1994, Question 9

Mechanics • Tautochrone and Collisions

STEP III 1994, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s3-q9

This question has a full guided treatment in *Beyond STEP*, Section 11.4, *The Tautochrone*: https://beyondstep.keithmcnulty.org/ch11/tautochrone.html

The key idea

Energy conservation along the surface gives a first integral, and the shape of the bowl, $y = s^2/(4k)$, turns it into the energy equation of simple harmonic motion in the arc-length s. The period of simple harmonic motion doesn't depend on the amplitude, so every particle released from rest reaches V after the same time, wherever it starts. Because of this, the two particles in the final part are certain to meet at V, and they meet again and again.

Solution

The constant

We model the particle as a point mass on a smooth surface. Two forces act on it: its weight and the normal reaction. The particle starts from rest, so its velocity starts with no horizontal component perpendicular to the vertical plane through its starting point and the axis. Neither force has a component perpendicular to that plane (the reaction of an axially symmetric surface lies in the plane through the axis), so the particle stays in that vertical cross-section. It moves along the curve $s = 2\sqrt{ky}$, with speed $|\mathrm{d}s/\mathrm{d}t|$.

The surface is smooth, so the normal reaction is perpendicular to the velocity and does no work. Only gravity does work, so mechanical energy is conserved:

$$\tfrac{1}{2}m\left(\frac{\mathrm{d}s}{\mathrm{d}t}\right)^2 + mgy = \text{constant}.$$

Multiplying by $2/m$ shows that $\left(\frac{\mathrm{d}s}{\mathrm{d}t}\right)^2 + 2gy$ is constant, as required. Released from rest at height h, the constant is $2gh$.

The time to reach V

On the curve, $y = s^2/(4k)$. Substituting,

$$\left(\frac{\mathrm{d}s}{\mathrm{d}t}\right)^2 + \frac{g}{2k}s^2 = 2gh.$$

Differentiating with respect to t and dividing by $2\,\mathrm{d}s/\mathrm{d}t$ (valid while the particle is moving) gives

$$\frac{\mathrm{d}^2 s}{\mathrm{d}t^2} = -\frac{g}{2k}s,$$

which is simple harmonic motion in s with $\omega^2 = \dfrac{g}{2k}$, centred on $s = 0$, the point V.

Ask yourself Why is the time from release to V a quarter of a period?

The particle starts from rest at the extreme position $s_0 = 2\sqrt{kh}$, so $s = s_0 \cos \omega t$. It first reaches $s = 0$ when $\omega t = \frac{\pi}{2}$, a quarter of the period $2\pi/\omega$. So the time to reach V is

$$T = \frac{\pi}{2\omega} = \frac{\pi}{2}\sqrt{\frac{2k}{g}} = \pi\sqrt{\frac{k}{2g}},$$

as required. This doesn't depend on h: the bowl is a *tautochrone*.

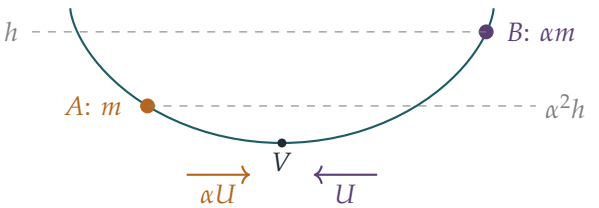
For later use, the speed at V is ωs_0, or directly from the energy equation with $y = 0$, $\sqrt{2gh}$.

The two particles

Treat the two particles as point masses that collide directly at V. The tangent there is horizontal, so the impact is along a horizontal line. Call the particle of mass m particle A (released at height $\alpha^2 h$) and the particle of mass αm particle B (released at height h on the opposite side).

They meet at V. By the previous part, each particle reaches V after time T, whatever its starting height. They are released together, so they arrive at V together and collide there at time T.

Speeds just before impact. Write $U = \sqrt{2gh}$. From the energy equation, A arrives with speed $\sqrt{2g\alpha^2 h} = \alpha U$ and B with speed U, travelling in opposite directions. Take the direction of A's motion as positive.



Momentum before impact is

$$m(\alpha U) + \alpha m(-U) = 0.$$

This is why the masses and heights were chosen as they were.

After the first impact. Let the velocities afterwards be v_A and v_B. Conservation of momentum and Newton's experimental law give

$$mv_A + \alpha m\, v_B = 0, \qquad v_B - v_A = \alpha\big(\alpha U - (-U)\big) = \alpha(1+\alpha)U.$$

From the first equation $v_B = -v_A/\alpha$. Substituting into the second gives $-v_A(1+\alpha)/\alpha = \alpha(1+\alpha)U$, so

$$v_A = -\alpha^2 U, \qquad v_B = \alpha U.$$

Both velocities are reversed and multiplied by α. (With zero total momentum, the relative velocity and each individual velocity are all scaled by $-\alpha$.) So each particle goes back up its *own* side of the bowl. By energy conservation, A rises to height $\alpha^4 h$ and B to height $\alpha^2 h$.

The pattern repeats. Each particle comes to rest at its new highest point at the same moment, a time T after the impact. Each then takes a further time T to return to V. So they collide again at V at time $3T$, with A from height $\alpha^4 h$ and B from height $\alpha^2 h$. This is exactly the starting configuration with h replaced by $\alpha^2 h$. Again the total momentum is zero, and again both velocities are reversed and multiplied by α.

By induction, the nth collision happens at V at time $(2n-1)T$, and after it

$$A \text{ rises to height } \alpha^{2n+2}h, \qquad B \text{ rises to height } \alpha^{2n}h,$$

each on the side of the bowl it started from. The particles oscillate on their own sides, with collisions at V at the equally spaced times $T, 3T, 5T, \ldots$. At each collision the speeds drop by the factor α, so the heights drop by the factor α^2 and a fraction $1-\alpha^2$ of the kinetic energy is lost. The collisions never stop, because the gaps between them are always $2T$. The amplitudes decrease geometrically, and both particles tend to rest at V as $t \to \infty$ without ever reaching it in finite time.

Checks

- Dimensions: k is a length (since $s^2 = 4ky$), so $\sqrt{k/g}$ is a time.
- Numerically, $\int_0^{s_0} \mathrm{d}s/\sqrt{2g(h-y)}$ was evaluated by quadrature for random k and h. With $k=1$, $g=9.81$ it gives $0.7092\ldots = \pi\sqrt{1/19.62}$, independent of h.
- We stepped the motion of both particles numerically (with $\alpha = 0.6$ and $\alpha = 0.3$) and applied the impact law at each meeting. The collisions came at $t/T = 1, 3, 5, 7$, always at V and with zero total momentum. After the first collision with $\alpha = 0.6$, the heights were $0.1296h = \alpha^4 h$ and $0.36h = \alpha^2 h$.
- Limiting case $\alpha \to 1$: equal masses from equal heights, perfectly elastic, so each bounces back to its starting height for ever, as expected.

Common slips

- Taking V as half a period from the start: the particle starts at an extreme of the oscillation, so V is a quarter-period away.
- Forgetting the reversal of direction in Newton's law: take the velocities with signs, not just speeds.
- Claiming the particles come to rest after finitely many collisions. The interval between collisions is constant, not shrinking (unlike a ball bouncing on a floor), so there are infinitely many collisions spread over infinite time.

Beyond STEP

The intrinsic equation $s^2 = 4ky$ describes a cycloid. This question rediscovers Huygens' tautochrone: on an inverted cycloid, the time to slide to the bottom doesn't depend on the starting point. Huygens used this to design pendulum clocks whose period doesn't depend on the swing. The same curve is the brachistochrone, the curve of fastest descent, found with the calculus of variations, a standard topic in second-year undergraduate mechanics. For more on oscillations and collisions, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP III 1994, Question 10

Mechanics • Relative Velocity

STEP III 1994, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

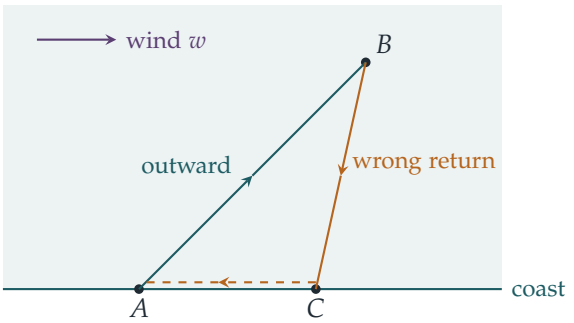
step.maths.org/questions/94-s3-q10

The key idea

Everything is vector addition: the plane's velocity over the ground is its air velocity plus the wind velocity. Put the coast along the x-axis with Alphaport at the origin, and write the outward air velocity in components. Reversing the heading reverses the air velocity but *not* the wind, so the northward and eastward parts of the motion behave very differently on the way back, and that is the whole story of the first part. For the second part, the two correct ground speeds come from the same quadratic, which lets Vieta's formulas do the work.

Solution

We model the plane as a particle moving with constant air velocity of magnitude v, relative to air moving with constant velocity w due east (a wind *from* the West blows *towards* the East). Take Alphaport A as the origin, x east, y north, so the coast is the line $y = 0$ and the island lies in $y > 0$. Betaville B is at $\left(\frac{d}{\sqrt{2}}, \frac{d}{\sqrt{2}}\right)$. Speeds are in $\mathrm{km\,h^{-1}}$ and times in hours. (The paper calls the places Alphabet/Alphaport and Betaville/Betaport; these are the same places.)



The outward flight

Let the outward air velocity be (a, b), so $a^2 + b^2 = v^2$. The ground velocity is $(a + w,\ b)$, and it must point north-east, so its two components are equal and positive:

$$a + w = b > 0.$$

She covers the y-distance $\frac{d}{\sqrt{2}}$ at y-speed b, and the outward time is t, so

$$bt = \frac{d}{\sqrt{2}} = (a + w)t. \tag{1}$$

The wrong return

With the opposite heading her air velocity is $(-a, -b)$, but the wind still adds $(w, 0)$, so her ground velocity is

$$(w - a,\ -b).$$

Ask yourself Where does she reach the coast, and on which side of Alphaport?

The northward motion is unaffected by the wind: her y-coordinate decreases at rate b from $y = \frac{d}{\sqrt{2}}$, so by (1) she reaches $y = 0$ after exactly time t, the same as the outward flight. In that time her x-coordinate changes by $(w - a)t$, so she crosses the coast at

$$x = \frac{d}{\sqrt{2}} + (w - a)t = (a + w)t + (w - a)t = 2wt,$$

using (1) again. So she lands at C, a distance $2wt$ *east* of Alphaport: going out, the wind pushed her east and her heading was chosen to cancel that, but coming back the heading's correction is reversed while the wind's push is not, so the drift doubles. Since $2wt > 0$, turning West along the coast is indeed the right way.

Flying due west into the wind at airspeed v, her ground speed is $v - w > 0$ (as $v > w$). The coastal leg takes $\dfrac{2wt}{v - w}$, so the whole return takes

$$t + \frac{2wt}{v - w} = \frac{(v - w) + 2w}{v - w}\, t = \left(\frac{v + w}{v - w}\right) t \text{ hours},$$

as required.

The correct return

Write a ground velocity along the line AB as $\pm(p, p)$, where $\sqrt{2}\,p$ is the ground speed. The air velocity is the ground velocity minus the wind, and has magnitude v.

Outward, ground velocity (p_1, p_1), air velocity $(p_1 - w,\ p_1)$:

$$(p_1 - w)^2 + p_1^2 = v^2 \iff 2p_1^2 - 2wp_1 + (w^2 - v^2) = 0.$$

Back, ground velocity $(-p_2, -p_2)$ with $p_2 > 0$, air velocity $(-p_2 - w, -p_2)$:

$$(p_2 + w)^2 + p_2^2 = v^2 \iff 2(-p_2)^2 - 2w(-p_2) + (w^2 - v^2) = 0.$$

So p_1 and $-p_2$ are both roots of the same quadratic $2p^2 - 2wp + (w^2 - v^2) = 0$. Its roots have product $\frac{w^2 - v^2}{2} < 0$, so one is positive and one negative; p_1 is the positive one and $-p_2$ the negative one, so they are the two distinct roots. By Vieta's formulas,

$$p_1 + (-p_2) = w, \qquad p_1(-p_2) = \frac{w^2 - v^2}{2},$$

that is, $p_1 - p_2 = w$ and $p_1 p_2 = \frac{1}{2}(v^2 - w^2)$.

The distance is d each way, so the outward time is $t = \dfrac{d}{\sqrt{2}\,p_1}$ and the correct return time is $\dfrac{d}{\sqrt{2}\,p_2}$. Hence

$$\begin{aligned}\frac{d}{\sqrt{2}\,p_2} &= t + \frac{d}{\sqrt{2}}\left(\frac{1}{p_2} - \frac{1}{p_1}\right) = t + \frac{d}{\sqrt{2}} \cdot \frac{p_1 - p_2}{p_1 p_2} \\ &= t + \frac{d}{\sqrt{2}} \cdot \frac{2w}{v^2 - w^2} = t + \frac{\sqrt{2}\,wd}{v^2 - w^2} \text{ hours,}\end{aligned}$$

which is the required result.

Checks

- If $w = 0$ both return times reduce to t, as they should in still air.
- With $v = 300$, $w = 60$, $d = 400$: $t \approx 1.1785$ h, the wrong return (found by stepping the motion numerically) takes $1.7678\,\text{h} = \frac{360}{240}t$, and the correct return takes 1.5713 h, matching $t + \frac{\sqrt{2}\cdot 60 \cdot 400}{86400}$. Random trials over many $v > w$ and d agree to rounding error.
- The wrong return is always slower than the correct one: this is the inequality $\frac{2wt}{v-w} > \frac{\sqrt{2}\,wd}{v^2 - w^2}$, i.e. $\sqrt{2}(v + w)t > d$, which holds because the outward ground speed $\frac{d}{t} = \sqrt{2}p_1 = \sqrt{2}(p_2 + w)$ is less than $\sqrt{2}(v + w)$, as $p_2 < v$.

Common slips

- Reversing the ground velocity instead of the air velocity: the wind does not turn round when the pilot does.
- Taking the wind as blowing *towards* the West. "From the West" means it carries the plane east.
- Solving each quadratic with the formula and subtracting the reciprocals of messy surds; spotting that p_1 and $-p_2$ are roots of one quadratic avoids all of that.

Beyond STEP

This is relative velocity in its simplest form: the velocity of the plane in the ground frame is its velocity in the air's frame plus the velocity of the air, a Galilean change of frame. The same bookkeeping, done carefully, underlies navigation, the river-crossing problems of first-year mechanics, and (when it breaks down at high speed) the motivation for special relativity. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats moving frames thoroughly.

WORKED SOLUTION

STEP III 1994, Question 11

Mechanics • Equilibrium of Rigid Bodies

STEP III 1994, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s3-q11

The key idea

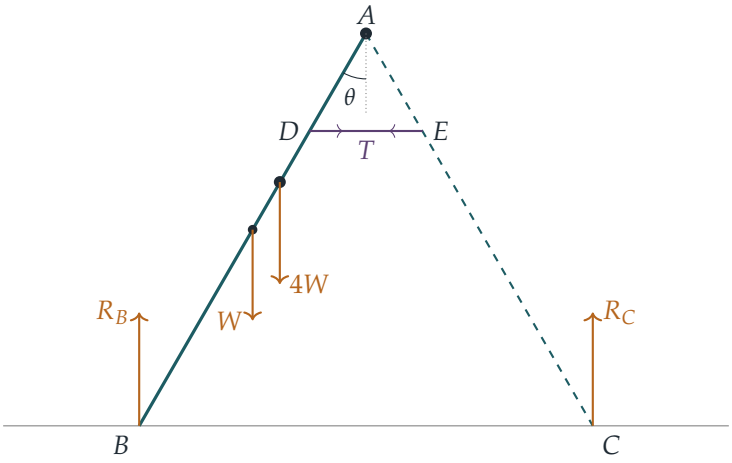
Because the floor is smooth, the only forces from the floor are vertical, so the whole ladder gives the reaction at C at once. The light leg AC then does the clever work: only three forces act on it (the floor reaction, the rope tension and the hinge force), so taking moments about the hinge A removes the unknown hinge force and links the rope tension to the reaction at C. Hooke's law ties the tension to the angle of the ladder, and that gives β as a function of the angle, which we maximise.

Solution

Setting up

We model AB as a uniform rod and AC as a light rod, both of length $4a$, with a smooth hinge at A, and the rope as light and elastic. The man is a particle. By symmetry ($AD = AE$ and $AB = AC$) the rope DE is horizontal. Let each leg make an angle θ with the vertical, with $0 < \theta < \frac{\pi}{2}$.

Since the floor is smooth, the reactions R_B and R_C at B and C are vertical. The weight W of AB acts at its midpoint, a horizontal distance $2a\sin\theta$ from B, and the man's weight $4W$ acts a horizontal distance $\beta a\sin\theta$ from B. The tension T pulls D and E towards each other.



The reaction at C

Taking moments about B for the whole step-ladder (the hinge force and the tension are internal, so they don't appear), with $BC = 8a\sin\theta$:

$$R_C \cdot 8a\sin\theta = W \cdot 2a\sin\theta + 4W \cdot \beta a\sin\theta \quad \Longrightarrow \quad R_C = \tfrac{1}{4}W(1+2\beta).$$

Resolving vertically, $R_B = 5W - R_C = \frac{1}{4}W(19-2\beta)$, which is positive since $\beta \leq 4$ (the rung is on the ladder). So the legs never lift off the floor.

The tension in the rope

The rope has length $DE = 2a\sin\theta$ and natural length $\frac{a}{4}$. If the rope were slack, $T = 0$ and moments about A for AC below would force $R_C = 0$, which is impossible. So the rope is stretched, $\sin\theta > \frac{1}{8}$, and Hooke's law gives

$$T = \frac{W\left(2a\sin\theta - \frac{a}{4}\right)}{a/4} = W(8\sin\theta - 1).$$

Equilibrium of the light leg AC

Ask yourself Why look at AC on its own rather than AB?

Leg AC is light, so only three forces act on it: R_C at C, the tension T at E and the hinge force at A. Moments about A remove the unknown hinge force. The tension is horizontal and acts at a vertical depth $a\cos\theta$ below A; the reaction R_C is vertical and acts a horizontal distance $4a\sin\theta$ from A. So

$$R_C \cdot 4a\sin\theta = T \cdot a\cos\theta.$$

Substituting for R_C and T:

$$\begin{aligned} W(1+2\beta)\sin\theta &= W(8\sin\theta - 1)\cos\theta, \\ \text{so} \quad 1+2\beta &= 8\cos\theta - \cot\theta, \\ \text{and} \quad \beta &= \tfrac{1}{2}\left(8\cos\theta - \cot\theta - 1\right). \end{aligned}$$

For each position β of the man, equilibrium is possible exactly when this equation has a solution θ (with $\sin\theta > \frac{1}{8}$). So the largest possible β is the maximum of $f(\theta) = 8\cos\theta - \cot\theta$.

Maximising

Differentiating,

$$f'(\theta) = -8\sin\theta + \operatorname{cosec}^2\theta = \frac{1 - 8\sin^3\theta}{\sin^2\theta}.$$

This is positive for $\sin\theta < \frac{1}{2}$ and negative for $\sin\theta > \frac{1}{2}$, since $\sin\theta$ increases on $(0, \frac{\pi}{2})$. So f has its greatest value when $\sin\theta = \frac{1}{2}$, that is, $\theta = \frac{\pi}{6}$, and then

$$f\left(\tfrac{\pi}{6}\right) = 8 \cdot \tfrac{\sqrt{3}}{2} - \sqrt{3} = 3\sqrt{3}, \qquad \beta_{\max} = \tfrac{1}{2}\left(3\sqrt{3} - 1\right) \approx 2.10.$$

This is less than 4, so the rung is on the ladder, and $\sin\theta = \frac{1}{2} > \frac{1}{8}$, so the rope is taut, as we assumed.

The height of the rung

The rung is a distance βa along BA from B, and BA makes angle θ with the vertical, so its height is

$$h = \beta a\cos\theta = \tfrac{1}{2}\left(3\sqrt{3} - 1\right) a \cdot \tfrac{\sqrt{3}}{2} = \frac{(9 - \sqrt{3})\,a}{4} \approx 1.82a.$$

Checks

- At $\theta = \frac{\pi}{6}$ with $W = a = 1$: $T = 8 \cdot \frac{1}{2} - 1 = 3$ and $R_C = \frac{1}{4}(1 + 3\sqrt{3} - 1) = \frac{3\sqrt{3}}{4} \approx 1.299$, and indeed $R_C \cdot 4\sin\theta = 2.598 = T\cos\theta$.
- A numerical check that solved the force and moment equations for both legs separately, and scanned θ over $(0, \frac{\pi}{2})$, found the largest β to be 2.09808 at $\theta \approx 30^\circ$, giving height $1.81699a$.
- Limiting cases make sense: as $\sin\theta \to \frac{1}{8}$ the rope goes slack and β would have to be negative, and as $\theta \to \frac{\pi}{2}$ the rope's moment arm $a\cos\theta$ vanishes, so again β must fall. A maximum in between is what we expect.

Common slips

- Taking the hinge force at A to be horizontal. It isn't, in general, because the loads on the two legs are different; moments about A avoid needing it at all.
- Using the whole length $DE = 2a\sin\theta$ as the extension. The extension is $2a\sin\theta - \frac{a}{4}$.
- Giving $\beta_{\max}$ (a distance along the ladder) instead of the height asked for, which needs the extra factor $a\cos\theta$.

Beyond STEP

This is a statics problem in which the configuration (θ) is not given but is set by the elastic rope, so equilibrium positions form a one-parameter family and the question becomes an optimisation over that family. At university this is the idea behind virtual work and potential energy methods: equilibria are stationary points of the potential energy, and asking which are stable leads to bifurcation and catastrophe theory. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good place to deepen the mechanics.

WORKED SOLUTION

STEP III 1994, Question 12

Probability and Statistics • First-Step Analysis

STEP III 1994, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s3-q12

The key idea

Look at the games in *pairs*. A lead of 2 can only appear after an even number of games, and after each pair of games one player has served once and the other once, so the serve is back where it started. The match then becomes a simple repeated experiment: each pair is won outright by A, won outright by B, or split, which puts us back at level scores. Standard Tennis is the same thing with the first two pairs treated specially.

Solution

Throughout, write

$$a = p_A p_B, \qquad b = q_A q_B, \qquad s = p_A q_B + q_A p_B.$$

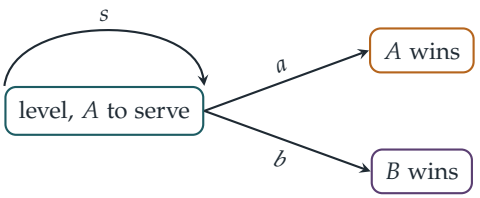
In a pair of consecutive games, one served by each player, a is the probability that A wins both, b that B wins both, and s that they win one each (either A holds serve and B holds serve, or both break). We assume, as the question intends, that the results of different games are independent. These are the only three outcomes, so

$$a + b + s = (p_A + q_A)(p_B + q_B) = 1.$$

Shortened Tennis

The difference in scores changes by ± 1 every game, so it is even after an even number of games and odd after an odd number. A lead of exactly 2 can therefore first occur only after games $2, 4, 6, \ldots$, and if the match is still going after an even number of games the scores are level.

So split the match into pairs (games 1–2, 3–4, ...). Starting from level scores with A to serve, the next pair ends the match in A's favour with probability a, in B's favour with probability b, and with probability s returns us to level scores with A to serve again. The situation is then exactly as at the start, so



$$P_{\text{short}} = a + sP_{\text{short}} \quad \Longrightarrow \quad P_{\text{short}} = \frac{a}{1-s} = \frac{a}{a+b} = \frac{p_A p_B}{p_A p_B + q_A q_B},$$

using $1 - s = a + b$. (Dividing by $1 - s$ is fine unless $s = 1$, the degenerate case where every pair is split and the match never ends.)

If B serves first, each pair consists of a B service game followed by an A service game. The probabilities that A wins both, B wins both, or they split are $p_B p_A$, $q_B q_A$ and $q_B p_A + p_B q_A$: the same a, b, s, because the order of the two games in a pair does not matter. The same argument gives the same answer, so P_{short} does not depend on who serves first.

Decided in 4 games

Nothing can be decided before 4 games have been played. After 4 games (two pairs, A serving first) the difference is even, and the match is decided exactly when the score is 4–0, 3–1, 1–3 or 0–4. In terms of the two pairs:

- 4–0 to A: A wins both pairs, probability $a^2 = p_A^2 p_B^2$;
- 0–4: B wins both pairs, probability $b^2 = q_A^2 q_B^2$;
- 3–1 to A: A wins one pair and the other is split, in either order, probability $2as$;
- 1–3: similarly $2bs$.

Adding,

$$\Pr(\text{decided in 4 games}) = p_A^2 p_B^2 + q_A^2 q_B^2 + 2(p_A p_B + q_A q_B)(p_A q_B + q_A p_B),$$

as required.

Standard Tennis with equal probabilities

If the match is not decided in 4 games, the score is 2–2. This happens with probability $s^2 + 2ab$ (both pairs split, or one pair to each player in either order). From 2–2 the rule "lead by 2" is exactly the Shortened Tennis rule from level scores, and A serves game 5; so from there A wins with probability P_{short}. Hence, in general,

$$P_{\text{stan}} = a^2 + 2as + (s^2 + 2ab)P_{\text{short}}.$$

Now put $p_A = p_B = p$ and $q_A = q_B = q$, so that $a = p^2$, $b = q^2$, $s = 2pq$ and $P_{\text{short}} = \dfrac{p^2}{p^2 + q^2}$. Then

$$P_{\text{stan}} = p^4 + 4p^3 q + \frac{6p^4 q^2}{p^2 + q^2}.$$

Ask yourself Is there a way to compare P_{stan} with P_{short} without expanding everything in powers of p and q?

Yes: unroll Shortened Tennis over the same first four games. Applying $P_{\text{short}} = a + sP_{\text{short}}$ twice,

$$P_{\text{short}} = a + as + s^2 P_{\text{short}}.$$

In words: A wins the first pair; or it is split and A wins the second; or both are split and we start again. Subtracting this from the formula for P_{stan}, the $s^2 P_{\text{short}}$ terms cancel:

$$\begin{aligned} P_{\text{stan}} - P_{\text{short}} &= a^2 + as - a + 2abP_{\text{short}} \\ &= a(a + s - 1) + 2abP_{\text{short}} \\ &= -ab + \frac{2a^2 b}{a+b} \qquad (\text{since } a + s - 1 = -b) \\ &= \frac{ab(a-b)}{a+b}. \end{aligned}$$

This holds for any p_A, p_B. With $a = p^2$ and $b = q^2$,

$$P_{\text{stan}} - P_{\text{short}} = \frac{p^2q^2(p^2 - q^2)}{p^2 + q^2} = \frac{p^2q^2(p-q)(p+q)}{p^2+q^2} = \frac{p^2q^2(p-q)}{p^2+q^2},$$

since $p + q = 1$. This is the required result.

The reason is visible in the working: the two formats differ only when A wins the first pair and B the second, or the other way round. Shortened Tennis awards that to whoever won the first pair; Standard Tennis plays on from 2–2. Both orders have probability ab, so the difference is $ab(2P_{\text{short}} - 1)$, which favours the stronger player.

Checks

- If $p = q = \frac{1}{2}$ the difference is 0, as it must be by symmetry: both probabilities are $\frac{1}{2}$.
- With $p = 0.6$: $P_{\text{short}} = 0.36/0.52 \approx 0.6923$ and $P_{\text{stan}} = 0.1296 + 0.3456 + 0.124416/0.52 \approx 0.7145$, whose difference 0.02215 agrees with $0.0576 \times 0.2/0.52$.
- The probability of a decision in 4 games, $a^2 + b^2 + 2(a+b)s$, and the probability $s^2 + 2ab$ of reaching 2–2 add to $(a+b+s)^2 = 1$, as they should.
- A computer simulation of the actual rules, with $p_A = 0.7$ and $p_B = 0.45$, gives A winning Shortened Tennis about 65.6% of the time whoever serves first, matching $0.315/0.48 = 0.65625$.

Common slips

- Forgetting that 3–1 can arise in two orders (split then win, or win then split), and so missing the factor 2.

- Treating the probability of winning a game as the same for every game: it depends on who serves, which is why grouping into pairs is so useful.
- Writing an infinite geometric series without noting that it is just the equation $P = a + sP$.

Beyond STEP

The pair-by-pair argument is a first example of a *Markov chain* with absorbing states, and the equation $P = a + sP$ is first-step analysis, the standard way to find absorption probabilities. The same ideas give the probability of winning a whole tennis set or a game of squash, and they lead on to gambler's ruin and random walks. Grimmett and Welsh, *Probability: An Introduction*, develops random walks and Markov chains at exactly this level.

WORKED SOLUTION

STEP III 1994, Question 13

Probability and Statistics • Exponential Lifetimes

STEP III 1994, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s3-q13

The key idea

Everything comes from the survival function $\mathrm{P}(T > t) = \mathrm{e}^{-\lambda t}$ of an exponential lifetime. Conditioning on survival to t_0 just divides two such exponentials, which is where the famous *memoryless* property comes from. The time until both ropes fail is the *maximum* of two independent lifetimes, and a maximum is at most t exactly when each lifetime is. So we find its distribution function first and then differentiate. For the last part, "both fail in the same performance" means the two ropes fail in the same interval $((n-1)h,\, nh]$, summed over n. That sum is a geometric series.

Solution

The rope already in use

Let T be the total lifetime of the rope, so for $t \geq 0$

$$\mathrm{P}(T > t) = \int_t^\infty \lambda \mathrm{e}^{-\lambda u}\, \mathrm{d}u = \mathrm{e}^{-\lambda t}.$$

We know the rope has survived t_0 hours, i.e. $T > t_0$. Let $S = T - t_0$ be the further time it lasts. For $s \geq 0$, the event $\{T > t_0 + s\}$ lies inside $\{T > t_0\}$, so

$$\mathrm{P}(S > s \mid T > t_0) = \frac{\mathrm{P}(T > t_0 + s)}{\mathrm{P}(T > t_0)} = \frac{\mathrm{e}^{-\lambda(t_0+s)}}{\mathrm{e}^{-\lambda t_0}} = \mathrm{e}^{-\lambda s}.$$

Hence the remaining time S has density $\lambda \mathrm{e}^{-\lambda s}$ for $s \geq 0$ (and 0 for $s < 0$). It is exponential with the *same* parameter λ, whatever t_0 is.

Interpretation. A used rope is exactly as good as a new one. The rope does not "remember" its past use, and its chance of failing in the next short interval δt is about $\lambda\, \delta t$ at every age (a constant failure rate). That fits failures caused by sudden random accidents. It is not realistic for real ropes, which fray and wear, so an old rope should be *more* likely to fail soon. A model with a failure rate that increases with age would be

better. The artist's insistence on new ropes is sensible in practice, even though under this model it makes no difference.

Time until both new ropes fail

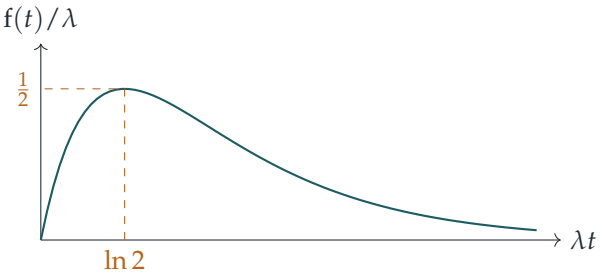
Let T_1, T_2 be the lifetimes of the two new ropes. They are independent, each with distribution function $\mathrm{P}(T_i \le t) = 1 - \mathrm{e}^{-\lambda t}$ for $t \ge 0$. Both have failed by time t exactly when the later failure has happened, so the time we want is $M = \max(T_1, T_2)$. Now $M \le t$ if and only if $T_1 \le t$ *and* $T_2 \le t$, so by independence

$$\mathrm{P}(M \le t) = \mathrm{P}(T_1 \le t)\,\mathrm{P}(T_2 \le t) = \left(1 - \mathrm{e}^{-\lambda t}\right)^2 \qquad (t \ge 0),$$

and $\mathrm{P}(M \le t) = 0$ for $t < 0$. Differentiating with the chain rule,

$$\mathrm{f}(t) = 2\left(1 - \mathrm{e}^{-\lambda t}\right) \cdot \lambda \mathrm{e}^{-\lambda t} = 2\lambda \mathrm{e}^{-\lambda t}\left(1 - \mathrm{e}^{-\lambda t}\right) \quad (t \ge 0),$$

and $\mathrm{f}(t) = 0$ for $t < 0$, as required.



Unlike a single exponential lifetime, M has density 0 at $t = 0$, because two failures cannot both happen at once straight away. The density peaks where $\mathrm{e}^{-\lambda t} = \frac{1}{2}$.

Both ropes failing during the nth performance

Performance n is the interval $\big((n-1)h,\, nh\big]$ of performance time. Write $q = \mathrm{e}^{-\lambda h}$. For one rope,

$$\mathrm{P}\big((n-1)h < T_i \le nh\big) = \mathrm{e}^{-\lambda(n-1)h} - \mathrm{e}^{-\lambda nh} = q^{n-1}(1-q).$$

? **Ask yourself** Should we integrate $\mathrm{f}(t)$ over the nth performance?

No. $\int_{(n-1)h}^{nh} \mathrm{f}(t)\,\mathrm{d}t$ is the probability that the *second* failure happens in performance n. The first rope could have failed in an earlier performance. We need *each* rope to fail in performance n, and the ropes are independent, so

$$p_n = \left(q^{n-1}(1-q)\right)^2 = \mathrm{e}^{-2\lambda(n-1)h}\left(1 - \mathrm{e}^{-\lambda h}\right)^2.$$

Both ropes failing during the same performance

The events "both fail in performance n" for $n = 1, 2, 3, \ldots$ are mutually exclusive, so we add them. This is a geometric series with ratio $q^2 < 1$ (since $\lambda h > 0$):

$$\sum_{n=1}^{\infty} p_n = (1-q)^2 \sum_{n=1}^{\infty} q^{2(n-1)} = \frac{(1-q)^2}{1-q^2} = \frac{1-q}{1+q}.$$

Now multiply top and bottom by $\mathrm{e}^{\lambda h/2}$:

$$\frac{1-\mathrm{e}^{-\lambda h}}{1+\mathrm{e}^{-\lambda h}} = \frac{\mathrm{e}^{\lambda h/2} - \mathrm{e}^{-\lambda h/2}}{\mathrm{e}^{\lambda h/2} + \mathrm{e}^{-\lambda h/2}} = \tanh\left(\frac{\lambda h}{2}\right),$$

which is the required result.

Checks

- $\int_0^\infty \mathrm{f}(t)\,\mathrm{d}t = \left[(1-\mathrm{e}^{-\lambda t})^2\right]_0^\infty = 1$, so f is a genuine density.
- Limits: as $h \to \infty$ (one endless performance) the answer $\tanh(\lambda h/2) \to 1$. As $h \to 0$ it tends to 0, since two continuous lifetimes almost never fall in a tiny common interval. For small h it is about $\lambda h/2$.
- Numerically, with $\lambda = 0.7$ and $h = 0.8$, the series and $\tanh(0.28)$ both give 0.27291. A simulation of 2×10^5 pairs of ropes gave 0.2746, and $p_1 \approx 0.1839$ matched the simulated 0.1848.

Common slips

- Using the *minimum* instead of the maximum. The minimum is the time until the *first* rope fails, and it is exponential with parameter 2λ.
- Integrating f over the nth performance to get p_n. Those probabilities add up to 1, not to $\tanh(\lambda h/2)$.

Beyond STEP

The memoryless property characterises the exponential distribution: it is the only continuous distribution with it. In reliability theory the "failure rate" (hazard function) $\mathrm{f}(t)/\mathrm{P}(T > t)$ is constant only for the exponential. Wear is modelled with increasing hazards, for example by the Weibull distribution. The maximum and minimum of independent variables are the extreme cases of *order statistics*, which you will meet in a first probability course. Grimmett and Welsh, *Probability: An Introduction*, covers all of this.

WORKED SOLUTION

STEP III 1994, Question 14

Probability and Statistics • Geometrical Probability

STEP III 1994, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s3-q14

The key idea

An angle inscribed in a circle is half the angle the opposite arc subtends at the centre. So the three angles of the triangle are just π times the three arcs into which P, Q and R cut the circle, measured as fractions of the circumference. The question becomes one about how a circle is broken into three random pieces, and that is a question about areas inside a triangle.

Solution

Setting up

Measure positions on the circle as fractions of the circumference, so the circle has "length" 1. By rotational symmetry the position of P doesn't matter, so put P at 0. Then Q and R are at independent positions uniformly distributed on $[0,1)$.

Let $U < V$ be the positions of Q and R in increasing order. The pair (U,V) is uniform on the region $0 < u < v < 1$, which has area $\frac{1}{2}$, so its joint density is 2 there. (Ties have probability 0, and so do all the boundary cases below.) The three arcs are

$$X = U, \qquad Y = V - U, \qquad Z = 1 - V, \qquad X + Y + Z = 1.$$

The change of variables $(u,v) \mapsto (x,y) = (u, v-u)$ has Jacobian 1, so (X,Y) has density 2 on the triangle

$$T = \{(x,y) : x > 0,\ y > 0,\ x + y < 1\},$$

which has area $\frac{1}{2}$. So every probability below is *twice an area* inside T.

Ask yourself Why is each angle of the triangle π times an arc?

The angle of the triangle at R is an inscribed angle standing on the arc PQ that does not contain R. By the inscribed angle theorem it is half the angle that arc subtends at the centre. An arc that is a fraction x of the circumference subtends $2\pi x$ at the centre,

so the angle at R is πx. The same holds at each vertex, so the angles of PQR are πX, πY and πZ (and their sum is π, as it should be). Therefore

$$\text{an angle exceeds } k\pi \iff \text{the corresponding arc exceeds } k.$$

Let A_X, A_Y and A_Z be the events $X > k$, $Y > k$ and $Z > k$. We want $\mathrm{P}(A_X \cup A_Y \cup A_Z)$.

One arc exceeding k

For $0 \le k \le 1$:

- A_X is the part of T with $x > k$: a triangle with legs $1-k$, area $\frac{1}{2}(1-k)^2$.
- A_Y is the part with $y > k$: by the same calculation, area $\frac{1}{2}(1-k)^2$.
- A_Z is the part with $x + y < 1 - k$: the corner triangle with legs $1-k$, area $\frac{1}{2}(1-k)^2$.

Doubling, $\mathrm{P}(A_X) = \mathrm{P}(A_Y) = \mathrm{P}(A_Z) = (1-k)^2$.

The case $\frac{1}{2} \le k \le 1$

At most one arc can exceed k: if two did, their total would be more than $2k \ge 1$, but all three arcs add up to 1. So the events A_X, A_Y, A_Z are mutually exclusive, and

$$\mathrm{P}(\text{some angle exceeds } k\pi) = 3(1-k)^2,$$

as required.

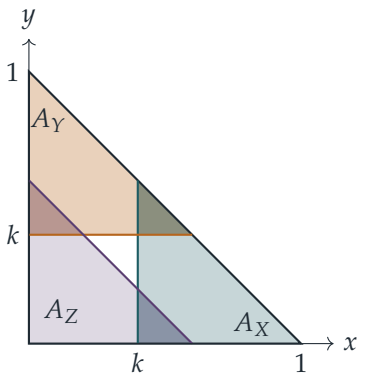
The case $\frac{1}{3} \le k \le \frac{1}{2}$

Now two arcs can both exceed k, so we use inclusion–exclusion.

Two arcs. $A_X \cap A_Y$ is the part of T with $x > k$, $y > k$, $x + y < 1$. Substituting $x = k + s$, $y = k + t$, this is $s, t > 0$, $s + t < 1 - 2k$, a triangle with legs $1 - 2k \ge 0$ and area $\frac{1}{2}(1-2k)^2$. So $\mathrm{P}(A_X \cap A_Y) = (1-2k)^2$. The other two pairs give the same: for example $A_X \cap A_Z$ is $x > k$, $y > 0$, $x + y < 1 - k$, again a triangle with legs $1 - 2k$.

Three arcs. All three exceeding k would need $X + Y + Z > 3k \ge 1$, which is impossible. So the triple intersection has probability 0.

The figure shows T for $k = 0.4$: the three corner triangles are A_X, A_Y, A_Z, and the darker overlaps are the pairwise intersections.



By inclusion–exclusion,

$$\begin{aligned}\mathrm{P}(A_X \cup A_Y \cup A_Z) &= 3(1-k)^2 - 3(1-2k)^2 + 0 \\ &= 3\big[(1-k)-(1-2k)\big]\big[(1-k)+(1-2k)\big] \\ &= 3k(2-3k).\end{aligned}$$

So for $\frac{1}{3} \le k \le \frac{1}{2}$ the probability that at least one angle exceeds $k\pi$ is

$$3k(2-3k) = 6k - 9k^2.$$

Checks

- At $k = \frac{1}{2}$ both formulae give $\frac{3}{4}$, so they join up continuously. This is the classic result that a random inscribed triangle is obtuse with probability $\frac{3}{4}$.
- At $k = \frac{1}{3}$ we get $3 \cdot \frac{1}{3} \cdot 1 = 1$. That is right: the largest angle is always at least $\frac{\pi}{3}$, since the three angles add up to π.
- At $k = 1$ we get 0, as no angle can exceed π.
- A simulation of 4×10^5 random triangles, computing the angles directly from the coordinates, gave 0.9602 at $k = 0.4$ (formula 0.96) and 0.4783 at $k = 0.6$ (formula 0.48).

Common slips

- Using the full arc PQ instead of the arc *not containing* the third vertex. Of the two arcs PQ, only one is the arc opposite R.
- Treating X, Y, Z as independent uniforms. They are constrained by $X + Y + Z = 1$, and the honest way to handle that is the area picture.
- Applying $3(1-k)^2$ below $k = \frac{1}{2}$. For $k < \frac{1}{2}$ it double-counts triangles with two large angles, and at $k = \frac{1}{3}$ it would give $\frac{4}{3} > 1$.

Beyond STEP

The arcs (X, Y, Z) are uniformly distributed on the simplex $x + y + z = 1$. This is the Dirichlet distribution, and it describes the "spacings" left when random points cut up an interval or a circle. With n points the same inclusion–exclusion gives $\mathrm{P}(\text{largest spacing} > k) = \sum_j (-1)^{j+1} \binom{n}{j} (1 - jk)_+^{n-1}$, a formula going back to Whitworth. Grimmett and Welsh, *Probability: An Introduction*, has a good treatment of geometrical probability and joint densities.

Beyond STEP

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